Volume III

Classical Analysis

Continuity

Self-Study Notes

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From Cantor to Ito

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Series. From Cantor to Ito

Volume. Volume III: Classical Analysis

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Chapter 1

Elementary Functions

elementary-functions

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1.1 Hyperbolic Functions

Toolkit — Hyperbolic Functions

- $\cosh x = (e^x + e^{-x})/2$: even part of e^x; $\cosh x \geq 1$; $\cosh 0 = 1$.
- $\sinh x = (e^x - e^{-x})/2$: odd part of e^x; bijective $\mathbb{R} \to \mathbb{R}$.
- $e^x = \cosh x + \sinh x$ for all x.
- Hyperbolic Pythagorean: $\cosh^2 x - \sinh^2 x = 1$ (note the minus sign, not plus).
- $(\sinh x)' = \cosh x$; $(\cosh x)' = \sinh x$ (no sign change, unlike trigonometric).
- $\tanh x \in (-1, 1)$ for all x; $\tanh x \to \pm 1$ as $x \to \pm\infty$.
- Inverse hyperbolic functions have explicit logarithmic formulas.

Hyperbolic Functions

Definition (Hyperbolic Functions)

Definition 1.1.1 (Hyperbolic Functions)**.** For all $x \in \mathbb{R}$:

$$\cosh x := \frac{e^x + e^{-x}}{2}, \qquad \sinh x := \frac{e^x - e^{-x}}{2}, \qquad \tanh x := \frac{\sinh x}{\cosh x}.$$

Additional: $\operatorname{sech} x := 1/\cosh x$, $\operatorname{csch} x := 1/\sinh x$, $\coth x := \cosh x/\sinh x$.

Remark (Standard quantified statement).

$$\forall x \in \mathbb{R}: \ \cosh x = \frac{e^x + e^{-x}}{2} \geq 1, \ \sinh x = \frac{e^x - e^{-x}}{2}, \ e^x = \cosh x + \sinh x.$$

Remark (Definition predicate reading).

$$\mathrm{HyperbolicCosh}(x,y) \;:\equiv\; y = \frac{e^x + e^{-x}}{2}. \qquad \mathrm{HyperbolicSinh}(x,y) \;:\equiv\; y = \frac{e^x - e^{-x}}{2}.$$

$$\mathrm{HyperbolicTanh}(x,y) \;:\equiv\; \cosh x \neq 0 \;\wedge\; y = \frac{\sinh x}{\cosh x}.$$

Remark (Definition predicate reading).

$$\mathrm{HyperbolicPythagorean}(x) \;:\equiv\; \mathrm{HyperbolicCosh}(x,c) \wedge \mathrm{HyperbolicSinh}(x,s) \wedge c^2 - s^2 = 1.$$

Remark (Negated quantified statement).

$$\neg\,\mathrm{HyperbolicTanh}(x,y) \iff y \neq \frac{\sinh x}{\cosh x}.$$

Note $\cosh x > 0$ for all x, so HyperbolicTanh is defined on all of $\mathbb{R}$.

Remark (Contrast with trigonometric functions).

Trigonometric	**Hyperbolic**
$\cos^2 x + \sin^2 x = 1$	$\cosh^2 x - \sinh^2 x = 1$
$(\cos x)' = -\sin x$	$(\cosh x)' = \sinh x$
$(\sin x)' = \cos x$	$(\sinh x)' = \cosh x$
$e^{ix} = \cos x + i \sin x$	$e^x = \cosh x + \sinh x$
Parametrises unit circle	Parametrises unit hyperbola

The minus sign in $\cosh^2 - \sinh^2 = 1$ versus $\cos^2 + \sin^2 = 1$ reflects the substitution $x \mapsto ix$ in Euler's formula.

Proposition (Hyperbolic Pythagorean Identity)

Remark (Dependencies).

- Exponential Function
- $\mathbb{R}$ Is a Field

Proposition 1.1.2 (Hyperbolic Pythagorean Identity). *For all $x \in \mathbb{R}$:*

$$\cosh^2 x - \sinh^2 x = 1.$$

Go to proof.

Remark (Standard quantified statement).

$$\forall x \in \mathbb{R} : \; \cosh^2 x - \sinh^2 x = 1.$$

Remark (Negated quantified statement).

$$\neg\,(\forall x : \cosh^2 x - \sinh^2 x = 1) \iff \exists x \in \mathbb{R} : \cosh^2 x - \sinh^2 x \neq 1.$$

The negation has no witness — the identity holds universally.

Remark (Proof). Direct computation:

$$\cosh^2 x - \sinh^2 x = \frac{(e^x + e^{-x})^2}{4} - \frac{(e^x - e^{-x})^2}{4} = \frac{4}{4} = 1.$$

Proposition (Addition Formulas for Hyperbolic Functions)

Remark (Dependencies).

- Hyperbolic Functions
- Exponential Function
- $\mathbb{R}$ Is a Field

Proposition 1.1.3 (Addition Formulas for Hyperbolic Functions)**.** *For all $x, y \in \mathbb{R}$:*

$$\sinh(x+y) = \sinh x \cosh y + \cosh x \sinh y, \qquad \cosh(x+y) = \cosh x \cosh y + \sinh x \sinh y. \blacksquare$$

Go to proof.

Remark (Standard quantified statement).

$$\forall x, y \in \mathbb{R} : \ \sinh(x+y) = \sinh x \cosh y + \cosh x \sinh y \land \cosh(x+y) = \cosh x \cosh y + \sinh x \sinh y. \blacksquare$$

Remark (Definition predicate reading).

$$\mathrm{HyperbolicAddition}(x, y) \ :\equiv \ \sinh(x+y) = \sinh x \cosh y + \cosh x \sinh y \land \cosh(x+y) = \cosh x \cosh y + \sin$$

Remark (Negated quantified statement).

$$\neg\,(\text{addition formula for } \sinh) \iff \exists x, y \in \mathbb{R} : \ \sinh(x+y) \neq \sinh x \cosh y + \cosh x \sinh y. \blacksquare$$

The negation has no witness.

Remark (Proof strategy). Substitute the definitions and expand, or note the formulas follow from $e^{x+y} = e^x e^y$ by taking even and odd parts.

Remark (Comparison with trigonometric addition formulas). The sinh formula matches $\sin(x+y)$ exactly. The cosh formula differs from $\cos(x+y)$ only in the sign of the last term: $+\sinh x \sinh y$ versus $-\sin x \sin y$. The sign difference is a direct consequence of $\cosh^2 - \sinh^2 = 1$ versus $\cos^2 + \sin^2 = 1$.

Proposition (Limits of $\tanh$)

Remark (Dependencies).

- Hyperbolic Functions
- Exponential Function
- Addition on $\mathbb{R}$
- Multiplication on $\mathbb{R}$

Proposition 1.1.4 (Limits of tanh)**.**

$$\lim_{x \to +\infty} \tanh x = 1, \qquad \lim_{x \to -\infty} \tanh x = -1.$$

Go to proof.

Remark (Standard quantified statement).

$$\mathrm{LimitAtInfinity}(\tanh x,\ +\infty,\ 1)\ \wedge\ \mathrm{LimitAtInfinity}(\tanh x,\ -\infty,\ -1).$$

Remark (Definition predicate reading).

$$\mathrm{LimitAtInfinity}(\tanh x,\ +\infty,\ 1)\ :\equiv\ \forall \varepsilon > 0,\ \exists M \in \mathbb{R},\ \forall x:\ x > M \Rightarrow \mathrm{OutputTolerance}(\tanh x, 1, \varepsilon).$$

Remark (Negated quantified statement).

$$\neg\left(\lim_{x\to+\infty} \tanh x = 1\right) \iff \exists \varepsilon_0 > 0,\ \forall M > 0,\ \exists x > M:\ |\tanh x - 1| \geq \varepsilon_0.$$

Remark (Proof strategy). Write $\tanh x = (1 - e^{-2x})/(1 + e^{-2x})$. As $x \to +\infty$, $e^{-2x} \to 0$, giving $\tanh x \to 1$. The limit at $-\infty$ follows by oddness: $\tanh(-x) = -\tanh(x)$.

Definition

Remark (Dependencies).

- Hyperbolic Functions
- Exponential Function
- Limit at Infinity
- Limit of a Quotient

Definition 1.1.5 (Inverse Hyperbolic Functions)**.** Since sinh is strictly increasing and bijective $\mathbb{R} \to \mathbb{R}$, its inverse arcsinh is defined on all of $\mathbb{R}$. Since cosh is strictly increasing on $[0, \infty)$ with range $[1, \infty)$, its inverse arccosh is defined on $[1, \infty)$. Since tanh is strictly increasing with range $(-1, 1)$, arctanh is defined on $(-1, 1)$.
Explicit logarithmic formulas:

$$\operatorname{arcsinh} x = \ln\left(x + \sqrt{x^2+1}\right), \quad x \in \mathbb{R}.$$
$$\operatorname{arccosh} x = \ln\left(x + \sqrt{x^2-1}\right), \quad x \geq 1.$$
$$\operatorname{arctanh} x = \tfrac{1}{2}\ln\left(\frac{1+x}{1-x}\right), \quad |x| < 1.$$

Remark (Standard quantified statement).

$$\forall x \in \mathbb{R}:\ \sinh(\operatorname{arcsinh} x) = x\ \wedge\ \operatorname{arcsinh}(\sinh x) = x.$$

$$\forall x \geq 1:\ \cosh(\operatorname{arccosh} x) = x. \qquad \forall |x| < 1:\ \tanh(\operatorname{arctanh} x) = x.$$

Remark (Definition predicate reading).

$$\mathrm{Arcsinh}(y, x)\ :\equiv\ x \in \mathbb{R}\ \wedge\ \sinh x = y\ :\equiv\ x = \ln\left(y + \sqrt{y^2+1}\right).$$

$$\mathrm{Arctanh}(y, x)\ :\equiv\ |y| < 1\ \wedge\ \tanh x = y\ :\equiv\ x = \tfrac{1}{2}\ln\left(\frac{1+y}{1-y}\right).$$

Remark (Negated quantified statement).

$$\neg\ \mathrm{Arctanh}(y,x) \iff |y| \geq 1\ \vee\ \tanh x \neq y.$$

Remark (Derivation of arcsinh). Set $y = \sinh x = (e^x - e^{-x})/2$ and let $u = e^x > 0$. Then $y = (u - 1/u)/2$, so $u^2 - 2yu - 1 = 0$. The positive root is $u = y + \sqrt{y^2+1}$, giving $x = \ln(y + \sqrt{y^2+1})$.

Remark (Derivatives). $(\operatorname{arcsinh} x)' = 1/\sqrt{x^2+1}$; $(\operatorname{arccosh} x)' = 1/\sqrt{x^2-1}$ for $x > 1$; $(\operatorname{arctanh} x)' = 1/(1-x^2)$ for $|x| < 1$. These follow from the inverse function theorem applied to sinh, cosh, and tanh respectively.

Remark (Dependencies).

- Hyperbolic Functions
- Natural Logarithm
- Continuous Inverse Theorem
- Absolute Value on $\mathbb{R}$

1.2 The Logarithm

Toolkit — Logarithmic Functions

- $\ln : (0,\infty) \to \mathbb{R}$: inverse of e^x. Equivalently, $\ln x = \int_1^x 1/t\,dt$.
- Functional equation: $\ln(xy) = \ln x + \ln y$. Converts products to sums.
- $(\ln x)' = 1/x$; $\ln 1 = 0$; $\ln e = 1$.
- Change of base: $\log_a x = \ln x / \ln a$.
- Fundamental logarithmic limit: $\lim_{x\to 0} \ln(1+x)/x = 1$.
- Growth dominance: for all $r > 0$, $(\ln x)/x^r \to 0$ as $x \to \infty$ and $x^r \ln x \to 0$ as $x \to 0^+$.

The Logarithm

Definition (Natural Logarithm)

Definition 1.2.1 (Natural Logarithm)**.** The **natural logarithm** $\ln : (0,\infty) \to \mathbb{R}$ is the inverse of the exponential function: for $x > 0$,

$$\ln x \ := \ \text{the unique } y \in \mathbb{R} \text{ such that } e^y = x.$$

Equivalently, $\ln x$ is defined by the integral

$$\ln x \ := \ \int_1^x \frac{1}{t}\,dt.$$

Remark (Standard quantified statement).

$$\forall x > 0 : \; e^{\ln x} = x. \qquad \forall y \in \mathbb{R} : \; \ln(e^y) = y.$$

Remark (Definition predicate reading).

$$\mathrm{NaturalLog}(x, y) \;:\equiv\; x > 0 \;\wedge\; e^y = x \;:\equiv\; x > 0 \;\wedge\; y = \int_1^x \frac{1}{t}\,dt.$$

Remark (Definition predicate reading).

$$\mathrm{FunctionLimit}\left(\frac{\ln(1+x)}{x},\, 0,\, 1\right) \;:\equiv\; \forall \varepsilon > 0,\; \exists \delta > 0,\; \forall x : \; \mathrm{InputTolerance}(x, 0, \delta) \Rightarrow \mathrm{OutputTolerance}$$

Remark (Negated quantified statement).

$$\neg\, \mathrm{NaturalLog}(x, y) \iff x \le 0 \;\vee\; e^y \ne x.$$

Remark (Two definitions, same function). The inverse-of-exponential definition requires e^x to be established first. The integral definition is self-contained and is preferred when building analysis from scratch. Both yield the same function.

Remark (Key properties). (i) $\ln 1 = 0$; $\ln e = 1$.

(ii) **Functional equation:** $\ln(xy) = \ln x + \ln y$ for all $x, y > 0$.

(iii) $\ln(x/y) = \ln x - \ln y$; $\ln(x^r) = r \ln x$ for $r \in \mathbb{R}$.

(iv) $\ln$ is strictly increasing on $(0, \infty)$.

(v) $\ln x \to +\infty$ as $x \to \infty$; $\ln x \to -\infty$ as $x \to 0^+$.

(vi) $(\ln x)' = 1/x$ for all $x > 0$.

Remark (General logarithm). For $a > 0$, $a \ne 1$, the **logarithm base** a is $\log_a x := \ln x / \ln a$. The common logarithm is $\log_{10}$; the binary logarithm is $\log_2$.

Proposition (Fundamental Logarithmic Limit)

Remark (Dependencies).

- Exponential Function
- Inverse Bijection
- Continuous Inverse Theorem

Proposition 1.2.2 (Fundamental Logarithmic Limit).

$$\lim_{x \to 0} \frac{\ln(1+x)}{x} = 1.$$

Go to proof.

Remark (Standard quantified statement).

$$\forall \varepsilon > 0,\ \exists \delta > 0 :\ 0 < |x| < \delta \Rightarrow \left| \frac{\ln(1+x)}{x} - 1 \right| < \varepsilon.$$

Remark (Negated quantified statement).

$$\neg \left(\lim_{x \to 0} \frac{\ln(1+x)}{x} = 1 \right) \iff \exists \varepsilon_0 > 0,\ \forall \delta > 0,\ \exists x :\ 0 < |x| < \delta \wedge \left| \frac{\ln(1+x)}{x} - 1 \right| \geq \varepsilon_0.$$

Remark (Equivalence with exponential limit). The fundamental exponential limit $\lim(e^x - 1)/x = 1$ and the fundamental logarithmic limit $\lim \ln(1+x)/x = 1$ are equivalent: each follows from the other via the substitution $x \leftrightarrow e^x - 1$. Both express the first-order Taylor approximation of the respective function at 0.

Remark (Proof strategy). From the series: $\ln(1+x) = x - x^2/2 + x^3/3 - \cdots$ (valid for $|x| < 1$). Dividing by x gives $1 - x/2 + x^2/3 - \cdots \to 1$ as $x \to 0$ by continuity of power series.

Proposition (Every Power Dominates the Logarithm)

Remark (Dependencies).

- Natural Logarithm
- Fundamental Exponential Limit
- Function Limit

Proposition 1.2.3 (Every Power Dominates the Logarithm)**.** *For every $r > 0$:*

$$\lim_{x \to \infty} \frac{\ln x}{x^r} = 0 \qquad \textit{and} \qquad \lim_{x \to 0^+} x^r \ln x = 0.$$

Go to proof.

Remark (Standard quantified statement).

$$\forall r > 0 :\ \lim_{x \to \infty} x^{-r} \ln x = 0 \ \wedge\ \lim_{x \to 0^+} x^r \ln x = 0.$$

Remark (Definition predicate reading).

$$\mathrm{LimitAtInfinity}(x^{-r} \ln x,\ +\infty,\ 0)\ :\equiv\ \forall \varepsilon > 0,\ \exists M \in \mathbb{R},\ \forall x :\ x > M \Rightarrow \mathrm{OutputTolerance}(x^{-r} \ln x, 0, \varepsilon)$$

Remark (Negated quantified statement).

$$\neg \left(\lim_{x \to \infty} \frac{\ln x}{x^r} = 0 \right) \iff \exists \varepsilon_0 > 0,\ \forall M > 0,\ \exists x > M :\ x^{-r} |\ln x| \geq \varepsilon_0.$$

Remark (Proof strategy). For the first: substitute $x = e^t$ to get $t/e^{rt} \to 0$ by exponential dominance. For the second: substitute $x = 1/t$ and reduce to the first: $x^r \ln x = (-\ln t)/t^r \to 0$.

Remark (Significance)*.* These limits establish the position of $\ln x$ in the growth hierarchy: $\ln x$ is dominated by every positive power. Combined with the exponential dominance result, the full ordering is $\ln x \ll x^r \ll e^x$ as $x \to \infty$ for any $r > 0$. The second limit says that near 0^+, the positive power x^r wins over the logarithmic blow-up $|\ln x| \to \infty$. Used in integrability arguments near 0.

Remark (Dependencies)*.*

- Natural Logarithm
- Power Function
- Exponential Dominates Every Power
- Limit at Infinity

1.3 Power Functions and the Exponential

Toolkit — Power and Exponential Functions

- $e^x := \sum_{n=0}^{\infty} x^n/n!$: absolutely convergent for all $x \in \mathbb{R}$; functional equation $e^{x+y} = e^x e^y$; self-derivative $(e^x)' = e^x$.
- $e := \sum_{n=0}^{\infty} 1/n! = \lim_{n\to\infty}(1 + 1/n)^n$.
- Real power: $x^r := e^{r \ln x}$ for $x > 0$, $r \in \mathbb{R}$. Unifies integer, rational, and irrational exponents.
- Growth hierarchy: $\ln x \ll x^r \ll e^x$ as $x \to \infty$ for any $r > 0$.
- Fundamental exponential limit: $\lim_{x\to 0}(e^x - 1)/x = 1$.

Power Functions and the Exponential

Definition (Power Function)

Definition 1.3.1 (Power Function)**.** Let $n \in \mathbb{N}$. The **integer power function** is $f : \mathbb{R} \to \mathbb{R}$, $f(x) := x^n$. For $r \in \mathbb{R}$ and $x > 0$, the **real power** is defined by

$$x^r \;:=\; e^{r \ln x}.$$

Remark (Standard quantified statement)*.*

$$\forall r \in \mathbb{R},\ \forall x > 0 :\ x^r = e^{r \ln x}.$$

For $r \in \mathbb{N}$ this reduces to the r-fold product $x \cdot x \cdots x$.

Remark (Definition predicate reading)*.*

$$\mathrm{RealPower}(x, r, y) \;:\equiv\; x > 0 \;\wedge\; y = e^{r \ln x}.$$

Remark (Laws of exponents). For $x, y > 0$ and $r, s \in \mathbb{R}$:

$$x^r \cdot x^s = x^{r+s}, \quad (x^r)^s = x^{rs}, \quad (xy)^r = x^r y^r, \quad x^0 = 1, \quad x^1 = x.$$

Each follows from the corresponding property of the exponential via the definition $x^r = e^{r \ln x}$.

Remark (Interpretation). Defining $x^r := e^{r \ln x}$ unifies all three cases — integer, rational, irrational exponents — into a single formula and makes differentiability of x^r immediate from that of e^x and $\ln x$ via the chain rule: $(x^r)' = rx^{r-1}$.

Definition (The Number e)

Remark (Dependencies).

- Integer Powers in a Field
- Exponential Function
- Natural Logarithm
- Multiplication on $\mathbb{R}$

Definition 1.3.2 (The Number e)**.** The **Euler number** is

$$e \;:=\; \sum_{n=0}^{\infty} \frac{1}{n!} \;=\; \lim_{n\to\infty} \left(1 + \frac{1}{n}\right)^n \;\approx\; 2.71828\ldots$$

Both expressions define the same real number.

Remark (Standard quantified statement).

$$e = \sum_{n=0}^{\infty} \frac{1}{n!} \;\wedge\; e = \lim_{n\to\infty} \left(1 + \frac{1}{n}\right)^n.$$

Remark (Interpretation). The series $\sum 1/n!$ converges rapidly by comparison with a geometric series. The limit $(1 + 1/n)^n$ arises from continuous compound interest and is proved equal to the series sum via the fundamental logarithmic limit. Both representations are in common use.

Remark (Dependencies).

- Sequence
- Convergent Sequence

Definition (Exponential Function)

Definition 1.3.3 (Exponential Function)**.** The **exponential function** $\exp : \mathbb{R} \to (0, \infty)$ is defined by the absolutely convergent power series

$$e^x \;:=\; \exp(x) \;:=\; \sum_{n=0}^{\infty} \frac{x^n}{n!}, \qquad x \in \mathbb{R}.$$

Remark (Standard quantified statement).

$$\forall x \in \mathbb{R}: \; e^x = \sum_{n=0}^{\infty} \frac{x^n}{n!}, \quad \text{convergent absolutely since } \sum \frac{|x|^n}{n!} < \infty.$$

Remark (Definition predicate reading).

$$\text{Exponential}(x, y) \; :\equiv \; y = \sum_{n=0}^{\infty} \frac{x^n}{n!}.$$

Remark (Key properties). (i) $e^0 = 1$; $e^1 = e$.

(ii) **Functional equation:** $e^{x+y} = e^x \cdot e^y$ for all $x, y \in \mathbb{R}$.

(iii) $e^x > 0$ for all x; $e^{-x} = 1/e^x$.

(iv) e^x is strictly increasing and bijective $\mathbb{R} \to (0, \infty)$.

(v) $(e^x)' = e^x$ — the exponential is its own derivative.

Remark (General exponential). For $a > 0$, $a \neq 1$: $a^x := e^{x \ln a}$. Strictly increasing if $a > 1$, strictly decreasing if $0 < a < 1$. $(a^x)' = a^x \ln a$. Only e^x is its own derivative.

Proposition (Fundamental Exponential Limit)

Remark (Dependencies).

- The Number e
- Integer Powers in a Field
- Sequence
- Convergent Sequence

Proposition 1.3.4 (Fundamental Exponential Limit)**.**

$$\lim_{x \to 0} \frac{e^x - 1}{x} = 1.$$

Go to proof.

Remark (Standard quantified statement).

$$\forall \varepsilon > 0, \; \exists \delta > 0: \; 0 < |x| < \delta \Rightarrow \left| \frac{e^x - 1}{x} - 1 \right| < \varepsilon.$$

Remark (Definition predicate reading).

$$\text{FunctionLimit}\left(\frac{e^x - 1}{x}, \; 0, \; 1\right) \; :\equiv \; \forall \varepsilon > 0, \; \exists \delta > 0, \; \forall x: \; \text{InputTolerance}(x, 0, \delta) \Rightarrow \text{OutputTolerance}\Bigg(\frac{e}{\;}$$

Remark (Negated quantified statement).

$$\neg \left(\lim_{x \to 0} \frac{e^x - 1}{x} = 1 \right) \iff \exists \varepsilon_0 > 0, \; \forall \delta > 0, \; \exists x: \; 0 < |x| < \delta \; \wedge \; \left| \frac{e^x - 1}{x} - 1 \right| \geq \varepsilon_0.$$

Remark (Interpretation). This is the statement $(e^x)'\big|_{x=0} = 1$. The series gives $(e^x - 1)/x = 1 + x/2! + x^2/3! + \cdots \to 1$ as $x \to 0$ by continuity of power series at their center. Combined with the functional equation, this implies $(e^x)' = e^x$ everywhere.

Proposition (Compound Interest Limit)

Remark (Dependencies).

- Exponential Function
- Function Limit
- Derivative at a Point

Proposition 1.3.5 (Compound Interest Limit)**.**

$$\lim_{x\to\infty}\left(1+\frac{1}{x}\right)^x = e.$$

Go to proof.

Remark (Standard quantified statement).

$$\forall \varepsilon > 0,\ \exists M > 0 :\ x > M \Rightarrow \left|\left(1+\frac{1}{x}\right)^x - e\right| < \varepsilon.$$

Remark (Definition predicate reading).

$$\mathrm{LimitAtInfinity}\big(\big(1+\tfrac{1}{x}\big)^x,\ +\infty,\ e\big)\ :\equiv\ \forall \varepsilon > 0,\ \exists M \in \mathbb{R},\ \forall x :\ x > M \Rightarrow \mathrm{OutputTolerance}\big(\big(1+\tfrac{1}{x}\big)^x, e$$

Remark (Negated quantified statement).

$$\neg\left(\lim_{x\to\infty}\big(1+\tfrac{1}{x}\big)^x = e\right) \iff \exists \varepsilon_0 > 0,\ \forall M > 0,\ \exists x > M :\ \left|\left(1+\frac{1}{x}\right)^x - e\right| \geq \varepsilon_0.$$

Remark (Proof strategy). Write $(1+1/x)^x = e^{x\ln(1+1/x)}$. Set $t = 1/x \to 0^+$ and use the fundamental logarithmic limit $\ln(1+t)/t \to 1$. Then $x\ln(1+1/x) = \ln(1+t)/t \to 1$, so $(1+1/x)^x = e^{x\ln(1+1/x)} \to e^1 = e$ by continuity of the exponential.

Proposition (Exponential Dominates Every Power)

Remark (Dependencies).

- The Number e
- Exponential Function
- Natural Logarithm
- Fundamental Logarithmic Limit
- Limit at Infinity

Proposition 1.3.6 (Exponential Dominates Every Power)**.** *For every $r > 0$:*

$$\lim_{x\to\infty} \frac{x^r}{e^x} = 0.$$

Go to proof.

Remark (Standard quantified statement).

$$\forall r > 0 : \lim_{x\to\infty} x^r e^{-x} = 0.$$

Remark (Definition predicate reading).

$$\mathrm{LimitAtInfinity}(x^r e^{-x},\ +\infty,\ 0) \ :\equiv\ \forall \varepsilon > 0,\ \exists M \in \mathbb{R},\ \forall x :\ x > M \Rightarrow \mathrm{OutputTolerance}(x^r e^{-x}, 0, \varepsilon).$$

Remark (Negated quantified statement).

$$\neg\left(\lim_{x\to\infty} \frac{x^r}{e^x} = 0\right) \iff \exists \varepsilon_0 > 0,\ \forall M > 0,\ \exists x > M :\ x^r e^{-x} \geq \varepsilon_0.$$

Remark (Proof strategy). Pick integer $n > r$. The series bound $e^x \geq x^n/n!$ gives $x^r/e^x \leq n!\,x^{r-n} \to 0$ since $r - n < 0$.

Remark (Growth hierarchy). For any $r > 0$:

$$\ln x \ \ll\ x^r \ \ll\ e^x \qquad \text{as } x \to \infty.$$

The exponential eventually dominates every polynomial and every power function. This ordering is used constantly in estimates, dominated convergence arguments, and comparison tests.

Remark (Dependencies).

- Power Function
- Exponential Function
- Limit at Infinity
- Squeeze Theorem for Function Limits

1.4 Trigonometric Functions

Toolkit — Trigonometric Functions

- $\sin x$ and $\cos x$ are defined by absolutely convergent power series; properties are derived analytically, not from geometry.
- Pythagorean identity: $\sin^2 x + \cos^2 x = 1$. Proof: the derivative of the left side is 0 and its value at 0 is 1.
- Addition formulas follow from $e^{i(x+y)} = e^{ix}e^{iy}$.
- Fundamental trigonometric limit: $\lim_{x\to 0} \sin x/x = 1$.
- Second fundamental limit: $\lim_{x\to 0}(1-\cos x)/x^2 = 1/2$.
- Inverse functions arcsin, arccos, arctan are defined by restricting sin, cos, tan to monotone subdomains.

Trigonometric Functions

Definition (Sine and Cosine)

Definition 1.4.1 (Sine and Cosine via Power Series)**.** The **sine** and **cosine** functions are defined for all $x \in \mathbb{R}$ by the absolutely convergent power series

$$\sin x := \sum_{n=0}^{\infty} \frac{(-1)^n x^{2n+1}}{(2n+1)!} = x - \frac{x^3}{3!} + \frac{x^5}{5!} - \cdots$$

$$\cos x := \sum_{n=0}^{\infty} \frac{(-1)^n x^{2n}}{(2n)!} = 1 - \frac{x^2}{2!} + \frac{x^4}{4!} - \cdots$$

Remark (Standard quantified statement).

$$\forall x \in \mathbb{R}: \ \sin x = \sum_{n=0}^{\infty} \frac{(-1)^n x^{2n+1}}{(2n+1)!}, \quad \cos x = \sum_{n=0}^{\infty} \frac{(-1)^n x^{2n}}{(2n)!},$$

both absolutely convergent by the ratio test.

Remark (Definition predicate reading).

$$\mathrm{Sine}(x,y) \ :\equiv \ y = \sum_{n=0}^{\infty} \frac{(-1)^n x^{2n+1}}{(2n+1)!}. \qquad \mathrm{Cosine}(x,y) \ :\equiv \ y = \sum_{n=0}^{\infty} \frac{(-1)^n x^{2n}}{(2n)!}.$$

Remark (Key properties). (i) $\sin 0 = 0$; $\cos 0 = 1$.

(ii) $\sin(-x) = -\sin x$ (odd); $\cos(-x) = \cos x$ (even).

(iii) $(\sin x)' = \cos x$; $(\cos x)' = -\sin x$.

(iv) $|\sin x| \le 1$; $|\cos x| \le 1$ for all x.

(v) $\sin^2 x + \cos^2 x = 1$ (Pythagorean identity).

(vi) Periodic: $\sin(x + 2\pi) = \sin x$; $\cos(x + 2\pi) = \cos x$.

Remark (Relationship to e^{ix}). Euler's formula: $e^{ix} = \cos x + i \sin x$, where e^{ix} is defined by the complex exponential series. This gives $\cos x = \mathrm{Re}(e^{ix})$ and $\sin x = \mathrm{Im}(e^{ix})$. Euler's identity $e^{i\pi} + 1 = 0$ follows immediately.

Proposition (Pythagorean Identity)

Remark (Dependencies).

- Real Numbers
- Integer Powers in a Field
- Sequence
- Convergent Sequence

Proposition 1.4.2 (Pythagorean Identity)**.** *For all $x \in \mathbb{R}$:*

$$\sin^2 x + \cos^2 x = 1.$$

Go to proof.

Remark (Standard quantified statement).

$$\forall x \in \mathbb{R} : \ \sin^2 x + \cos^2 x = 1.$$

Remark (Definition predicate reading).

$$\mathrm{PythagoreanIdentity}(x) \ :\equiv \ \mathrm{Sine}(x, s) \ \wedge \ \mathrm{Cosine}(x, c) \ \wedge \ s^2 + c^2 = 1.$$

Remark (Negated quantified statement).

$$\neg \left(\forall x : \sin^2 x + \cos^2 x = 1\right) \iff \exists x \in \mathbb{R} : \sin^2 x + \cos^2 x \neq 1.$$

The negation has no witness — the identity holds universally.

Remark (Proof strategy). Define $f(x) := \sin^2 x + \cos^2 x$. Then $f'(x) = 2 \sin x \cos x + 2 \cos x(-\sin x) = 0$, so f is constant. Since $f(0) = 0 + 1 = 1$, the identity follows.

Proposition (Addition Formulas)

Remark (Dependencies).

- Sine and Cosine
- Derivative at a Point
- Zero Derivative Implies Constant

Proposition 1.4.3 (Addition Formulas)**.** *For all $x, y \in \mathbb{R}$:*

$$\sin(x + y) = \sin x \cos y + \cos x \sin y, \qquad \cos(x + y) = \cos x \cos y - \sin x \sin y.$$

Go to proof.

Remark (Standard quantified statement).

$$\forall x, y \in \mathbb{R} : \ \sin(x+y) = \sin x \cos y + \cos x \sin y \ \wedge \ \cos(x+y) = \cos x \cos y - \sin x \sin y.$$

Remark (Definition predicate reading).

$$\mathrm{TrigAddition}(x, y) \ :\equiv \ \sin(x+y) = \sin x \cos y + \cos x \sin y \wedge \cos(x+y) = \cos x \cos y - \sin x \sin y.$$

Remark (Negated quantified statement).

$$\neg\,(\text{addition formula for } \sin) \iff \exists x, y \in \mathbb{R} : \ \sin(x+y) \neq \sin x \cos y + \cos x \sin y.$$

The negation has no witness.

Remark (Proof strategy). From the Cauchy product: $e^{i(x+y)} = e^{ix}e^{iy}$. Taking real and imaginary parts yields both addition formulas simultaneously.

Remark (Derived identities).

$$\begin{aligned}
\sin(2x) &= 2 \sin x \cos x \\
\cos(2x) &= \cos^2 x - \sin^2 x = 1 - 2\sin^2 x = 2\cos^2 x - 1 \\
\sin^2 x &= \tfrac{1}{2}(1 - \cos 2x) \\
\cos^2 x &= \tfrac{1}{2}(1 + \cos 2x)
\end{aligned}$$

Definition

Remark (Dependencies).

- Sine and Cosine
- Addition on $\mathbb{R}$
- Multiplication on $\mathbb{R}$

Definition 1.4.4 (Tangent, Cotangent, Secant, Cosecant)**.**

$$\tan x := \frac{\sin x}{\cos x}, \quad \cot x := \frac{\cos x}{\sin x}, \quad \sec x := \frac{1}{\cos x}, \quad \csc x := \frac{1}{\sin x}.$$

$\tan x$ and $\sec x$ are defined where $\cos x \neq 0$, i.e., $x \neq \pi/2 + k\pi$ for $k \in \mathbb{Z}$.

Remark (Standard quantified statement).

$$\forall x \in \mathbb{R}, \ \cos x \neq 0 : \ \tan x = \frac{\sin x}{\cos x}.$$

Remark (Definition predicate reading).

$$\mathrm{Tangent}(x, y) \ :\equiv \ \cos x \neq 0 \ \wedge \ y = \frac{\sin x}{\cos x}.$$

Remark (Pythagorean identities for tan). Dividing $\sin^2 x + \cos^2 x = 1$ by $\cos^2 x$: $1 + \tan^2 x = \sec^2 x$. Dividing by $\sin^2 x$: $\cot^2 x + 1 = \csc^2 x$.

Definition

Remark (Dependencies).

- Sine and Cosine
- $\mathbb{R}$ Is a Field

Definition 1.4.5 (Inverse Trigonometric Functions).

- $\arcsin : [-1,1] \to [-\pi/2, \pi/2]$ is the inverse of $\sin\big|_{[-\pi/2,\pi/2]}$.
- $\arccos : [-1,1] \to [0,\pi]$ is the inverse of $\cos\big|_{[0,\pi]}$.
- $\arctan : \mathbb{R} \to (-\pi/2, \pi/2)$ is the inverse of $\tan\big|_{(-\pi/2,\pi/2)}$.

Remark (Standard quantified statement).

$$\forall y \in [-1,1] :\ \sin(\arcsin y) = y. \qquad \forall x \in [-\pi/2, \pi/2] :\ \arcsin(\sin x) = x.$$

Remark (Definition predicate reading).

$$\operatorname{Arctan}(y,x) \ :\equiv\ x \in (-\pi/2, \pi/2) \ \wedge\ \tan x = y.$$

Remark (Definition predicate reading).

$$\operatorname{FunctionLimit}\left(\frac{\sin x}{x},\, 0,\, 1\right) \ :\equiv\ \forall \varepsilon > 0,\ \exists \delta > 0,\ \forall x :\ \operatorname{InputTolerance}(x, 0, \delta) \Rightarrow \operatorname{OutputTolerance}\left(\frac{\sin x}{x}\right.$$

Remark (Negated quantified statement).

$$\neg \operatorname{Arctan}(y,x) \iff x \notin (-\pi/2, \pi/2) \ \vee\ \tan x \neq y.$$

Remark (Key limits of arctan).

$$\lim_{x\to+\infty} \arctan x = \frac{\pi}{2}, \qquad \lim_{x\to-\infty} \arctan x = -\frac{\pi}{2}.$$

arctan is bounded, strictly increasing, and continuous on $\mathbb{R}$; the Monotone Limit Theorem gives the limits as the supremum and infimum of its range.

Theorem (Fundamental Trigonometric Limit)

Remark (Dependencies).

- Sine and Cosine
- Tangent, Cotangent, Secant, Cosecant
- Continuous Inverse Theorem
- Closed Interval

Theorem 1.4.6 (Fundamental Trigonometric Limit).

$$\lim_{x\to 0} \frac{\sin x}{x} = 1.$$

Go to proof.

Remark (Standard quantified statement).

$$\forall \varepsilon > 0,\ \exists \delta > 0 :\ 0 < |x| < \delta \Rightarrow \left| \frac{\sin x}{x} - 1 \right| < \varepsilon.$$

Remark (Negated quantified statement).

$$\neg \left(\lim_{x \to 0} \frac{\sin x}{x} = 1 \right) \iff \exists \varepsilon_0 > 0,\ \forall \delta > 0,\ \exists x :\ 0 < |x| < \delta \ \wedge\ \left| \frac{\sin x}{x} - 1 \right| \geq \varepsilon_0.$$

Remark (Proof strategy — series). From the series: $\sin x/x = 1 - x^2/3! + x^4/5! - \cdots \to 1$ as $x \to 0$ by continuity of power series at their center.

Remark (Proof strategy — geometric squeeze). For $0 < x < \pi/2$, area comparison of two triangles and a circular sector gives $\cos x \leq \sin x/x \leq 1$. Since $\cos x \to 1$ as $x \to 0^+$, the Squeeze Theorem gives $\sin x/x \to 1$. Even symmetry extends this to $x \to 0^-$.

Remark (Significance). This is the statement $(\sin x)'\big|_{x=0} = 1$, which implies $(\sin x)' = \cos x$ everywhere. It underlies all trigonometric differentiation.

Proposition (Second Fundamental Trigonometric Limit)

Remark (Dependencies).

- Sine and Cosine
- Function Limit
- Squeeze Theorem for Function Limits

Proposition 1.4.7 (Second Fundamental Trigonometric Limit)**.**

$$\lim_{x \to 0} \frac{1 - \cos x}{x^2} = \frac{1}{2}.$$

Go to proof.

Remark (Standard quantified statement).

$$\forall \varepsilon > 0,\ \exists \delta > 0 :\ 0 < |x| < \delta \Rightarrow \left| \frac{1 - \cos x}{x^2} - \frac{1}{2} \right| < \varepsilon.$$

Remark (Definition predicate reading).

$$\mathrm{FunctionLimit}\left(\frac{1 - \cos x}{x^2},\ 0,\ \tfrac{1}{2} \right) \ :\equiv\ \forall \varepsilon > 0,\ \exists \delta > 0,\ \forall x :\ \mathrm{InputTolerance}(x, 0, \delta) \Rightarrow \mathrm{OutputTolerance}$$

Remark (Negated quantified statement).

$$\neg \left(\lim_{x \to 0} \frac{1 - \cos x}{x^2} = \frac{1}{2} \right) \iff \exists \varepsilon_0 > 0,\ \forall \delta > 0,\ \exists x :\ 0 < |x| < \delta \wedge \left| \frac{1 - \cos x}{x^2} - \frac{1}{2} \right| \geq \varepsilon_0.$$

Remark (Proof strategy). Use the half-angle identity $1 - \cos x = 2\sin^2(x/2)$:

$$\frac{1 - \cos x}{x^2} = \frac{2\sin^2(x/2)}{x^2} = \frac{1}{2} \left(\frac{\sin(x/2)}{x/2} \right)^2 \to \frac{1}{2} \cdot 1^2 = \frac{1}{2}.$$

Remark (Dependencies).

- Sine and Cosine
- Trigonometric Addition Formulas
- Fundamental Trigonometric Limit
- Function Limit

Chapter 2

Continuity

continuity

Bounds $\to$ **Continuity** $\to$ Differentiation

2.1 Limits

Toolkit — One-Sided Limits

- RightLimit(f, c, L): approach from $x > c$ only. Input condition: $0 < x - c < \delta$.
- LeftLimit(f, c, L): approach from $x < c$ only. Input condition: $0 < c - x < \delta$.
- Sequential criteria hold for each direction separately.
- FunctionLimit$(f, c, L) \iff$ RightLimit$(f, c, L) \wedge$ LeftLimit(f, c, L).

Toolkit — Limits of Functions

- $\lim_{x \to c} f(x) = L$ requires c to be a cluster point of $\mathrm{dom}(f)$. The value $f(c)$ is irrelevant; f need not be defined at c.
- The input condition is punctured: $0 < |x - c| < \delta$ means $0 < |x - c| < \delta$. This is what separates limits from continuity, where Within(x, c, δ) permits $x = c$.
- Three equivalent forms: ε-δ, neighbourhood, sequential. The Sequential Criterion is the primary working tool.
- Bridge principle: algebra of sequence limits transfers automatically via the Sequential Criterion.
- A finite limit implies $\exists \delta > 0 \; \exists M > 0 \; \forall x \in A \; (0 < |x - c| < \delta \Rightarrow |f(x)| \le M)$.
- Limits are local: behavior far from c is irrelevant.

Toolkit — Algebra of Limits

- All five rules follow from the Sequential Criterion plus algebra of sequence limits. No new ε-δ work.
- Bridge: $(x_n) \subseteq A \setminus \{c\}$, $x_n \to c \Rightarrow f(x_n) \to L$, $g(x_n) \to M \Rightarrow$ apply sequence algebra $\Rightarrow$ lift back by Sequential Criterion.
- Quotient: $M \neq 0$ forces BoundedAwayFromZeroNear(g, c), making f/g well-defined on a punctured neighbourhood.

Limits of Functions

Definition (ε-Closeness of a Function to a Value)

Definition 2.1.1 (ε-Closeness of a Function to a Value)**.** Let $X \subseteq \mathbb{R}$, let $f : X \to \mathbb{R}$, let $L \in \mathbb{R}$, and let $\varepsilon > 0$. The function f is **ε-close to L** if $|f(x) - L| \leq \varepsilon$ for all $x \in X$.

Remark (Standard quantified statement).

$$f \ \varepsilon\text{-close to } L \iff \forall x \in X \ \big(|f(x) - L| \leq \varepsilon\big).$$

Remark (Interpretation). A global condition: every output lies within ε of L, regardless of where in X the input lies. This is Tao's scaffolding toward the limit definition, which localises the same output condition to a punctured neighbourhood of c.

Definition (Local δ-Closeness Near a Point)

Definition 2.1.2 (Local δ-Closeness Near a Point)**.** Let $f : X \to \mathbb{R}$, let $L \in \mathbb{R}$, let x_0 be adherent to X, and let $\varepsilon, \delta > 0$. The function f is **ε-close to L near x_0 within δ** if

$$\forall x \in X \ \big(|x - x_0| < \delta \Rightarrow |f(x) - L| \leq \varepsilon\big).$$

Remark (Standard quantified statement).

$$\forall x \in X : \ |x - x_0| < \delta \ \Rightarrow \ |f(x) - L| \leq \varepsilon.$$

Remark (Interpretation). A local condition: only outputs from the δ-ball around x_0 need lie within ε of L. The input condition here is Within — unpunctured. The limit definition replaces this with InputTolerance (punctured) and asserts that for every ε such a δ exists.

Remark (Dependencies).

- ε-Closeness of a Function to a Value

Definition (Limit of a Function)

Definition 2.1.3 (Limit of a Function)**.** Let $f : A \to \mathbb{R}$ and let c be a cluster point of A. A real number L is the **limit of f at** c, written $\lim_{x \to c} f(x) = L$, if

$$\forall \varepsilon > 0 \ \exists \delta > 0 \ \forall x \in A \ 0 < |x - c| < \delta \Rightarrow |f(x) - L| < \varepsilon.$$

Go to uniqueness proof.

Remark (Standard quantified statement).

$$\forall \varepsilon > 0 \; \exists \delta > 0 \; \forall x \in A \; 0 < |x - c| < \delta \;\Rightarrow\; |f(x) - L| < \varepsilon.$$

Remark (Definition predicate reading). Define

$$\begin{aligned} \mathrm{OutputTol}(\varepsilon) &\iff \varepsilon > 0, \\ \mathrm{InputTol}(\delta) &\iff \delta > 0, \\ \mathrm{PuncturedNbhd}(x, c, \delta) &\iff 0 < |x - c| < \delta, \\ \mathrm{OutputNbhd}(y, L, \varepsilon) &\iff |y - L| < \varepsilon. \end{aligned}$$

Then

$$\mathrm{FunctionLimit}(f, A, c, L) \iff \forall \varepsilon \Big(\mathrm{OutputTol}(\varepsilon) \Rightarrow \exists \delta \left(\mathrm{InputTol}(\delta) \wedge \forall x \in A \, [\mathrm{PuncturedNbhd}(x, c, \delta) \right.$$

Remark (Negated quantified statement).

$$\exists \varepsilon_0 > 0 \; \forall \delta > 0 \; \exists x \in A \; 0 < |x - c| < \delta \;\wedge\; |f(x) - L| \geq \varepsilon_0.$$

Remark (Negation predicate reading).

$$\begin{aligned} \neg\, \mathrm{FunctionLimit}(f, A, c, L) &\iff \exists \varepsilon_0 \Big(\mathrm{OutputTol}(\varepsilon_0) \wedge \forall \delta \left(\mathrm{InputTol}(\delta) \Rightarrow \exists x \in A \, [\mathrm{PuncturedNbhd}(x, c \right. \\ &\iff \exists \varepsilon_0 > 0 \; \forall \delta > 0 \; \exists x \in A \; \big(0 < |x - c| < \delta \wedge |f(x) - L| \geq \varepsilon_0\big). \end{aligned}$$

Remark (Failure modes).

- The proposed value L is the wrong target: the function approaches some value $M \neq L$ near c.
- The function does not settle near any single value as x approaches c through A.
- The outputs repeatedly stay at least a fixed distance from L no matter how tightly the inputs are restricted around c.

Remark (Failure mode decomposition).

$$\neg\, \mathrm{FunctionLimit}(f, A, c, L) \iff \exists \varepsilon_0 > 0 \; \forall \delta > 0 \; \exists x \in A \; \big(\underbrace{0 < |x - c| < \delta}_{\text{arbitrarily close to } c} \wedge \underbrace{|f(x) - L| \geq \varepsilon_0}_{\text{fixed output gap from } L}\big).$$

Remark (Interpretation). The quantifier structure is asymmetric by design. The output tolerance ε is a *demand*: the caller specifies how precise the output must be. The input tolerance δ is a *response*: the definition asserts one always exists. The input condition is punctured (InputTolerance, not Within) because f may not be defined at c, and because continuity — not limits — is the concept that includes c itself. This is the deepest structural difference between the two definitions. Four critical features: (i) f need not be defined at c; (ii) c must be a cluster point of A else the condition is vacuously satisfied by every L; (iii) δ may depend on ε; (iv) the condition is about all x near c simultaneously, not just one.

Remark (Dependencies).

- Cluster Point

Definition (Sequential Definition of Limit)

Definition 2.1.4 (Sequential Definition of Limit)**.** Let $f : A \to \mathbb{R}$ and let c be a cluster point of A. The limit $\lim_{x\to c} f(x) = L$ in the **sequential sense** holds if for every sequence $(x_n) \subseteq A \setminus \{c\}$ with $x_n \to c$, the sequence $(f(x_n))$ converges to L.

Remark (Standard quantified statement).

$$\forall (x_n) \subseteq A \setminus \{c\} \ \big(x_n \to c \Rightarrow f(x_n) \to L\big).$$

Remark (Definition predicate reading).

$$\mathrm{FunctionLimit}_{\mathrm{seq}}(f, c, L) \ :\equiv \ \forall (x_n) \subseteq A\backslash\{c\} : \ \mathrm{ConvergentSequence}(x_n, c) \ \Rightarrow \ \mathrm{ConvergentSequence}(f($$

Remark (Negated quantified statement).

$$\exists (x_n) \subseteq A \setminus \{c\} \ \big(x_n \to c \ \wedge \ f(x_n) \not\to L\big).$$

Remark (Interpretation)*.* The sequence (x_n) must lie in $A \setminus \{c\}$: the term $x_n = c$ is excluded, mirroring the $0 <$ in the ε-δ form. The sequential form is the primary tool for both proving limits (via the bridge principle) and disproving them (exhibit one bad sequence). The ε-δ form is better for constructing explicit δ functions.

Remark (Dependencies).

- Limit of a Function
- Cluster Point

Definition (Neighbourhood Form of Limit)

Definition 2.1.5 (Neighbourhood Form of Limit)**.** Let $f : A \to \mathbb{R}$ and let c be a cluster point of A. The limit $\lim_{x\to c} f(x) = L$ in the **neighbourhood sense** holds if for every ε-neighbourhood $V_\varepsilon(L)$ there exists a deleted δ-neighbourhood $V_\delta^*(c)$ such that

$$x \in V_\delta^*(c) \cap A \quad \Longrightarrow \quad f(x) \in V_\varepsilon(L).$$

Remark (Standard quantified statement).

$$\forall \varepsilon > 0 \ \exists \delta > 0 \ \forall x \in V_\delta^*(c) \cap A \ f(x) \in V_\varepsilon(L).$$

Remark (Interpretation)*.* The neighbourhood form packages the same local limit condition in terms of sets around the input point and output value. The domain is still restricted to A, and the neighbourhood of c is deleted so the value of f at c itself is irrelevant.

Remark (Dependencies).

- Limit of a Function
- ε-Neighbourhood
- Deleted ε-Neighbourhood

Proposition

Proposition 2.1.6 (Neighbourhood Form of the Limit)**.** *Let $f : A \to \mathbb{R}$ and let c be a cluster point of A. Then $\lim_{x\to c} f(x) = L$ if and only if for every ε-neighbourhood $V_\varepsilon(L)$ there exists a δ-neighbourhood $V_\delta(c)$ such that $x \in V_\delta^*(c) \cap A$ implies $f(x) \in V_\varepsilon(L)$.*

Go to proof.

Remark (Standard quantified statement)*.*

$$\Big[\forall \varepsilon > 0\ \exists \delta > 0\ \forall x \in A\ 0 < |x - c| < \delta \Rightarrow |f(x) - L| < \varepsilon\Big]$$
$$\iff \Big[\forall \varepsilon > 0\ \exists \delta > 0\ \forall x \in V_\delta^*(c) \cap A\ f(x) \in V_\varepsilon(L)\Big].$$

Remark (Interpretation)*.* The equivalence is purely notational: $V_\delta^*(c)\cap A$ is exactly the set of $x \in A$ satisfying $0 < |x-c| < \delta$, and $f(x) \in V_\varepsilon(L)$ is exactly $\mathrm{OutputTolerance}(f(x), L, \varepsilon)$. The neighbourhood form makes the geometric content explicit: the image of every punctured δ-slice of A must fit inside the ε-ball around L. This is the form of continuity used in metric space topology.

Remark (Dependencies)*.*

- Limit of a Function
- Neighbourhood Form of Limit
- ε-Neighbourhood
- Deleted ε-Neighbourhood

Theorem

Theorem 2.1.7 (Uniqueness of Limits of Functions)**.** *Let $f : A \to \mathbb{R}$ and let c be a cluster point of A. Then f has at most one limit at c.*

Go to proof.

Remark (Standard quantified statement)*.*

$$\lim_{x\to c} f(x) = L\ \wedge\ \lim_{x\to c} f(x) = L'\ \Rightarrow\ L = L'.$$

Remark (Failure modes)*.* Failure of uniqueness would mean that two distinct real numbers both satisfy the limit definition at the same cluster point.

Remark (Failure mode decomposition)*.* Non-uniqueness would require two values $L \neq L'$ each satisfying the limit definition. The $\varepsilon = |L - L'|/2$ argument shows any x within $\min(\delta_1, \delta_2)$ of c satisfies both $|f(x) - L| < \varepsilon$ and $|f(x) - L'| < \varepsilon$, but then $|L - L'| \leq |L - f(x)| + |f(x) - L'| < 2\varepsilon = |L - L'|$, a contradiction. The cluster point condition is essential: it guarantees such an x exists.

Remark (Interpretation)*.* The limit, when it exists, is uniquely determined by the local behavior of f near c. Uniqueness is what licenses writing $\lim_{x\to c} f(x) = L$ as an equation rather than an assertion.

Remark (Dependencies)*.*

- Limit of a Function

Proposition

Proposition 2.1.8 (Sequential Criterion for Function Limits)**.** *Let $f : A \to \mathbb{R}$ and let c be a cluster point of A. Then $\lim_{x\to c} f(x) = L$ if and only if for every sequence $(x_n) \subseteq A \setminus \{c\}$ with $x_n \to c$, one has $f(x_n) \to L$.*

Go to proof.

Remark (Standard quantified statement)*.*

$$\Big[\forall \varepsilon > 0\ \exists \delta > 0\ \forall x \in A\ 0 < |x - c| < \delta \Rightarrow |f(x) - L| < \varepsilon\Big]$$
$$\iff \Big[\forall (x_n) \subseteq A \setminus \{c\}\ x_n \to c \Rightarrow f(x_n) \to L\Big].$$

Remark (Negated quantified statement)*.*

$$\neg \lim_{x\to c} f(x) = L \iff \exists (x_n) \subseteq A \setminus \{c\} : \ x_n \to c \ \land \ f(x_n) \not\to L.$$

Remark (Interpretation)*.* The Sequential Criterion is the engine of the algebra of limits. Once it is in hand:

$$\text{algebra of sequence limits} \xRightarrow{\text{Sequential Criterion}} \text{algebra of function limits.}$$

Every sequence result — sum, product, quotient, squeeze — transfers to function limits by feeding a sequence into the criterion and reading the result back. The negation is equally powerful: to disprove $\lim_{x\to c} f(x) = L$, exhibit one sequence $(x_n) \subseteq A \setminus \{c\}$ with $x_n \to c$ and $f(x_n) \not\to L$.

Remark (Dependencies)*.*

- Limit of a Function
- Sequential Definition of Limit

Corollary

Corollary 2.1.9 (Three Equivalent Forms of the Function Limit)**.** *Let $f : A \to \mathbb{R}$ and let c be a cluster point of A. The ε-δ form of $\lim_{x\to c} f(x) = L$, the neighbourhood form, and the sequential form are equivalent descriptions of the same limit condition.*

Go to proof.

Remark (Standard quantified statement)*.*

$$\Big[\forall \varepsilon > 0\ \exists \delta > 0\ \forall x \in A\ 0 < |x - c| < \delta \Rightarrow |f(x) - L| < \varepsilon\Big]$$
$$\iff \Big[\forall \varepsilon > 0\ \exists \delta > 0\ \forall x \in V_\delta^*(c) \cap A\ f(x) \in V_\varepsilon(L)\Big]$$
$$\iff \Big[\forall (x_n) \subseteq A \setminus \{c\}\ x_n \to c \Rightarrow f(x_n) \to L\Big].$$

Remark (Interpretation)*.* All three encodings capture the same mathematical condition. The ε-δ form is best for proving explicit bounds. The neighbourhood form is best for geometric reasoning and metric space generalisation. The sequential form is best for importing sequence results and constructing counterexamples.

Remark (Dependencies).

- Limit of a Function
- Neighbourhood Form of Limit
- Sequential Definition of Limit
- Neighbourhood Form of the Limit
- Sequential Criterion for Function Limits

Theorem

Theorem 2.1.10 (Limit Implies Local Boundedness). *Let $f : A \to \mathbb{R}$ and let c be a cluster point of A. If $\lim_{x\to c} f(x) = L$ for some $L \in \mathbb{R}$, then there exist $\delta, M > 0$ such that $|f(x)| \leq M$ for all $x \in A$ with $0 < |x - c| < \delta$.*
Go to proof.

Remark (Standard quantified statement).

$$\lim_{x\to c} f(x) = L \;\Rightarrow\; \exists\delta > 0\ \exists M > 0\ \forall x \in A\ \big(0 < |x - c| < \delta \Rightarrow |f(x)| \leq M\big).$$

Remark (Proof sketch). Apply the definition with $\varepsilon = 1$: get $\delta > 0$ such that $|f(x)-L| < 1$ holds for all x satisfying $0 < |x - c| < \delta$. The triangle inequality then gives $|f(x)| \leq |f(x) - L| + |L| < 1 + |L|$. Set $M = 1 + |L|$.

Remark (Interpretation). A finite limit does more than control the output near L: it prevents the function from growing without bound near c. Local boundedness is a coarser statement than limit existence, but it is what is needed for the product and quotient rules.

Remark (Dependencies).

- Limit of a Function

Theorem

Theorem 2.1.11 (Bounded Limits). *Let $f : A \to \mathbb{R}$, let c be a cluster point of A, and let $a, b \in \mathbb{R}$. If $a \leq f(x) \leq b$ for all $x \in A$ with $x \neq c$, and $\lim_{x\to c} f(x) = L$, then $a \leq L \leq b$.*
Go to proof.

Remark (Standard quantified statement).

$$\big(\forall x \in A \setminus \{c\} :\ a \leq f(x) \leq b\big) \;\wedge\; \lim_{x\to c} f(x) = L \;\Rightarrow\; a \leq L \leq b.$$

Remark (Interpretation). If every nearby value of f is trapped in $[a, b]$, the limit cannot escape outside $[a, b]$. This is the function-limit analogue of the sequence result that limits preserve non-strict inequalities. The proof applies the Sequential Criterion and that sequence result directly.

Remark (Dependencies).

- Limit of a Function

Theorem

Theorem 2.1.12 (Squeeze Theorem for Function Limits)**.** *Let $f, g, h : A \to \mathbb{R}$ and let c be a cluster point of A. Suppose $\lim_{x\to c} f(x) = L = \lim_{x\to c} g(x) = L$, and suppose there exists $\delta_0 > 0$ such that*

$$\forall x \in A,\ 0 < |x - c| < \delta_0 \ \Rightarrow\ f(x) \le h(x) \le g(x).$$

Then $\lim_{x\to c} h(x) = L$.
Go to proof.

Remark (Standard quantified statement).

$$\lim_{x\to c} f(x) = L = \lim_{x\to c} g(x) = L \ \wedge\ f \le h \le g \text{ near } c \ \Rightarrow\ \lim_{x\to c} h(x) = L.$$

Remark (Interpretation). The proof routes entirely through the Sequential Criterion: convert to sequences, apply the sequential squeeze theorem, convert back. No new ε-δ argument is needed. The condition $f \le h \le g$ is only required near c, not globally.

Remark (Dependencies).

- Limit of a Function
- Bounded Limits

Theorem

Theorem 2.1.13 (Limits Are Local)**.** *Let $E \subseteq X \subseteq \mathbb{R}$, let x_0 be a cluster point of E, let $f : X \to \mathbb{R}$, and let $\delta > 0$. Then*

$$\lim_{x\to x_0;\, x\in E} f(x) = L \iff \lim_{x\to x_0;\, x\in E\cap(x_0-\delta, x_0+\delta)} f(x) = L.$$

Go to proof.

Remark (Standard quantified statement).

$$\lim_{x\to x_0;\, x\in E} f(x) = L \iff \lim_{x\to x_0;\, x\in E\cap(x_0-\delta,\, x_0+\delta)} f(x) = L.$$

Remark (Interpretation). The existence and value of FunctionLimit(f, x_0, L) depend only on the behavior of f within any fixed neighbourhood of x_0. Points outside that neighbourhood are irrelevant. This justifies focusing all limit arguments on small balls around c without any loss of generality.

Remark (Dependencies).

- Limit of a Function

Proposition

Proposition 2.1.14 (Limit of Absolute Value)**.** *Let $f : A \to \mathbb{R}$ and let c be a cluster point of A. If $\lim_{x\to c} f(x) = L$, then $\lim_{x\to c} |f(x)| = |L|$.*
Go to proof.

Remark (Standard quantified statement).

$$\lim_{x\to c} f(x) = L \;\Rightarrow\; \lim_{x\to c} |f(x)| = |L|.$$

Remark (Negated quantified statement).

$$\lim_{x\to c} |f(x)| = |L| \;\not\Rightarrow\; \lim_{x\to c} f(x) = L.$$

Remark (Failure modes). The absolute values may converge while the signs of $f(x)$ oscillate or approach the opposite sign from the proposed limiting value.

Remark (Proof strategy). The reverse triangle inequality $\big||f(x)| - |L|\big| \le |f(x) - L|$ feeds the ε-δ definition directly: the same δ that delivers $|f(x) - L| < \varepsilon$ also delivers $\big||f(x)| - |L|\big| < \varepsilon$.

Remark (Interpretation). Taking absolute values can collapse sign information, so convergence of f forces convergence of $|f|$, but convergence of $|f|$ alone does not recover the original limiting sign.

Remark (Dependencies).

- Limit of a Function

Theorem

Theorem 2.1.15 (Composition of Limits)**.** *Let $f : A \to B$, let $g : B \to \mathbb{R}$, let c_1 be a cluster point of A, and let $c_2 \in \mathbb{R}$ be a cluster point of B. Suppose $\lim_{x\to c_1} f(x) = c_2$ and $\lim_{y\to c_2} g(y) = L$. If $c_2 \in B$, assume also that $g(c_2) = L$. Then $\lim_{x\to c_1} g(f(x)) = L$.*

Go to proof.

Remark (Standard quantified statement).

$$\lim_{x\to c_1} f(x) = c_2 \;\wedge\; \lim_{y\to c_2} g(y) = L \;\wedge\; \big(c_2 \in B \Rightarrow g(c_2) = L\big) \;\Rightarrow\; \lim_{x\to c_1} g(f(x)) = L.$$

Remark (Failure modes). The composition conclusion can fail if the outer function is not controlled at the intermediate value reached by the inner limit.

Remark (Failure mode decomposition). The hypothesis $g(c_2) = L$ is essential. Without it the composition can fail: take $f \equiv c_2$ (constant) and $g(c_2) \neq L$. Then $g(f(x)) = g(c_2) \neq L$ for all x, so $\lim_{x\to c_1} g(f(x)) = L$ fails even though both component limits exist. The failure is that $f(x) = c_2$ is excluded from the punctured condition on g, but f hits c_2 constantly.

Remark (Interpretation). Composition of limits requires care precisely because the input condition is punctured. The extra hypothesis $g(c_2) = L$ bridges the gap: when $f(x) = c_2$, the output condition $g(f(x)) = g(c_2) = L$ is satisfied directly. This hypothesis is equivalent to requiring g to be continuous at c_2 (within B).

Remark (Dependencies).

- Limit of a Function

Theorem

Theorem 2.1.16 (Monotone Limit Theorem). *Let $a < b$ and let $f : (a,b) \to \mathbb{R}$ be monotone and bounded. Then both one-sided limits exist at every interior point. For increasing f and every $x_0 \in (a,b)$:*

$$f(x_0^-) = \sup\{\, f(x) : a < x < x_0 \,\}, \qquad f(x_0^+) = \inf\{\, f(x) : x_0 < x < b \,\}.$$

Go to proof.

Remark (Standard quantified statement).

$$f \text{ is monotone on } (a,b) \wedge \exists B > 0\ \forall x \in (a,b)\ (|f(x)| \le B) \ \Rightarrow\ \forall x_0 \in (a,b) :\ \lim_{x \to x_0^-} f(x) = f(x_0^-) \wedge \lim_{x \to x_0^+}$$

Remark (Interpretation). Boundedness guarantees the supremum and infimum used as candidate limits exist. Monotonicity forces convergence toward them from each side: for increasing f, the values $f(x)$ for $x < x_0$ form a bounded increasing net approaching their supremum. No oscillation is possible. The theorem establishes that monotone functions can only have jump discontinuities — never oscillatory ones.

Remark (Dependencies).

- Monotone Function
- Bounded Function
- Left-Hand Limit
- Right-Hand Limit

Proposition

Proposition 2.1.17 (Cauchy Criterion for Function Limits). *Let $f : A \to \mathbb{R}$ and let c be a cluster point of A. Then $\lim_{x \to c} f(x) = L$ exists for some $L \in \mathbb{R}$ if and only if*

$$\forall\ \text{OutputTolerance}(\varepsilon),\ \exists\ \text{InputTolerance}(\delta),\ \forall x, y \in A :\ 0 < |x - c| < \delta \wedge 0 < |y - c| < \delta\ \Rightarrow\ |f(x) -$$

Go to proof.

Remark (Standard quantified statement).

$$\exists L \in \mathbb{R}\ \Big(\lim_{x \to c} f(x) = L\Big) \iff \forall \varepsilon > 0,\ \exists \delta > 0,\ \forall x, y \in A :\ 0 < |x - c| < \delta \wedge 0 < |y - c| < \delta \Rightarrow |f(x) - f($$

Remark (Interpretation). The Cauchy criterion tests limit existence without knowing L. Two outputs from any sufficiently small punctured ball must lie within ε of each other — the outputs cluster even if their common limit is unknown. This is the function-limit analogue of the Cauchy completeness criterion for sequences, and relies on the completeness of $\mathbb{R}$ in exactly the same way.

Remark (Dependencies).

- Limit of a Function

Algebra of Limits

Theorem

Theorem 2.1.18 (Sum Rule for Function Limits). *Let $f, g : A \to \mathbb{R}$ and let c be a cluster point of A. If $\lim_{x\to c} f(x) = L$ and $\lim_{x\to c} g(x) = M$, then*

$$\lim_{x\to c}(f+g)(x) = L + M.$$

Go to proof.

Remark (Standard quantified statement).

$$\Big[\forall\varepsilon > 0\ \exists\delta_f > 0\ \forall x \in A\ 0 < |x-c| < \delta_f \Rightarrow |f(x) - L| < \varepsilon\Big]$$
$$\wedge \Big[\forall\varepsilon > 0\ \exists\delta_g > 0\ \forall x \in A\ 0 < |x-c| < \delta_g \Rightarrow |g(x) - M| < \varepsilon\Big]$$
$$\Rightarrow \forall\varepsilon > 0\ \exists\delta > 0\ \forall x \in A\ 0 < |x-c| < \delta \Rightarrow |(f+g)(x) - (L+M)| < \varepsilon.$$

Remark (Definition predicate reading).

$$\mathrm{FunctionLimit}(f, c, L) \wedge \mathrm{FunctionLimit}(g, c, M) \implies \mathrm{FunctionLimit}(f+g, c, L+M).$$

Remark (Interpretation). Limits respect addition: sufficiently close values of f and g add to values of $f+g$ close to the sum of the two limits.

Remark (Dependencies).

- Limit of a Function
- Pointwise Sum of Functions

Theorem

Theorem 2.1.19 (Difference Rule for Function Limits). *Let $f, g : A \to \mathbb{R}$ and let c be a cluster point of A. If $\lim_{x\to c} f(x) = L$ and $\lim_{x\to c} g(x) = M$, then*

$$\lim_{x\to c}(f-g)(x) = L - M.$$

Go to proof.

Remark (Standard quantified statement).

$$\Big[\lim_{x\to c} f(x) = L\Big] \wedge \Big[\lim_{x\to c} g(x) = M\Big]$$
$$\Rightarrow \forall\varepsilon > 0\ \exists\delta > 0\ \forall x \in A\ 0 < |x-c| < \delta \Rightarrow |(f-g)(x) - (L-M)| < \varepsilon.$$

Remark (Definition predicate reading).

$$\mathrm{FunctionLimit}(f, c, L) \wedge \mathrm{FunctionLimit}(g, c, M) \implies \mathrm{FunctionLimit}(f-g, c, L-M).$$

Remark (Interpretation). The difference rule is the sum rule applied to $f + (-g)$.

Remark (Dependencies).

- Limit of a Function
- Pointwise Difference of Functions

Theorem

Theorem 2.1.20 (Scalar Multiple Rule for Function Limits). *Let $f : A \to \mathbb{R}$, let c be a cluster point of A, and let $\alpha \in \mathbb{R}$. If $\lim_{x\to c} f(x) = L$, then*

$$\lim_{x\to c}(\alpha f)(x) = \alpha L.$$

Go to proof.

Remark (Standard quantified statement).

$$\left[\lim_{x\to c} f(x) = L\right] \Rightarrow \forall\varepsilon > 0\ \exists\delta > 0\ \forall x \in A\ 0 < |x - c| < \delta \Rightarrow |\alpha f(x) - \alpha L| < \varepsilon.$$

Remark (Definition predicate reading).

$$\mathrm{FunctionLimit}(f, c, L) \implies \mathrm{FunctionLimit}(\alpha f, c, \alpha L).$$

Remark (Interpretation). Multiplying a function by a fixed scalar multiplies its limiting value by that same scalar.

Remark (Dependencies).

- Limit of a Function
- Pointwise Scalar Multiple of a Function

Theorem

Theorem 2.1.21 (Product Rule for Function Limits). *Let $f, g : A \to \mathbb{R}$ and let c be a cluster point of A. If $\lim_{x\to c} f(x) = L$ and $\lim_{x\to c} g(x) = M$, then*

$$\lim_{x\to c}(fg)(x) = LM.$$

Go to proof.

Remark (Standard quantified statement).

$$\left[\lim_{x\to c} f(x) = L\right] \wedge \left[\lim_{x\to c} g(x) = M\right]$$
$$\Rightarrow \forall\varepsilon > 0\ \exists\delta > 0\ \forall x \in A\ 0 < |x - c| < \delta \Rightarrow |f(x)g(x) - LM| < \varepsilon.$$

Remark (Definition predicate reading).

$$\mathrm{FunctionLimit}(f, c, L) \wedge \mathrm{FunctionLimit}(g, c, M) \implies \mathrm{FunctionLimit}(fg, c, LM).$$

Remark (Interpretation). The product rule combines closeness of f to L with local boundedness of g near c.

Remark (Dependencies).

- Limit of a Function
- Pointwise Product of Functions

Theorem

Theorem 2.1.22 (Quotient Rule for Function Limits)**.** *Let $f, g : A \to \mathbb{R}$ and let c be a cluster point of A. If $\lim_{x\to c} f(x) = L$, $\lim_{x\to c} g(x) = M$, and $M \neq 0$, then there exists $\delta_0 > 0$ such that $g(x) \neq 0$ whenever $x \in A$ and $0 < |x - c| < \delta_0$, and*

$$\lim_{x\to c} \frac{f(x)}{g(x)} = \frac{L}{M}.$$

Go to proof.

Remark (Standard quantified statement).

$$\begin{aligned}
&\left[\lim_{x\to c} f(x) = L\right] \wedge \left[\lim_{x\to c} g(x) = M\right] \wedge M \neq 0 \\
&\quad \Rightarrow \exists \delta_0 > 0\ \forall x \in A\ 0 < |x - c| < \delta_0 \Rightarrow g(x) \neq 0, \\
&\quad \text{and} \quad \forall \varepsilon > 0\ \exists \delta > 0\ \forall x \in A\ 0 < |x - c| < \delta \Rightarrow \left|\frac{f(x)}{g(x)} - \frac{L}{M}\right| < \varepsilon.
\end{aligned}$$

Remark (Definition predicate reading).

$$\mathrm{FunctionLimit}(f, c, L) \wedge \mathrm{FunctionLimit}(g, c, M) \wedge M \neq 0 \implies \mathrm{FunctionLimit}(f/g, c, L/M).$$ ∎

Remark (Interpretation). A nonzero limiting denominator stays away from zero near the limit point, so division becomes locally legitimate and the quotient limit follows.

Remark (Dependencies).

- Limit of a Function
- Pointwise Quotient of Functions

One-Sided Limits

Definition (Right-Hand Limit)

Definition 2.1.23 (Right-Hand Limit)**.** Let $A \subseteq \mathbb{R}$, let $f : A \to \mathbb{R}$, let $c \in \mathbb{R}$, and suppose c is a cluster point of $A \cap (c, \infty)$. A real number L is the **right-hand limit** of f at c if

$$\forall \varepsilon > 0\ \exists \delta > 0\ \forall x \in A\ 0 < x - c < \delta \Rightarrow |f(x) - L| < \varepsilon.$$

Remark (Standard quantified statement).

$$\forall \varepsilon > 0\ \exists \delta > 0\ \forall x \in A\ c < x < c + \delta \Rightarrow |f(x) - L| < \varepsilon.$$

Remark (Definition predicate reading).

$$\mathrm{RightLimit}(f, c, L) :\equiv \forall \varepsilon > 0\ \exists \delta > 0\ \forall x \in A\ 0 < x - c < \delta \Rightarrow |f(x) - L| < \varepsilon.$$

Remark (Negated quantified statement).

$$\exists \varepsilon_0 > 0\ \forall \delta > 0\ \exists x \in A\ 0 < x - c < \delta\ \wedge\ |f(x) - L| \geq \varepsilon_0.$$

Remark (Negation predicate reading).

$$\neg\ \mathrm{RightLimit}(f, c, L).$$

Remark (Failure modes). A proposed right-hand limit fails when some positive output tolerance is violated arbitrarily close to c from the right.

Remark (Failure mode decomposition).

$$\exists \varepsilon_0 > 0\ \forall \delta > 0\ \exists x \in A\ 0 < x - c < \delta\ \wedge\ |f(x) - L| \geq \varepsilon_0.$$

Remark (Interpretation). Only inputs to the right of c are tested. The value of f at c, if it is defined, is irrelevant to the right-hand limit.

Remark (Dependencies).

- Limit of a Function
- Cluster Point

Definition (Left-Hand Limit)

Definition 2.1.24 (Left-Hand Limit)**.** Let $A \subseteq \mathbb{R}$, let $f : A \to \mathbb{R}$, let $c \in \mathbb{R}$, and suppose c is a cluster point of $A \cap (-\infty, c)$. A real number L is the **left-hand limit** of f at c if

$$\forall \varepsilon > 0\ \exists \delta > 0\ \forall x \in A\ 0 < c - x < \delta \Rightarrow |f(x) - L| < \varepsilon.$$

Remark (Standard quantified statement).

$$\forall \varepsilon > 0\ \exists \delta > 0\ \forall x \in A\ c - \delta < x < c \Rightarrow |f(x) - L| < \varepsilon.$$

Remark (Definition predicate reading).

$$\mathrm{LeftLimit}(f, c, L)\ :=\ \forall \varepsilon > 0\ \exists \delta > 0\ \forall x \in A\ 0 < c - x < \delta \Rightarrow |f(x) - L| < \varepsilon.$$

Remark (Negated quantified statement).

$$\exists \varepsilon_0 > 0\ \forall \delta > 0\ \exists x \in A\ 0 < c - x < \delta\ \wedge\ |f(x) - L| \geq \varepsilon_0.$$

Remark (Negation predicate reading).

$$\neg\ \mathrm{LeftLimit}(f, c, L).$$

Remark (Failure modes). A proposed left-hand limit fails when some positive output tolerance is violated arbitrarily close to c from the left.

Remark (Failure mode decomposition).

$$\exists \varepsilon_0 > 0\ \forall \delta > 0\ \exists x \in A\ 0 < c - x < \delta\ \wedge\ |f(x) - L| \geq \varepsilon_0.$$

Remark (Interpretation). Only inputs to the left of c are tested. This is the mirror image of the right-hand limit definition.

Remark (Dependencies).

- Limit of a Function
- Cluster Point

Theorem

Theorem 2.1.25 (Sequential Criterion for Right-Hand Limits). *Let $A \subseteq \mathbb{R}$, let $f : A \to \mathbb{R}$, and suppose c is a cluster point of $A \cap (c, \infty)$. Then $\lim_{x \to c^+} f(x) = L$ if and only if for every sequence $(x_n) \subseteq A \cap (c, \infty)$ with $x_n \to c$, one has $f(x_n) \to L$.*
Go to proof.

Remark (Standard quantified statement).

$$\Big[\forall \varepsilon > 0\ \exists \delta > 0\ \forall x \in A\ 0 < x - c < \delta \Rightarrow |f(x) - L| < \varepsilon\Big] \\ \iff \Big[\forall (x_n) \subseteq A \cap (c, \infty)\ x_n \to c \Rightarrow f(x_n) \to L\Big].$$

Remark (Definition predicate reading).

$$\mathrm{RightLimit}(f, c, L) \iff \forall (x_n) \subseteq A \cap (c, \infty)\ x_n \to c \Rightarrow f(x_n) \to L.$$

Remark (Interpretation). Right-hand limits can be tested by sequences approaching c from above. Every such sequence must force the corresponding output sequence toward L.

Remark (Dependencies).

- Right-Hand Limit
- Sequential Definition of Limit

Corollary

Corollary 2.1.26 (Sequential Criterion for Left-Hand Limits). *Let $A \subseteq \mathbb{R}$, let $f : A \to \mathbb{R}$, and suppose c is a cluster point of $A \cap (-\infty, c)$. Then $\lim_{x \to c^-} f(x) = L$ if and only if for every sequence $(x_n) \subseteq A \cap (-\infty, c)$ with $x_n \to c$, one has $f(x_n) \to L$.*
Go to proof.

Remark (Standard quantified statement).

$$\Big[\forall \varepsilon > 0\ \exists \delta > 0\ \forall x \in A\ 0 < c - x < \delta \Rightarrow |f(x) - L| < \varepsilon\Big] \\ \iff \Big[\forall (x_n) \subseteq A \cap (-\infty, c)\ x_n \to c \Rightarrow f(x_n) \to L\Big].$$

Remark (Definition predicate reading).

$$\mathrm{LeftLimit}(f, c, L) \iff \forall (x_n) \subseteq A \cap (-\infty, c)\ x_n \to c \Rightarrow f(x_n) \to L.$$

Remark (Interpretation). Left-hand limits can be tested by sequences approaching c from below. The criterion is the left-sided analogue of the right-hand sequential criterion.

Remark (Dependencies).

- Left-Hand Limit
- Sequential Definition of Limit

Theorem

Theorem 2.1.27 (Two-Sided Limit via One-Sided Limits)**.** *Let $A \subseteq \mathbb{R}$ and let $f : A \to \mathbb{R}$. If c is a cluster point of both $A \cap (c, \infty)$ and $A \cap (-\infty, c)$, then $\lim_{x\to c} f(x) = L$ if and only if both $\lim_{x\to c^+} f(x) = L$ and $\lim_{x\to c^-} f(x) = L$.*
Go to proof.

Remark (Standard quantified statement).

$$\begin{aligned}&\Big[\forall \varepsilon > 0\ \exists \delta > 0\ \forall x \in A\ 0 < |x - c| < \delta \Rightarrow |f(x) - L| < \varepsilon\Big]\\ &\iff \Big[\forall \varepsilon > 0\ \exists \delta_+ > 0\ \forall x \in A\ 0 < x - c < \delta_+ \Rightarrow |f(x) - L| < \varepsilon\Big]\\ &\quad \wedge \Big[\forall \varepsilon > 0\ \exists \delta_- > 0\ \forall x \in A\ 0 < c - x < \delta_- \Rightarrow |f(x) - L| < \varepsilon\Big].\end{aligned}$$

Remark (Definition predicate reading).

$$\mathrm{FunctionLimit}(f, c, L) \iff \mathrm{RightLimit}(f, c, L) \wedge \mathrm{LeftLimit}(f, c, L).$$

Remark (Negated quantified statement).

$$\begin{aligned}&\exists \varepsilon_0 > 0\ \forall \delta > 0\ \exists x \in A\ 0 < |x - c| < \delta\ \wedge\ |f(x) - L| \geq \varepsilon_0\\ &\iff \Big[\exists \varepsilon_0 > 0\ \forall \delta > 0\ \exists x \in A\ 0 < x - c < \delta\ \wedge\ |f(x) - L| \geq \varepsilon_0\Big]\\ &\quad \vee \Big[\exists \varepsilon_0 > 0\ \forall \delta > 0\ \exists x \in A\ 0 < c - x < \delta\ \wedge\ |f(x) - L| \geq \varepsilon_0\Big].\end{aligned}$$

Remark (Negation predicate reading).

$$\neg\,\mathrm{FunctionLimit}(f, c, L) \iff \neg\,\mathrm{RightLimit}(f, c, L) \vee \neg\,\mathrm{LeftLimit}(f, c, L).$$

Remark (Failure modes). A two-sided limit fails precisely when the right-hand limit fails, the left-hand limit fails, or the two one-sided behaviours do not meet at the same value.

Remark (Failure mode decomposition).

$$\neg\,\mathrm{FunctionLimit}(f, c, L) \iff \neg\,\mathrm{RightLimit}(f, c, L) \vee \neg\,\mathrm{LeftLimit}(f, c, L).$$

Remark (Interpretation). A two-sided limit is exactly the agreement of the two one-sided tests. Both sides must approach the same value.

Remark (Dependencies).

- Limit of a Function
- Right-Hand Limit
- Left-Hand Limit

2.2 Point Continuity

Toolkit — Oscillation

- OscillationOn(f, E, ω): $\omega(f; E) = \sup_{x_1,x_2 \in E} |f(x_1) - f(x_2)| = \operatorname{diam}(f(E))$.
- OscillationAt(f, x_0, ω): $\omega(f; x_0) = \lim_{\delta \to 0^+} \omega(f; U_\delta(x_0) \cap E)$. The limit exists because the function $\delta \mapsto \omega(f; U_\delta(x_0) \cap E)$ is nonincreasing.
- ContinuousAt$(f, x_0) \iff \omega(f; x_0) = 0$.
- Discontinuity set: $D(f) = \{x : \omega(f; x) > 0\} = \bigcup_{n=1}^{\infty} \{x : \omega(f; x) \geq 1/n\}$.
- Each threshold set is closed; $D(f)$ is F_σ.

Toolkit — Discontinuity

- DiscontinuousAt(f, a) is the negation of ContinuousAt(f, a).
- Quantifier reversal governs failure of continuity.
- Three equivalent formulations: ε-δ, sequential, neighbourhood.
- Taxonomy: removable, jump, essential.

Toolkit — Continuity

- Three equivalent forms: ε-δ, neighbourhood, sequential. The ε-δ form is unpunctured ($x = c$ allowed), unlike function limits.
- δ may depend on both ε and c. If δ depends only on ε, the condition is uniform continuity.
- Isolated points are automatically continuous.
- Algebra: sums, differences, products, scalar multiples, and quotients (denominator nonzero) of continuous functions are continuous.
- Composition preserves continuity.

Continuity

Definition (Relative Neighborhood)

Definition 2.2.1 (Relative Neighborhood)**.** Let $E \subseteq \mathbb{R}$ and $a \in E$. A set $U \subseteq \mathbb{R}$ is a neighborhood of a in E if and only if there exists $\delta > 0$ such that $U = E \cap (a - \delta, a + \delta)$. Equivalently, every neighborhood of a in E is a set of the form $U_E(a) := E \cap (a - \delta, a + \delta)$ for some $\delta > 0$.

Remark (Standard quantified statement).

$$\begin{aligned}
&\text{Let } E \subseteq \mathbb{R},\ a \in E,\ \text{and } U \subseteq \mathbb{R}. \\
&U \text{ is a neighborhood of } a \text{ in } E \iff \exists \delta > 0\ \big(U = E \cap (a - \delta, a + \delta)\big).
\end{aligned}$$

Remark (Definition predicate reading).

$$\text{RelativeNeighborhood}(U, E, a) :\equiv \exists \delta > 0 \ \big(U = E \cap (a-\delta, a+\delta)\big).$$

Remark (Negated quantified statement).

$$\begin{aligned}&\text{Let } E \subseteq \mathbb{R},\ a \in E,\ \text{and } U \subseteq \mathbb{R}.\\ &U \text{ fails to be a neighborhood of } a \text{ in } E \iff \forall \delta > 0 \ \big(U \neq E \cap (a-\delta, a+\delta)\big).\end{aligned}$$

Remark (Negation predicate reading).

$$\neg\,\text{RelativeNeighborhood}(U, E, a) \iff \forall \delta > 0 \ \big(U \neq E \cap (a-\delta, a+\delta)\big).$$

Remark (Failure modes). Failure occurs precisely when no radius δ makes U coincide with the relative interval slice: for every $\delta > 0$ either U contains a point outside $E \cap (a-\delta, a+\delta)$ or the slice contains a point missing from U.

Remark (Failure mode decomposition).

$$\forall \delta > 0 \ \underbrace{\big(\exists x \in U :\ x \notin E \cap (a-\delta, a+\delta)\big)}_{\text{extraneous point}} \vee \underbrace{\big(\exists y \in E \cap (a-\delta, a+\delta) :\ y \notin U\big)}_{\text{missing point}}.$$

Remark (Interpretation). Relative neighborhoods capture the local behavior of E around a by intersecting E with an ordinary open interval centered at a. The construction restricts the ambient notion of openness to the geometry of E, so it is suited to relative topologies. Failure simply means that U never matches the precise slice of E at any scale, typically because U omits some nearby points of E or includes points that fall outside E near a.

Remark (Dependencies).

- Subset
- Epsilon Neighbourhood
- Order on $\mathbb{R}$

Definition (Continuity at a Point)

Definition 2.2.2 (Continuity at a Point)**.** Let $A \subseteq \mathbb{R}$, let $f : A \to \mathbb{R}$, and let $c \in A$. The function f is **continuous at** c if and only if

$$\forall \varepsilon > 0 \ \exists \delta > 0 \ \forall x \in A : \big(|x - c| < \delta \Rightarrow |f(x) - f(c)| < \varepsilon\big).$$

Remark (Standard quantified statement).

$$\begin{aligned}&\text{Let } A \subseteq \mathbb{R},\ f : A \to \mathbb{R},\ c \in A.\\ &f \text{ is continuous at } c \iff \forall \varepsilon > 0 \ \exists \delta > 0 \ \forall x \in A \ \big(|x - c| < \delta \Rightarrow |f(x) - f(c)| < \varepsilon\big).\end{aligned}$$

Remark (Definition predicate reading).

$$\text{ContinuousAt}(f, A, c) :\equiv \forall \varepsilon > 0 \ \exists \delta > 0 \ \forall x \in A \ \big(|x - c| < \delta \Rightarrow |f(x) - f(c)| < \varepsilon\big).$$

Remark (Negated quantified statement).

$$\begin{aligned}&\text{Let } A \subseteq \mathbb{R},\ f : A \to \mathbb{R},\ c \in A.\\ &f \text{ is not continuous at } c \iff \exists \varepsilon_0 > 0\ \forall \delta > 0\ \exists x \in A\ \big(|x-c| < \delta \wedge |f(x)-f(c)| \geq \varepsilon_0\big).\end{aligned}$$

Remark (Negation predicate reading).

$$\neg\,\mathrm{ContinuousAt}(f, A, c) \iff \exists \varepsilon_0 > 0\ \forall \delta > 0\ \exists x \in A\ \big(|x-c| < \delta \wedge |f(x)-f(c)| \geq \varepsilon_0\big).$$

Remark (Failure modes). Continuity at c fails precisely when there exists a persistent output gap $\varepsilon_0 > 0$ such that no matter how small the chosen neighborhood of c is, a point of A within that neighborhood keeps the function values at least ε_0 away from $f(c)$. The single failure branch records this inescapable oscillation near c.

Remark (Failure mode decomposition).

$$\underbrace{\exists \varepsilon_0 > 0}_{\text{persistent gap}}\quad \underbrace{\forall \delta > 0}_{\text{every neighborhood}}\quad \underbrace{\exists x \in A}_{\text{nearby witness}}\ :\ |x-c| < \delta \wedge |f(x)-f(c)| \geq \varepsilon_0.$$

Remark (Interpretation). Continuity at c means that the output $f(x)$ can be kept arbitrarily close to $f(c)$ by taking x sufficiently close to c while staying in the domain A. The negated form highlights that a discontinuity arises only when there is a fixed positive separation in output that persists no matter how tightly one zooms in on c, so nearby inputs cannot force the outputs to settle near $f(c)$. This captures the geometric intuition that the graph of a continuous function has no abrupt jumps or breaks at c.

Remark (Dependencies).

- Definition: Limit of a Function

Definition (Continuity at a Point — Neighbourhood Form)

Definition 2.2.3 (Continuity at a Point — Neighbourhood Form)**.** Let $A \subseteq \mathbb{R}$, let $f : A \to \mathbb{R}$, and let $c \in A$. The function f is continuous at c if and only if for every $\varepsilon > 0$ there exists $\delta > 0$ such that

$$f\big(A \cap (c-\delta, c+\delta)\big) \subseteq (f(c)-\varepsilon, f(c)+\varepsilon).$$

Remark (Standard quantified statement).

$$\begin{aligned}&\text{Let } A \subseteq \mathbb{R},\ f : A \to \mathbb{R},\ c \in A.\\ &f \text{ is continuous at } c \iff \forall \varepsilon > 0\ \exists \delta > 0 :\\ &\qquad f\big(A \cap (c-\delta, c+\delta)\big) \subseteq (f(c)-\varepsilon, f(c)+\varepsilon).\end{aligned}$$

Remark (Definition predicate reading).

$$\mathrm{ContinuousAt}(f, c) :\!= \forall \varepsilon > 0\ \exists \delta > 0 :\ f\big(A \cap (c-\delta, c+\delta)\big) \subseteq (f(c)-\varepsilon, f(c)+\varepsilon).$$

Remark (Negated quantified statement).

$$\begin{aligned}&\text{Let } A \subseteq \mathbb{R},\ f : A \to \mathbb{R},\ c \in A.\\ &\neg\big(f \text{ is continuous at } c\big) \iff \exists \varepsilon_0 > 0\ \forall \delta > 0 :\\ &\qquad f\big(A \cap (c-\delta, c+\delta)\big) \nsubseteq (f(c)-\varepsilon_0, f(c)+\varepsilon_0).\end{aligned}$$

Remark (Negation predicate reading).

$$\text{DiscontinuousAt}(f,c) \iff \exists \varepsilon_0 > 0\ \forall \delta > 0 : \ f\big(A \cap (c-\delta, c+\delta)\big) \not\subseteq (f(c)-\varepsilon_0, f(c)+\varepsilon_0).$$

Remark (Failure modes). Failure occurs precisely when some fixed output tolerance cannot be matched by any neighborhood about c. This manifests by finding points of A arbitrarily close to c whose images either exceed the upper endpoint $f(c)+\varepsilon_0$ or fall below the lower endpoint $f(c)-\varepsilon_0$.

Remark (Failure mode decomposition).

$$\exists \varepsilon_0 > 0\ \forall \delta > 0\ \underbrace{\Big(\exists x \in A \cap (c-\delta, c+\delta) : f(x) \geq f(c)+\varepsilon_0\Big)}_{\text{overshoot}} \vee \underbrace{\Big(\exists x \in A \cap (c-\delta, c+\delta) : f(x) \leq f(c) -}_{\text{undershoot}}$$

Remark (Interpretation). Continuity at c demands that every output tolerance around $f(c)$ has a matching neighborhood around c whose image remains confined to the tolerance band. The negation says that one tolerance band resists every neighborhood choice, so some sequence of points from A approaches c while their images either spike above or dip below $f(c)$ by at least a fixed amount. Geometrically, arbitrarily tight vertical strips around the graph above c must be achievable by shrinking the horizontal window; failure means the graph keeps escaping that strip no matter how closely the input variable hugs c.

Definition (Continuity at a Point — Sequential Form)

Definition 2.2.4 (Continuity at a Point — Sequential Form). Let $A \subseteq \mathbb{R}$, let $c \in A$, and let $f\colon A \to \mathbb{R}$. The function f is continuous at c if and only if for every sequence $(x_n)_{n\in\mathbb{N}}$ in A, the convergence $\lim_{n\to\infty} x_n = c$ implies $\lim_{n\to\infty} f(x_n) = f(c)$.

Remark (Standard quantified statement).

$$\text{Let } A \subseteq \mathbb{R},\ c \in A,\ f\colon A \to \mathbb{R}.$$
$$f \text{ is continuous at } c \iff \forall (x_n)_{n\in\mathbb{N}} \subseteq A\ \Big(\lim_{n\to\infty} x_n = c \Rightarrow \lim_{n\to\infty} f(x_n) = f(c)\Big).$$

Remark (Definition predicate reading).

$$\text{ContinuousAt}(f,c) :\equiv \forall x_n\ \Big(\text{Sequence}(x_n, A) \wedge \text{ConvergentSequence}(x_n, c) \Rightarrow \text{ConvergentSequence}(f(x_n$$

Remark (Negated quantified statement).

$$\text{Let } A \subseteq \mathbb{R},\ c \in A,\ f\colon A \to \mathbb{R}.$$
$$f \text{ is not continuous at } c \iff \exists (x_n)_{n\in\mathbb{N}} \subseteq A\ \Big(\lim_{n\to\infty} x_n = c \wedge \lim_{n\to\infty} f(x_n) \neq f(c)\Big).$$

Remark (Negation predicate reading).

$$\neg\, \text{ContinuousAt}(f,c) \iff \exists x_n\ \Big(\text{Sequence}(x_n, A) \wedge \text{ConvergentSequence}(x_n, c) \wedge \neg\, \text{ConvergentSequence}(f$$

Remark (Failure modes). Continuity at c fails sequentially when there exists a sequence $(x_n)_{n\in\mathbb{N}}$ in A with $x_n \to c$ but $f(x_n)$ does not converge to $f(c)$. This manifests in two distinct ways: either $f(x_n)$ converges to some limit different from $f(c)$, or $f(x_n)$ fails to converge entirely (oscillating without settling or diverging to infinity).

Remark (Failure mode decomposition).

$$\neg(f \text{ continuous at } c) \iff \exists (x_n)_{n\in\mathbb{N}} \subseteq A : \underbrace{\lim_{n\to\infty} x_n = c}_{\text{valid approach}} \wedge \underbrace{\lim_{n\to\infty} f(x_n) \neq f(c)}_{\text{image sequence fails}}.$$

Remark (Interpretation). The sequential characterization asserts that information about f near c is completely encoded by its action on limits of sequences taken within the domain. Any approach to c through admissible points must transport convergent input sequences to convergent output sequences with the expected limit; conversely, a single pathological sequence suffices to expose a discontinuity. Geometrically, this captures the idea that f cannot introduce jumps or oscillations that persist along every infinitesimal path toward c.

Remark (Dependencies).

Definition 2.2.2.

Characterization of Discontinuity

Definition (Point of Discontinuity)

Definition 2.2.5 (Point of Discontinuity)**.** Let $A \subseteq \mathbb{R}$, let $f : A \to \mathbb{R}$, and let $a \in A$. The function f is discontinuous at a if and only if f is not continuous at a.

Remark (Standard quantified statement).

Let $A \subseteq \mathbb{R},\ f : A \to \mathbb{R},\ a \in A$.

f is discontinuous at $a \iff \neg\Big(\forall \varepsilon > 0\ \exists \delta > 0\ \forall x \in A\ \big(|x-a| < \delta \Rightarrow |f(x) - f(a)| < \varepsilon\big)\Big)$.

Remark (Definition predicate reading).

$$\text{DiscontinuousAt}(f,a) := \neg\,\text{ContinuousAt}(f,a).$$

Remark (Negated quantified statement).

Let $A \subseteq \mathbb{R},\ f : A \to \mathbb{R},\ a \in A$.

f is discontinuous at $a \iff \exists \varepsilon > 0\ \forall \delta > 0\ \exists x \in A\ \big(|x-a| < \delta \wedge |f(x) - f(a)| \geq \varepsilon\big)$.

Remark (Negation predicate reading).

$$\neg\,\text{DiscontinuousAt}(f,a) \iff \text{ContinuousAt}(f,a) \iff \forall \varepsilon > 0\ \exists \delta > 0\ \forall x \in A\ \big(|x-a| < \delta \Rightarrow |f(x)-f(a)$$

Remark (Failure modes). Discontinuity arises either because the nearby values of f do not approach any single real number as $x \to a$, or because a well-defined limit exists but disagrees with the actual value $f(a)$.

Remark (Failure mode decomposition).

$$\underbrace{\neg\exists L \in \mathbb{R}\ \text{FunctionLimit}(f,A,a,L)}_{\text{no limit exists}} \vee \underbrace{\big(\exists L \in \mathbb{R}\ \text{FunctionLimit}(f,A,a,L) \wedge L \neq f(a)\big)}_{\text{limit-value mismatch}}.$$

Remark (Interpretation). A point of discontinuity is detected by the breakdown of the usual epsilon-delta control. Understanding removable, jump, and essential cases guides how to remedy or classify singular behavior in proofs.

Remark (Dependencies).
Definition 2.2.2.

Definition (Sequential Characterization of Discontinuity)

Definition 2.2.6 (Sequential Characterization of Discontinuity)**.** Let $A \subseteq \mathbb{R}$, let $f : A \to \mathbb{R}$, and let $c \in A$. The function f is discontinuous at c if and only if there exists a sequence $(x_n)_{n\in\mathbb{N}} \subseteq A$ such that $x_n \to c$ while $f(x_n) \not\to f(c)$.

Remark (Standard quantified statement).

$$f \text{ is discontinuous at } c \iff \exists (x_n) \subseteq A \left[x_n \to c \;\wedge\; f(x_n) \not\to f(c)\right].$$

Remark (Definition predicate reading).

$$\text{DiscontinuousAt}(f, c) := \exists (x_n) \subseteq A : \; x_n \to c \;\wedge\; \neg\big(f(x_n) \to f(c)\big).$$

Remark (Negated quantified statement).

$$f \text{ is continuous at } c \iff \forall (x_n) \subseteq A : \; x_n \to c \;\Rightarrow\; f(x_n) \to f(c).$$

Remark (Negation predicate reading).

$$\neg\,\text{DiscontinuousAt}(f, c) \iff \text{ContinuousAt}(f, c) \iff \forall (x_n) \subseteq A : \; x_n \to c \;\Rightarrow\; f(x_n) \to f(c).\blacksquare$$

Remark (Failure modes). The characterization fails — meaning f is NOT sequentially discontinuous — when every sequence (x_n) in A converging to c has $f(x_n) \to f(c)$. No sequential witness exists.

Remark (Failure mode decomposition).

$$\neg\,\text{DiscontinuousAt}(f, c) \iff \underbrace{\forall (x_n) \subseteq A : \; x_n \to c \;\Rightarrow\; f(x_n) \to f(c)}_{\text{every approach preserves the limit}}.$$

Remark (Interpretation). A discontinuity at c can be certified by a single sequence approaching c whose images refuse to settle at $f(c)$.

Remark (Dependencies).
Definition 2.2.4; Definition 2.2.5.

Definition (Neighbourhood Characterization of Discontinuity)

Definition 2.2.7 (Neighbourhood Characterization of Discontinuity)**.** Let $A \subseteq \mathbb{R}$, let $f\colon A \to \mathbb{R}$, and let $c \in A$. The function f is discontinuous at c if and only if there exists a neighborhood V of $f(c)$ such that for every neighborhood U of c relative to A there exists $x \in U \cap A$ with $f(x) \notin V$.

Remark (Standard quantified statement).

$$f \text{ is discontinuous at } c \iff \exists V \text{ nbhd of } f(c)\; \forall U \text{ nbhd of } c\; \exists x \in U \cap A \;\big(f(x) \notin V\big).$$

Remark (Definition predicate reading).

$$\mathrm{DiscontinuousAt}(f,c) :\equiv \exists V \text{ nbhd of } f(c)\ \forall U \text{ nbhd of } c\ \exists x \in U \cap A :\ f(x) \notin V.$$

Remark (Negated quantified statement).

$$f \text{ is continuous at } c \iff \forall V \text{ nbhd of } f(c)\ \exists U \text{ nbhd of } c\ \forall x \in U \cap A\ \big(f(x) \in V\big).$$

Remark (Negation predicate reading).

$$\neg\,\mathrm{DiscontinuousAt}(f,c) \iff \mathrm{ContinuousAt}(f,c) \iff \forall V \text{ nbhd of } f(c)\ \exists U \text{ nbhd of } c\ \forall x \in U \cap A\ \big(f(x)$$

Remark (Failure modes). The neighbourhood discontinuity characterization fails — meaning f is actually continuous — when every output neighbourhood V of $f(c)$ can be matched by some input neighbourhood U such that f maps all of $U \cap A$ into V. No perpetually-missed output neighbourhood exists.

Remark (Failure mode decomposition).

$$\neg\,\mathrm{DiscontinuousAt}(f,c) \iff \forall V \text{ nbhd of } f(c)\ \underbrace{\exists U \text{ nbhd of } c}_{\text{some input window}} :\ \underbrace{\forall x \in U \cap A :\ f(x) \in V}_{\text{all outputs inside tolerance}}.$$

Remark (Interpretation). Discontinuity means a fixed output neighborhood V is perpetually missed no matter how tightly the input is restricted around c.

Remark (Dependencies).

Definition 2.2.1.

Lemma (Equivalent Formulations of Discontinuity at a Point)

Lemma 2.2.8 (Equivalent Formulations of Discontinuity at a Point)**.** *Let $A \subseteq \mathbb{R}$, let $f : A \to \mathbb{R}$, and let $c \in A$. The following are equivalent:*

1. *c is a point of discontinuity in the ε-δ sense.*
2. *c is a sequential point of discontinuity.*
3. *c is a neighbourhood point of discontinuity.*

Go to proof.

Remark (Standard quantified statement).

$$\begin{aligned} & (1)\ \exists \varepsilon_0 > 0\ \forall \delta > 0\ \exists x \in A :\ 0 < |x - c| < \delta \wedge |f(x) - f(c)| \geq \varepsilon_0 \\ \iff & (2)\ \exists (x_n) \subseteq A :\ x_n \to c \wedge \neg\big(f(x_n) \to f(c)\big) \\ \iff & (3)\ \exists V \text{ nbhd of } f(c)\ \forall U \text{ nbhd of } c\ \exists x \in U \cap A :\ f(x) \notin V. \end{aligned}$$

Remark (Predicate reading).

$$\begin{aligned} \mathrm{DiscontinuousAt}(f,c) &\iff \exists \varepsilon_0 > 0\ \forall \delta > 0\ \exists x \in A :\ 0 < |x - c| < \delta \wedge |f(x) - f(c)| \geq \varepsilon_0 \\ &\iff \exists (x_n) \subseteq A :\ x_n \to c \wedge \neg\big(f(x_n) \to f(c)\big) \\ &\iff \exists V \text{ nbhd of } f(c)\ \forall U \text{ nbhd of } c\ \exists x \in U \cap A :\ f(x) \notin V. \end{aligned}$$

Remark (Failure modes). The equivalence cannot fail for functions at limit points of A: the proof that all three characterizations are equivalent is constructive in both directions. In the degenerate case where c is not a limit point of A, the punctured-neighbourhood conditions are vacuously satisfied by every L, and the equivalence becomes trivial rather than informative.

Remark (Interpretation). The three characterizations of discontinuity are mutually convertible. Analysts select whichever viewpoint simplifies a particular argument.

Remark (Dependencies).

- Point of Discontinuity
- Sequential Characterization of Discontinuity
- Neighbourhood Characterization of Discontinuity

Definition (Types of Discontinuity)

Definition 2.2.9 (Types of Discontinuity)**.** Let $A \subseteq \mathbb{R}$, let $f : A \to \mathbb{R}$, and let $x_0 \in A$ with f discontinuous at x_0.

(i) *Removable* iff $\exists L \in \mathbb{R} : \lim_{x\to x_0} f(x) = L$ and $L \neq f(x_0)$.

(ii) *Jump* iff $\exists L_-, L_+ \in \mathbb{R} : \lim_{x\to x_0^-} f(x) = L_-$, $\lim_{x\to x_0^+} f(x) = L_+$, and $L_- \neq L_+$.

(iii) *Essential* iff it is not removable.

Remark (Standard quantified statement).

$$\begin{aligned}
\text{Removable} &\iff \exists L \in \mathbb{R}\ \big(\lim_{x\to x_0} f(x) = L \wedge L \neq f(x_0)\big).\\
\text{Jump} &\iff \exists L_-, L_+ \in \mathbb{R}\ \big(\lim_{x\to x_0^-} f(x) = L_- \wedge \lim_{x\to x_0^+} f(x) = L_+ \wedge L_- \neq L_+\big).\\
\text{Essential} &\iff \neg\exists L \in \mathbb{R}\ \big(\lim_{x\to x_0} f(x) = L \wedge L \neq f(x_0)\big).
\end{aligned}$$

Remark (Definition predicate reading).

$$\begin{aligned}
\text{RemovableDiscontinuity}(f, x_0) &:= \exists L \in \mathbb{R}\ \big(\text{FunctionLimit}(f, A, x_0, L) \wedge L \neq f(x_0)\big).\\
\text{JumpDiscontinuity}(f, x_0) &:= \exists L_-, L_+ \in \mathbb{R}\ \big(\text{LeftLimit}(f, A, x_0, L_-) \wedge \text{RightLimit}(f, A, x_0, L_+) \wedge L
\\
\text{EssentialDiscontinuity}(f, x_0) &:= \neg\, \text{RemovableDiscontinuity}(f, x_0).
\end{aligned}$$

Remark (Negated quantified statement).

$$\begin{aligned}
\neg\text{Removable} &\iff \forall L \in \mathbb{R}\ \big(\lim_{x\to x_0} f(x) \neq L \vee L = f(x_0)\big),\\
\neg\text{Jump} &\iff \forall L_-, L_+ \in \mathbb{R}\ \big(\lim_{x\to x_0^-} f(x) \neq L_- \vee \lim_{x\to x_0^+} f(x) \neq L_+ \vee L_- = L_+\big),\\
\neg\text{Essential} &\iff \text{Removable}.
\end{aligned}$$

Remark (Negation predicate reading).

$$\begin{aligned}
\neg\, \text{RemovableDiscontinuity}(f, x_0) &\iff \forall L \in \mathbb{R}\ \big(\neg\, \text{FunctionLimit}(f, A, x_0, L) \vee L = f(x_0)\big).\\
\neg\, \text{JumpDiscontinuity}(f, x_0) &\iff \forall L_-, L_+ \in \mathbb{R}\ \big(\neg\, \text{LeftLimit}(f, A, x_0, L_-) \vee \neg\, \text{RightLimit}(f, A, x_0,
\\
\neg\, \text{EssentialDiscontinuity}(f, x_0) &\iff \text{RemovableDiscontinuity}(f, x_0).
\end{aligned}$$

Remark (Failure modes).

- Removable discontinuity fails when no bilateral limit exists at x_0, or when the bilateral limit exists but equals $f(x_0)$ (making the function actually continuous).
- Jump discontinuity fails when at least one of the one-sided limits does not exist, or when both one-sided limits exist but are equal (making it potentially removable or continuous).
- Essential discontinuity fails when the discontinuity is removable (the bilateral limit exists and differs from $f(x_0)$).

Remark (Failure mode decomposition).

$$\neg\,\mathrm{RemovableDiscontinuity}(f, x_0) \iff \underbrace{\neg\exists L \in \mathbb{R}\ \lim_{x\to x_0} f(x) = L}_{\text{no bilateral limit exists}} \lor \underbrace{\lim_{x\to x_0} f(x) = f(x_0)}_{\text{limit equals function value}}.$$

$$\neg\,\mathrm{JumpDiscontinuity}(f, x_0) \iff \underbrace{\neg\exists L_- \in \mathbb{R}\ \lim_{x\to x_0^-} f(x) = L_-}_{\text{no left limit}} \lor \underbrace{\neg\exists L_+ \in \mathbb{R}\ \lim_{x\to x_0^+} f(x) = L_+}_{\text{no right limit}}$$
$$\lor \underbrace{\lim_{x\to x_0^-} f(x) = \lim_{x\to x_0^+} f(x)}_{\text{one-sided limits agree}}.$$

Remark (Interpretation). Removable discontinuities can be healed by redefining $f(x_0)$. Jump discontinuities provide well-defined one-sided limits useful in integration theory. Essential discontinuities are pathological and require more global tools.

Remark (Dependencies).

- Point of Discontinuity
- Limit of a Function
- Left-Hand Limit
- Right-Hand Limit

Oscillation

Definition (Oscillation on a Set)

Definition 2.2.10 (Oscillation on a Set)**.** Let $A \subseteq \mathbb{R}$, let $f : A \to \mathbb{R}$, and let $E \subseteq A$. The oscillation of f on E is the extended real number

$$\omega(f; E) := \sup\{\, |f(x_1) - f(x_2)| : x_1, x_2 \in E \,\} = \mathrm{diam}(f(E)).$$

Remark (Standard quantified statement).

$$\text{Let } A \subseteq \mathbb{R},\ f : A \to \mathbb{R},\ E \subseteq A,\ \omega \in [0, \infty].$$
$$\omega(f; E) = \omega \iff \omega = \sup\{\, |f(x_1) - f(x_2)| : x_1, x_2 \in E \,\}.$$

Remark (Definition predicate reading).

$$\mathrm{OscillationOn}(f, E, \omega) := \omega = \sup\{\, |f(x_1) - f(x_2)| : x_1, x_2 \in E \,\}.$$

Remark (Negated quantified statement).

$$\begin{aligned}&\text{Let } A \subseteq \mathbb{R},\ f : A \to \mathbb{R},\ E \subseteq A,\ \omega \in [0,\infty].\\ &\omega(f;E) \neq \omega \iff \omega \neq \sup\{\,|f(x_1) - f(x_2)| : x_1, x_2 \in E\,\}.\end{aligned}$$

Remark (Negation predicate reading).

$$\neg\,\mathrm{OscillationOn}(f, E, \omega) \iff \omega \neq \sup\{\,|f(x_1) - f(x_2)| : x_1, x_2 \in E\,\}.$$

Remark (Failure modes). The proposed value ω fails either because it is not an upper bound for pairwise differences of function values on E, or because it is an upper bound but not the least such upper bound.

Remark (Failure mode decomposition).

$$\underbrace{\exists x_1, x_2 \in E :\ |f(x_1) - f(x_2)| > \omega}_{\text{not an upper bound}} \vee \underbrace{\exists \eta < \omega\ \forall x_1, x_2 \in E :\ |f(x_1) - f(x_2)| \leq \eta}_{\text{not least}}.$$

Remark (Interpretation). Oscillation on a set records the spread of the image — the diameter of $f(E)$. Small oscillation means all values of f on E lie close together. Zero oscillation means f is constant on E.

Remark (Dependencies).

- Function
- Subset
- Supremum
- Absolute Value on $\mathbb{R}$

Definition (Oscillation at a Point)

Definition 2.2.11 (Oscillation at a Point)**.** Let $E \subseteq \mathbb{R}$, let $f : E \to \mathbb{R}$, and let $x_0 \in \mathbb{R}$ satisfy $U_\delta(x_0) \cap E \neq \varnothing$ for every $\delta > 0$, where $U_\delta(x_0) = \{x \in \mathbb{R} : |x - x_0| < \delta\}$. The oscillation of f at x_0 is the extended real number

$$\omega(f;x_0) = \lim_{\delta \to 0^+} \omega\big(f;\, U_\delta(x_0) \cap E\big),$$

with $\omega(f;A)$ denoting the oscillation of f on the set A. Because the function $\delta \mapsto \omega\big(f;\, U_\delta(x_0) \cap E\big)$ is nonincreasing, this right-hand limit exists (possibly $+\infty$).

Remark (Standard quantified statement).

$$\begin{aligned}&\text{Let } E \subseteq \mathbb{R},\ f : E \to \mathbb{R},\ x_0 \in \mathbb{R},\ \omega \in [0,\infty].\\ &\big[(\forall \delta > 0 :\ U_\delta(x_0) \cap E \neq \varnothing)\ \wedge\ \omega(f;x_0) = \omega\big]\\ &\iff \forall \varepsilon > 0\ \exists \delta > 0\ \forall r\ (0 < r < \delta \Rightarrow |\omega(f;U_r(x_0) \cap E) - \omega| < \varepsilon).\end{aligned}$$

Remark (Definition predicate reading).

$$\mathrm{OscillationAt}(f, x_0, \omega) :\!= \lim_{\delta \to 0^+} \omega\big(f;\, U_\delta(x_0) \cap E\big) = \omega, \quad U_\delta(x_0) = \{x \in \mathbb{R} : |x - x_0| < \delta\}.$$

Remark (Negated quantified statement).

$$\begin{aligned}
&\text{Let } E \subseteq \mathbb{R},\ f : E \to \mathbb{R},\ x_0 \in \mathbb{R},\ \omega \in [0, \infty]. \\
&\neg\big[(\forall \delta > 0 : U_\delta(x_0) \cap E \neq \varnothing) \ \wedge \ \omega(f; x_0) = \omega\big] \\
&\iff \Big[\exists \delta_0 > 0 : U_{\delta_0}(x_0) \cap E = \varnothing\Big] \\
&\qquad \vee \Big(\exists \varepsilon_0 > 0\ \forall \delta > 0\ \exists r\ (0 < r < \delta \wedge |\omega(f; U_r(x_0) \cap E) - \omega| \geq \varepsilon_0)\Big).
\end{aligned}$$

Remark (Negation predicate reading).

$$\neg\,\mathrm{OscillationAt}(f, x_0, \omega) \iff \Big[\exists \delta_0 > 0 : U_{\delta_0}(x_0) \cap E = \varnothing\Big] \vee \Big(\exists \varepsilon_0 > 0\ \forall \delta > 0\ \exists r\ (0 < r < \delta \wedge |\omega(f; U_r(x_0)$$

Remark (Failure modes). The definition fails in two distinct ways: either the point x_0 has a punctured neighborhood missing E, so there are no arbitrarily small samples to measure, or the local oscillation values near x_0 refuse to stabilize, preventing the limit from existing.

Remark (Failure mode decomposition).

$$\neg\,\mathrm{OscillationAt}(f, x_0, \omega) = \underbrace{\exists \delta_0 > 0 : U_{\delta_0}(x_0) \cap E = \varnothing}_{\text{no nearby sample points}} \vee \underbrace{\exists \varepsilon_0 > 0\ \forall \delta > 0\ \exists r\ (0 < r < \delta \wedge |\omega(f; U_r(x_0) \cap}_{\text{oscillation limit does not exist}}$$

Remark (Interpretation). Oscillation at x_0 measures the largest possible jump in function values that survives after one zooms in arbitrarily tightly around x_0. The shrinking neighborhoods $U_r(x_0) \cap E$ supply finer and finer samples, and the oscillation on each such set records the spread of f there. If these spreads settle to a single number, that number captures the local variability; it is zero precisely when f becomes arbitrarily close to a single value, i.e., when f is continuous at x_0. Failure occurs either because x_0 has no nearby domain points or because the spreads fluctuate indefinitely, reflecting wild local behavior.

Remark (Dependencies).

Definition 2.2.10

Theorem (Continuity Characterised by Oscillation)

Theorem 2.2.12 (Continuity Characterised by Oscillation)**.** *Let $A \subseteq \mathbb{R}$, let $f\colon A \to \mathbb{R}$, and let $c \in A$. Then f is continuous at c if and only if $\omega(f; c) = 0$. Go to proof.*

Remark (Standard quantified statement).

$$\begin{aligned}
&\text{Let } A \subseteq \mathbb{R},\ f\colon A \to \mathbb{R},\ c \in A. \\
&(\forall \varepsilon > 0\ \exists \delta > 0\ \forall x \in A\ \big(|x - c| < \delta \Rightarrow |f(x) - f(c)| < \varepsilon\big)) \\
&\iff \\
&\left(\inf_{\delta > 0} \sup\big\{|f(x) - f(y)| : x, y \in A,\ |x - c| < \delta,\ |y - c| < \delta\big\} = 0\right).
\end{aligned}$$

Remark (Predicate reading).

$$\mathrm{ContinuousAt}(f, c) \iff \mathrm{OscillationAt}(f, c, 0).$$

Remark (Negated quantified statement).

Let $A \subseteq \mathbb{R},\ f\colon A \to \mathbb{R},\ c \in A$.

$$\Big(\forall \varepsilon > 0\ \exists \delta > 0\ \forall x \in A\ (|x-c| < \delta \Rightarrow |f(x) - f(c)| < \varepsilon)\Big) \wedge \Big(\inf_{\delta>0} \sup\{|f(x) - f(y)| : x, y \in A,\ |x-c| <$$
$$\vee$$
$$\Big(\exists \varepsilon_0 > 0\ \forall \delta > 0\ \exists x \in A\ (|x-c| < \delta \wedge |f(x) - f(c)| \geq \varepsilon_0)\Big) \wedge \Big(\inf_{\delta>0} \sup\{|f(x) - f(y)| : x, y \in A,\ |x-c|$$

Remark (Negation predicate reading).

$$\neg\big(\mathrm{ContinuousAt}(f, c) \iff \mathrm{OscillationAt}(f, c, 0)\big).$$

Remark (Failure modes). The equivalence fails precisely in two mutually exclusive ways: either f remains continuous at c while the oscillation retains a positive lower bound, or $\omega(f; c)$ vanishes even though f does not satisfy the ε-δ continuity condition at c.

Remark (Failure mode decomposition).

$$\underbrace{\mathrm{ContinuousAt}(f, c) \wedge \neg\,\mathrm{OscillationAt}(f, c, 0)}_{\text{Positive oscillation despite continuity}} \vee \underbrace{\neg\,\mathrm{ContinuousAt}(f, c) \wedge \mathrm{OscillationAt}(f, c, 0)}_{\text{Zero oscillation without continuity}}.$$

Remark (Interpretation). The theorem states that measuring the oscillation of f near c captures exactly the same local regularity as the classical ε-δ definition of continuity. If oscillation collapses to zero, every pair of nearby function values becomes arbitrarily close, forcing $f(x)$ to approach $f(c)$. Conversely, any discontinuity manifests as a nonzero oscillation, because nearby points can produce separated outputs. The standard failure therefore arises from detecting a residual jump or oscillatory behavior in arbitrarily small neighborhoods of c.

Remark (Dependencies).

Definition 2.2.2, Definition 2.2.11.

Proposition

Proposition 2.2.13 (Discontinuity Set via Oscillation). *Let $f : E \to \mathbb{R}$. Then*

$$D(f) := \{x \in E : f \text{ is discontinuous at } x\} = \{x \in E : \omega(f; x) > 0\} = \bigcup_{n=1}^{\infty} \{x \in E : \omega(f; x) \geq 1/n\}.$$

Go to proof.

Remark (Standard quantified statement).

Let $f : E \to \mathbb{R}$.
$\forall x \in E :\ x \in D(f) \iff \omega(f; x) > 0 \iff \exists n \in \mathbb{N} :\ \omega(f; x) \geq \frac{1}{n}$.

Remark (Predicate reading).

$$\begin{aligned} &x \in D(f) \iff \mathrm{DiscontinuousAt}(f, x) \\ &\iff \exists \omega\ (\mathrm{OscillationAt}(f, x, \omega) \wedge \omega > 0) \\ &\iff \exists n \in \mathbb{N}\ \exists \omega\ (\mathrm{OscillationAt}(f, x, \omega) \wedge \omega \geq \tfrac{1}{n}). \end{aligned}$$

Remark (Negated quantified statement).

$$\begin{aligned}
&\text{Let } f : E \to \mathbb{R}. \\
&\neg\Big[\forall x \in E \ \big(x \in D(f) \iff \omega(f;x) > 0 \iff \exists n \in \mathbb{N}\ \omega(f;x) \geq \tfrac{1}{n}\big)\Big] \\
&\iff \exists x \in E\ \Big((x \in D(f) \wedge \omega(f;x) = 0) \vee (x \notin D(f) \wedge \omega(f;x) > 0) \\
&\qquad\qquad \vee (\omega(f;x) > 0 \wedge \forall n \in \mathbb{N}\ \omega(f;x) < \tfrac{1}{n})\Big).
\end{aligned}$$

Remark (Negation predicate reading).

$$\begin{aligned}
\exists x \in E\ \Big(&\exists \omega\ (\mathrm{DiscontinuousAt}(f,x) \wedge \mathrm{OscillationAt}(f,x,\omega) \wedge \omega = 0) \vee \\
&\exists \omega\ (\neg\,\mathrm{DiscontinuousAt}(f,x) \wedge \mathrm{OscillationAt}(f,x,\omega) \wedge \omega > 0) \vee \\
&\exists \omega\ (\mathrm{OscillationAt}(f,x,\omega) \wedge \omega > 0 \wedge \forall n \in \mathbb{N}\ \omega < \tfrac{1}{n})\Big).
\end{aligned}$$

Remark (Failure modes). The proposition fails precisely when some $x \in E$ is marked as discontinuous although its oscillation vanishes, or when x is declared continuous despite a positive oscillation, or when a positive oscillation never exceeds any reciprocal integer, contradicting the Archimedean step that converts $\omega(f;x) > 0$ into the countable union description.

Remark (Failure mode decomposition).

$$\exists x \in E \ \begin{cases} x \in D(f) \wedge \omega(f;x) = 0 & \text{(Type I)}, \\ x \notin D(f) \wedge \omega(f;x) > 0 & \text{(Type II)}, \\ \omega(f;x) > 0 \wedge \forall n \in \mathbb{N}\ \omega(f;x) < \frac{1}{n} & \text{(Type III)}. \end{cases}$$

Remark (Interpretation). Oscillation detects discontinuity quantitatively: every discontinuity forces $\omega(f;x)$ to stay above some positive threshold, and conversely the vanishing of oscillation characterises continuity. Because each set $\{x \in E : \omega(f;x) \geq 1/n\}$ is closed, the discontinuity set $D(f)$ is an F_σ, a countable union of closed sets. This structural description is the gateway to measure-theoretic criteria such as the Lebesgue characterisation of Riemann integrability, where the size of $D(f)$ controls integrability.

Remark (Dependencies).

Definition 2.2.5; Definition 2.2.11.

2.3 Global Theorems

Toolkit — Boundedness, Extrema, and Intermediate Values

- Implication chain on $[a,b]$: $\mathrm{ContinuousOn}(f,[a,b]) \Rightarrow \mathrm{BoundedOn}(f,[a,b]) \Rightarrow$ extrema attained $\Rightarrow$ all intermediate values attained $\Rightarrow$ $f([a,b])$ is a closed bounded interval.
- All three hypotheses are essential: closed, bounded, continuous. Drop any one and the conclusion fails.
- IVT on a general interval: continuous image of any interval is an interval (Preservation of Intervals).
- Darboux property: a continuous function cannot skip a value.

Intermediate Value Theorem

Definition (Bounded on a Set)

Definition 2.3.1 (Bounded on a Set)**.** Let $A \subseteq \mathbb{R}$ and $f\colon A \to \mathbb{R}$. The function f is bounded on A if and only if there exists a constant $M > 0$ such that $|f(x)| \le M$ for every $x \in A$.

Remark (Standard quantified statement).

$$\begin{aligned}&\text{Let } A \subseteq \mathbb{R} \text{ and } f\colon A \to \mathbb{R}.\\ &f \text{ is bounded on } A \iff \exists M > 0\ \forall x \in A\ (|f(x)| \le M).\end{aligned}$$

Remark (Definition predicate reading).

$$\mathrm{BoundedOn}(f,A) \coloneqq \exists M > 0\ \forall x \in A\ (|f(x)| \le M).$$

Remark (Predicate reading).

$$\mathrm{ContinuousOn}(f,[a,b]) \wedge \forall x \in [a,b]\ (f(x) \ne c) \implies \Big[\big(\forall x \in [a,b]\ f(x) > c\big) \vee \big(\forall x \in [a,b]\ f(x) < c\big)\Big].$$

Remark (Negated quantified statement).

$$\begin{aligned}&\text{Let } A \subseteq \mathbb{R} \text{ and } f\colon A \to \mathbb{R}.\\ &\neg(f \text{ bounded on } A) \iff \forall M > 0\ \exists x \in A\ (|f(x)| > M).\end{aligned}$$

Remark (Negation predicate reading).

$$\neg\,\mathrm{BoundedOn}(f,A) \iff \forall M > 0\ \exists x \in A\ (|f(x)| > M).$$

Remark (Failure modes)*.* Failure occurs precisely when every proposed positive bound M is exceeded by some value of $|f(x)|$ with $x \in A$.

Remark (Failure mode decomposition).

$$\underbrace{\forall M > 0}_{\text{no admissible bound}}\quad \underbrace{\exists x \in A\ (|f(x)| > M)}_{\text{value exceeding } M}.$$

Remark (Interpretation). The definition packages the informal idea that the values of f stay within a fixed vertical strip over A. The witness M serves as a uniform ceiling for the magnitudes $|f(x)|$, so once such an M is known the graph of f never escapes the band $[-M, M]$. Failure means that the outputs of f grow without limit along some sequence of points inside A, so no finite strip can contain the entire image. Boundedness on a set is the foundational regularity hypothesis that underlies the boundedness and extreme value theorems for continuous functions on compact domains.

Remark (Dependencies).

- Function
- Subset
- Absolute Value on $\mathbb{R}$
- Order on $\mathbb{R}$

Theorem (Boundedness Theorem)

Theorem 2.3.2 (Boundedness Theorem). *Let $a < b$ and let $f : [a, b] \to \mathbb{R}$ be continuous on $[a, b]$. Then there exists $M > 0$ such that $|f(x)| \le M$ for every $x \in [a, b]$.* *Go to proof.*

Remark (Standard quantified statement).

$$\begin{aligned}&\text{Let } a < b \text{ and } f : [a, b] \to \mathbb{R}.\\&\Big(\forall c \in [a, b]\ \forall \varepsilon > 0\ \exists \delta > 0\ \forall x \in [a, b]\big(|x - c| < \delta \Rightarrow |f(x) - f(c)| < \varepsilon\big)\Big)\\&\qquad \Longrightarrow \exists M > 0\ \forall x \in [a, b]\ (|f(x)| \le M).\end{aligned}$$

Remark (Predicate reading).

$$\mathrm{ContinuousOn}(f, [a, b]) \Longrightarrow \mathrm{BoundedOn}(f, [a, b]).$$

Remark (Negated quantified statement).

$$\begin{aligned}&\text{Let } a < b \text{ and } f : [a, b] \to \mathbb{R}.\\&\Big(\forall c \in [a, b]\ \forall \varepsilon > 0\ \exists \delta > 0\ \forall x \in [a, b]\big(|x - c| < \delta \Rightarrow |f(x) - f(c)| < \varepsilon\big)\Big)\\&\qquad \wedge \Big(\forall M > 0\ \exists x \in [a, b]\ (|f(x)| > M)\Big).\end{aligned}$$

Remark (Negation predicate reading).

$$\mathrm{ContinuousOn}(f, [a, b]) \wedge \neg\,\mathrm{BoundedOn}(f, [a, b]).$$

Remark (Failure modes). Each component hypothesis is indispensable: without closedness (for instance $f(x) = 1/x$ on $(0, 1]$) continuity does not prevent divergence near the missing endpoint; without bounded domain (for example $f(x) = x$ on $[0, \infty)$) linear growth forces unbounded values; without continuity (take $f(0) = 0$ and $f(x) = 1/x$ for $x \in (0, 1]$) the discontinuity at 0 again creates arbitrarily large magnitudes.

Remark (Failure mode decomposition).

$$\neg\big[\mathrm{ClosedInterval}(a,b) \wedge \mathrm{BoundedSet}([a,b]) \wedge \mathrm{ContinuousOn}(f,[a,b])\big] = \underbrace{\neg\,\mathrm{ClosedInterval}(a,b)}_{\text{missing endpoint}} \vee \underbrace{\neg\,\mathrm{Bou}}_{\text{unbo}}$$

Remark (Interpretation). Continuity ties nearby inputs to nearby outputs, and the compact interval $[a,b]$ can be covered by finitely many neighborhoods in which the oscillation of f is controlled; taking the maximal oscillation among this finite cover supplies a global bound. Failure typically arises when the domain lacks compactness: an omitted endpoint allows the function to escape control near the hole, an unbounded interval lets values diverge along the ray, and a point of discontinuity can spike the function without upsetting nearby behavior elsewhere.

Remark (Dependencies).

Definition 2.2.2; Definition 2.3.1.

Definition (Absolute Maximum and Absolute Minimum)

Definition 2.3.3 (Absolute Maximum and Absolute Minimum)**.** Let $A \subseteq \mathbb{R}$ and let $f\colon A \to \mathbb{R}$. A point $x^* \in A$ is an absolute maximum of f on A if and only if $f(x^*) \ge f(x)$ for every $x \in A$. A point $x_* \in A$ is an absolute minimum of f on A if and only if $f(x_*) \le f(x)$ for every $x \in A$.

Remark (Standard quantified statement).

$$\begin{aligned}
&\text{Let } A \subseteq \mathbb{R},\ f\colon A \to \mathbb{R},\ x^*, x_* \in A.\\
&x^* \text{ is an absolute maximum of } f \text{ on } A \iff \forall x \in A\ (f(x) \le f(x^*)).\\
&x_* \text{ is an absolute minimum of } f \text{ on } A \iff \forall x \in A\ (f(x_*) \le f(x)).
\end{aligned}$$

Remark (Definition predicate reading).

$$\begin{aligned}
(x^* \in A) \wedge \mathrm{Maximum}(f(x^*), f(A)) &\iff (x^* \in A) \wedge \forall x \in A\ (f(x) \le f(x^*)),\\
(x_* \in A) \wedge \mathrm{Minimum}(f(x_*), f(A)) &\iff (x_* \in A) \wedge \forall x \in A\ (f(x_*) \le f(x)).
\end{aligned}$$

Remark (Negated quantified statement).

$$\begin{aligned}
&\text{Let } A \subseteq \mathbb{R},\ f\colon A \to \mathbb{R},\ x^*, x_* \in \mathbb{R}.\\
&x^* \text{ fails to be an absolute maximum of } f \text{ on } A\\
&\iff (x^* \notin A) \vee \big(\exists x \in A\ (f(x) > f(x^*))\big).\\
&x_* \text{ fails to be an absolute minimum of } f \text{ on } A\\
&\iff (x_* \notin A) \vee \big(\exists x \in A\ (f(x_*) > f(x))\big).
\end{aligned}$$

Remark (Negation predicate reading).

$$\begin{aligned}
\neg\big[(x^* \in A) \wedge \mathrm{Maximum}(f(x^*), f(A))\big] &\iff (x^* \notin A) \vee \big(\exists x \in A\ (f(x) > f(x^*))\big),\\
\neg\big[(x_* \in A) \wedge \mathrm{Minimum}(f(x_*), f(A))\big] &\iff (x_* \notin A) \vee \big(\exists x \in A\ (f(x_*) > f(x))\big).
\end{aligned}$$

Remark (Failure modes). An alleged absolute maximum can fail because the proposed point lies outside the domain A, or because some $x \in A$ produces a strictly larger function value. An alleged absolute minimum fails for the analogous reasons, either by leaving the domain or by admitting a point in A with a strictly smaller value.

Remark (Failure mode decomposition).

$$\neg\big[(x^* \in A) \wedge \mathrm{Maximum}(f(x^*), f(A))\big] = \underbrace{(x^* \notin A)}_{\text{external point}} \vee \underbrace{\big(\exists x \in A\ (f(x) > f(x^*))\big)}_{\text{better value}},$$

$$\neg\big[(x_* \in A) \wedge \mathrm{Minimum}(f(x_*), f(A))\big] = \underbrace{(x_* \notin A)}_{\text{external point}} \vee \underbrace{\big(\exists x \in A\ (f(x_*) > f(x))\big)}_{\text{better value}}.$$

Remark (Interpretation). An absolute maximum identifies the highest value that f attains on its domain A, together with a point of attainment, and the absolute minimum identifies the lowest attained value. These extrema describe the global ceiling and floor of the graph restricted to A, so they are the endpoints of the vertical range. Failure occurs either when the candidate point is not part of the domain or when another domain point produces a superior value, reflecting that global optimization is sensitive to both membership and comparative size. Geometrically the absolute maximum sits at the tallest point of the graph above A, while the absolute minimum sits at the lowest point.

Remark (Dependencies).

- Function
- Subset
- Order on $\mathbb{R}$
- Maximum
- Minimum

Theorem (Extreme Value Theorem)

Theorem 2.3.4 (Extreme Value Theorem)**.** *Let $a, b \in \mathbb{R}$ with $a < b$, and let $f : [a, b] \to \mathbb{R}$ be continuous. There exist points $x_m, x_M \in [a, b]$ such that $f(x_m) \le f(x) \le f(x_M)$ for every $x \in [a, b]$. Equivalently, f attains both an absolute maximum and an absolute minimum on $[a, b]$. Go to proof.*

Remark (Standard quantified statement).

$$\begin{aligned}&\text{Let } a, b \in \mathbb{R} \text{ with } a < b, \text{ and let } f : [a, b] \to \mathbb{R} \text{ be continuous.}\\ &\exists x_m, x_M \in [a, b]\ \forall x \in [a, b]\ (f(x_m) \le f(x) \le f(x_M)).\end{aligned}$$

Remark (Predicate reading).

$$\mathrm{ContinuousOn}(f, [a, b]) \implies \exists x_m, x_M \in [a, b]\ \big(\mathrm{Maximum}(f(x_M), f([a, b])) \wedge \mathrm{Minimum}(f(x_m), f([a, b]))\big).$$

Remark (Failure modes). The theorem can fail if the compact interval hypothesis is lost, or if the function is not continuous on the whole interval. Without compactness an extremal value may exist only as a limiting value, and without continuity the graph may jump over its supremum or infimum.

Remark (Failure mode decomposition).

$$\underbrace{\neg(a < b)}_{\text{invalid interval}} \vee \underbrace{\neg\big(f : [a, b] \to \mathbb{R}\big)}_{\text{wrong function type}} \vee \underbrace{\exists c \in [a, b]\ \exists \varepsilon > 0\ \forall \delta > 0\ \exists x \in [a, b]\ \big(|x - c| < \delta \wedge |f(x) - f(c)| \ge \varepsilon\big)}_{\text{loss of continuity}}$$

Remark (Interpretation). Continuity on a closed and bounded interval forces the image of the interval to inherit the compact features of the domain. The function cannot escape to infinity or oscillate without settling, so it must reach both its highest and lowest values somewhere in the interval. When the hypotheses fail, the typical obstruction is the loss of compactness: on an open interval the function might drift off without achieving its supremum, and without continuity it may jump over candidate extrema.

Remark (Dependencies).

Uses Definition 2.3.3 and Theorem 2.3.2.

Theorem (Location of Roots)

Theorem 2.3.5 (Location of Roots)**.** *Let $a, b \in \mathbb{R}$ with $a < b$ and let $f : [a, b] \to \mathbb{R}$ be continuous. If either $f(a) < 0 < f(b)$ or $f(a) > 0 > f(b)$, then there exists $c \in (a, b)$ such that $f(c) = 0$. Go to proof.*

Remark (Standard quantified statement).

$$\text{Let } a, b \in \mathbb{R} \text{ with } a < b \text{ and } f : [a, b] \to \mathbb{R} \text{ continuous.}$$
$$\big(f(a) < 0 < f(b) \vee f(a) > 0 > f(b)\big) \Rightarrow \exists c \in (a, b)\ \big(f(c) = 0\big).$$

Remark (Failure modes). The root conclusion can fail if f is not continuous on $[a, b]$, or if the endpoint values do not have opposite signs.

Remark (Failure mode decomposition).

$$\underbrace{\exists c \in [a, b]\ \exists \varepsilon > 0\ \forall \delta > 0\ \exists x \in [a, b]\ \big(|x - c| < \delta \wedge |f(x) - f(c)| \geq \varepsilon\big)}_{\text{loss of continuity}} \vee \underbrace{\big(f(a)f(b) \geq 0\big)}_{\text{no sign change}}.$$

Remark (Interpretation). Continuity forces the image of the interval $[a, b]$ to contain every intermediate value between $f(a)$ and $f(b)$. A sign change therefore compels the graph to cross the horizontal axis, producing a root strictly inside the interval. Failure occurs when the endpoint values share the same sign or when f is not continuous, because then the image need not sweep across zero. Geometrically, the theorem asserts that a continuous curve connecting opposite sides of the x-axis must intersect the axis somewhere in between.

Remark (Dependencies).

Bolzano's Intermediate Value Theorem.

Theorem (Bolzano's Intermediate Value Theorem)

Theorem 2.3.6 (Bolzano's Intermediate Value Theorem)**.** *Let $I \subseteq \mathbb{R}$ be an interval, let $f : I \to \mathbb{R}$ be continuous on I, and choose $a, b \in I$ with $f(a) < k < f(b)$. Then there exists $c \in I$ such that $\min\{a, b\} < c < \max\{a, b\}$ and $f(c) = k$. Go to proof.*

Remark (Standard quantified statement).

$$\forall I \subseteq \mathbb{R}\ \forall f : I \to \mathbb{R}\ \forall a, b \in I\ \forall k \in \mathbb{R}\ \Big[\big(\forall x, y \in I\ \forall z \in \mathbb{R}\ (x < z < y \Rightarrow z \in I)\big) \wedge \big(\forall c \in I\ \forall \varepsilon > 0\ \exists \delta > 0\ \forall$$

Remark (Failure modes). The conclusion can fail only if the continuity-on-an-interval hypothesis breaks down, or if the proposed level k is not strictly between the two attained endpoint values.

Remark (Failure mode decomposition).

$$\underbrace{\exists x, y \in I \; \exists z \in \mathbb{R} \; \big(x < z < y \land z \notin I\big)}_{\text{not an interval}}$$
$$\lor \; \underbrace{\exists c \in I \; \exists \varepsilon > 0 \; \forall \delta > 0 \; \exists x \in I \; \big(|x - c| < \delta \land |f(x) - f(c)| \geq \varepsilon\big)}_{\text{loss of continuity}}$$
$$\lor \quad \underbrace{\neg\big(f(a) < k < f(b)\big)}_{\text{level not between endpoint values}} \quad .$$

Remark (Interpretation). A continuous real-valued function on an interval cannot skip any intermediate scalar between two attained values, so the graph must cross every horizontal level that lies strictly between $f(a)$ and $f(b)$. The conclusion provides an interior point c whose image equals the prescribed level k, ensuring root-finding for $f-k$. If the hypothesis fails, either f is not continuous or the level k does not lie between the endpoint values, and no such interior point need exist.

Remark (Dependencies).

- Function
- Subset
- Continuity at a Point
- Order on $\mathbb{R}$
- Maximum
- Minimum

Theorem (Image of Closed Bounded Interval Is Closed Bounded Interval)

Theorem 2.3.7 (Image of Closed Bounded Interval Is Closed Bounded Interval)**.** *Let $f : [a, b] \to \mathbb{R}$ be continuous. Then $f([a, b]) = [\min_{[a,b]} f, \max_{[a,b]} f]$. Go to proof.*

Remark (Standard quantified statement).

$\forall a, b \in \mathbb{R} \; \forall f : [a, b] \to \mathbb{R} :$
$$\Big(a < b \land \forall c \in [a, b] \; \forall \varepsilon > 0 \; \exists \delta > 0 \; \forall x \in [a, b] \; \big(|x - c| < \delta \Rightarrow |f(x) - f(c)| < \varepsilon\big)\Big) \implies f([a, b]) = [\mathrm{m}$$

Remark (Predicate reading).

$$\mathrm{ContinuousOn}(f, [a, b]) \implies f([a, b]) = [\min_{[a,b]} f, \max_{[a,b]} f].$$

Remark (Interpretation). The statement combines the extreme value theorem, which supplies both a minimum and a maximum on $[a, b]$, with the intermediate value theorem, which fills in every intermediate value between them. Consequently the image $f([a, b])$ is a closed interval with endpoints given by the absolute extremal values of f on $[a, b]$, showing that continuous functions map compact intervals onto compact intervals without introducing gaps.

Remark (Dependencies).

Extreme Value Theorem, Bolzano's Intermediate Value Theorem

Theorem (Preservation of Intervals)

Theorem 2.3.8 (Preservation of Intervals)**.** *Let $I \subseteq \mathbb{R}$ be an interval and let $f : I \to \mathbb{R}$ be continuous. Then $f(I)$ is an interval. Go to proof.*

Remark (Standard quantified statement).

$$\begin{aligned}&\forall I \subseteq \mathbb{R}\ \forall f : I \to \mathbb{R} :\\ &\qquad \Big[\big(\forall x_1, x_2 \in I\ \forall z \in \mathbb{R}\ (x_1 < z < x_2 \Rightarrow z \in I)\big)\\ &\qquad \wedge \big(\forall c \in I\ \forall \varepsilon > 0\ \exists \delta > 0\ \forall x \in I\ (|x-c| < \delta \Rightarrow |f(x) - f(c)| < \varepsilon)\big)\Big] \implies \forall y_1, y_2 \in f(I)\ \forall y \in \mathbb{R}\ \big(y_1\end{aligned}$$

Remark (Predicate reading).

$$\mathrm{ContinuousOn}(f, I) \implies \mathrm{Interval}(f(I)).$$

Remark (Negated quantified statement).

$$\begin{aligned}&\exists I \subseteq \mathbb{R}\ \exists f : I \to \mathbb{R}\ \exists y_1, y_2 \in f(I)\ \exists y \in \mathbb{R} :\\ &\qquad \big(\forall x_1, x_2 \in I\ \forall z \in \mathbb{R}\ (x_1 < z < x_2 \Rightarrow z \in I)\big)\\ &\qquad \wedge \big(\forall c \in I\ \forall \varepsilon > 0\ \exists \delta > 0\ \forall x \in I\ (|x-c| < \delta \Rightarrow |f(x) - f(c)| < \varepsilon)\big)\\ &\qquad \wedge y_1 < y < y_2 \wedge y \notin f(I).\end{aligned}$$

Remark (Negation predicate reading).

$$\mathrm{Interval}(I) \wedge \mathrm{ContinuousOn}(f, I) \wedge \neg\,\mathrm{Interval}(f(I)).$$

Remark (Interpretation). Continuous functions cannot tear gaps open inside their images of intervals. Once the function achieves two output values, continuity and the Intermediate Value Theorem force every intermediate value to occur as well, so the image remains an interval even when I is unbounded or half-open. Failure occurs precisely when continuity breaks, because discontinuities can skip intermediate heights and leave disconnected image sets.

Remark (Dependencies).

thm:bolzano-intermediate-value

Theorem (Darboux Property)

Theorem 2.3.9 (Darboux Property)**.** *Let $a, b \in \mathbb{R}$ satisfy $a < b$, let $f : [a, b] \to \mathbb{R}$ be continuous, and fix $c \in \mathbb{R}$. If $f(x) \neq c$ for every $x \in [a, b]$, then either $f(x) > c$ for all $x \in [a, b]$ or $f(x) < c$ for all $x \in [a, b]$. Go to proof.*

Remark (Standard quantified statement).

$$\begin{aligned}&\text{Let } a < b,\ f : [a, b] \to \mathbb{R} \text{ be continuous, and } c \in \mathbb{R}.\\ &\Big(\forall x \in [a, b]\ f(x) \neq c\Big) \Rightarrow \Big[\big(\forall x \in [a, b]\ f(x) > c\big) \vee \big(\forall x \in [a, b]\ f(x) < c\big)\Big].\end{aligned}$$

Remark (Negated quantified statement).

$$\text{Let } a<b,\ f:[a,b]\to\mathbb{R} \text{ be continuous, and } c\in\mathbb{R}.$$
$$\Big(\forall x\in[a,b]\ f(x)\neq c\Big)\wedge\Big(\exists u\in[a,b]\ f(u)\le c\Big)\wedge\Big(\exists v\in[a,b]\ f(v)\ge c\Big).$$

Remark (Negation predicate reading).

$$\mathrm{ContinuousOn}(f,[a,b])\wedge\Big(\forall x\in[a,b]\ f(x)\neq c\Big)\wedge\neg\Big[\big(\forall x\in[a,b]\ f(x)>c\big)\vee\big(\forall x\in[a,b]\ f(x)<c\big)\Big].$$

Remark (Contrapositive quantified statement).

$$\text{Let } a<b,\ f:[a,b]\to\mathbb{R} \text{ be continuous, and } c\in\mathbb{R}.$$
$$\Big(\exists x\in[a,b]\ f(x)\le c\Big)\wedge\Big(\exists y\in[a,b]\ f(y)\ge c\Big)\Rightarrow\exists z\in[a,b]\ f(z)=c.$$

Remark (Contrapositive predicate reading).

$$\mathrm{ContinuousOn}(f,[a,b])\wedge\mathrm{MeetsBothSides}(f,[a,b],c)\Longrightarrow\exists z\in[a,b]\ (f(z)=c).$$

Remark (Interpretation). A continuous function on a compact interval cannot oscillate across a level c without actually attaining that level. Stated contrapositive fashion, if c is never taken then the entire graph lies strictly above or strictly below the horizontal line y=c. The only way this guarantee can fail is when the function produces values on both sides of c, in which case continuity forces a crossing point. Geometrically the conclusion describes the image of a connected set under a continuous map: it is an interval, so omitting c forces the image to stay in one of the open half-lines determined by c.

Remark (Dependencies).

Bolzano's Intermediate Value Theorem.

2.4 Uniform Continuity

Toolkit — Uniform Continuity

- Key distinction: $\delta=\delta(\varepsilon)$ only, not $\delta(\varepsilon,c)$. One δ serves all pairs in A.
- Input condition: $\mathrm{Within}(x,u,\delta)$ — both x and u range freely, no puncturing.
- Implication chain: $\mathrm{Lipschitz}(f,A)\Rightarrow\mathrm{UniformlyContinuous}(f,A)\Rightarrow\mathrm{ContinuousOn}(f,A)$. Both implications are strict.
- Heine–Cantor: $\mathrm{ContinuousOn}(f,[a,b])\Rightarrow\mathrm{UniformlyContinuous}(f,[a,b])$.
- Failure criterion: sequences $(x_n),(u_n)$ with $x_n-u_n\to0$ but $|f(x_n)-f(u_n)|\ge\varepsilon_0$.
- $\mathrm{UniformlyContinuous}(f,A)$ preserves Cauchy sequences — the key lemma for continuous extensions.

Uniform Continuity

Definition (Uniform Continuity)

Definition 2.4.1 (Uniform Continuity)**.** Let $A \subseteq \mathbb{R}$ and let $f : A \to \mathbb{R}$. The function f is uniformly continuous on A if and only if for every $\varepsilon > 0$ there exists $\delta > 0$ such that for all $x, u \in A$ the implication $|x - u| < \delta \Rightarrow |f(x) - f(u)| < \varepsilon$ holds.

Remark (Standard quantified statement).

$$\begin{aligned}&(A \subseteq \mathbb{R}) \wedge (f : A \to \mathbb{R})\\ &\qquad \Longrightarrow \forall \varepsilon > 0\ \exists \delta > 0\ \forall x, u \in A\ (|x - u| < \delta \Rightarrow |f(x) - f(u)| < \varepsilon).\end{aligned}$$

Remark (Definition predicate reading).

$$\mathrm{UniformlyContinuous}(f, A) := \forall \varepsilon > 0\ \exists \delta > 0\ \forall x, u \in A\ (|x - u| < \delta \Rightarrow |f(x) - f(u)| < \varepsilon).$$

Remark (Negated quantified statement).

$$\begin{aligned}&\neg(A \subseteq \mathbb{R}) \vee \neg(f : A \to \mathbb{R})\\ &\qquad \vee \exists \varepsilon > 0\ \forall \delta > 0\ \exists x, u \in A\ (|x - u| < \delta \wedge |f(x) - f(u)| \geq \varepsilon).\end{aligned}$$

Remark (Negation predicate reading).

$$\neg\,\mathrm{UniformlyContinuous}(f, A) \Longleftrightarrow \exists \varepsilon > 0\ \forall \delta > 0\ \exists x, u \in A\ (|x - u| < \delta \wedge |f(x) - f(u)| \geq \varepsilon).$$

Remark (Failure modes). The definition fails exactly when a single positive tolerance ε_0 exposes pairs of inputs arbitrarily close together whose function values remain at least ε_0 apart, so no universal δ works for that ε_0.

Remark (Failure mode decomposition).

$$\neg\,\mathrm{UniformlyContinuous}(f, A) = \underbrace{\exists \varepsilon_0 > 0}_{\text{frozen tolerance}} \wedge \underbrace{\forall \delta > 0}_{\text{no global control}} \wedge \underbrace{\exists x, u \in A\ (|x - u| < \delta \wedge |f(x) - f(u)| \geq \varepsilon}_{\text{bad witness pair}}$$

Remark (Interpretation). Uniform continuity demands that a single proximity scale δ answers any prescribed accuracy ε simultaneously across the whole domain. The negation shows what goes wrong: a fixed output tolerance resists all choices of δ because pairs of nearby inputs can always be found that stretch the function values too far apart. Geometrically, the graph of a uniformly continuous function can be traversed with a ruler of length δ horizontally and ε vertically without ever overstepping the vertical tolerance, whereas a nonuniformly continuous function exhibits spikes that defeat every fixed ruler length.

Remark (Dependencies).

- Function
- Subset
- Absolute Value on $\mathbb{R}$
- Order on $\mathbb{R}$

Theorem (Algebra of Uniform Continuity on a General Domain)

Theorem 2.4.2 (Algebra of Uniform Continuity on a General Domain)**.** *(a) For every set $A \subseteq \mathbb{R}$, every pair of functions $f, g : A \to \mathbb{R}$ that are uniformly continuous on A, and every scalar $c \in \mathbb{R}$, the functions $f + g$, $f - g$, and cf are uniformly continuous on A.*

(b) There exist a set $A \subseteq \mathbb{R}$ and uniformly continuous functions $f, g : A \to \mathbb{R}$ such that the product fg is not uniformly continuous on A.

Go to proof.

Remark (Standard quantified statement).

$$\forall A \subseteq \mathbb{R}\ \forall f, g : A \to \mathbb{R}\ \forall c \in \mathbb{R} :$$
$$\Big[\big(\forall \varepsilon > 0\ \exists \delta_f > 0\ \forall x, y \in A\ (|x - y| < \delta_f \Rightarrow |f(x) - f(y)| < \varepsilon)\big)$$
$$\wedge \big(\forall \varepsilon > 0\ \exists \delta_g > 0\ \forall x, y \in A\ (|x - y| < \delta_g \Rightarrow |g(x) - g(y)| < \varepsilon)\big)\Big] \Rightarrow \Big[\forall \varepsilon > 0\ \exists \delta_1 > 0\ \forall x, y \in A\ (|x$$
$$\wedge \forall \varepsilon > 0\ \exists \delta_2 > 0\ \forall x, y \in A\ (|x - y| < \delta_2 \Rightarrow |(f - g)(x) - (f - g)(y)| < \varepsilon)$$
$$\wedge \forall \varepsilon > 0\ \exists \delta_3 > 0\ \forall x, y \in A\ (|x - y| < \delta_3 \Rightarrow |cf(x) - cf(y)| < \varepsilon)\Big],$$
$$\exists A \subseteq \mathbb{R}\ \exists f, g : A \to \mathbb{R} :\ \Big[\forall \varepsilon > 0\ \exists \delta_f > 0\ \forall x, y \in A\ (|x - y| < \delta_f \Rightarrow |f(x) - f(y)| < \varepsilon)\Big]$$
$$\wedge \Big[\forall \varepsilon > 0\ \exists \delta_g > 0\ \forall x, y \in A\ (|x - y| < \delta_g \Rightarrow |g(x) - g(y)| < \varepsilon)\Big]$$
$$\wedge \exists \varepsilon_0 > 0\ \forall \delta > 0\ \exists x, y \in A\ \big(|x - y| < \delta \wedge |f(x)g(x) - f(y)g(y)| \geq \varepsilon_0\big).$$

Remark (Predicate reading).

$$\forall A \subseteq \mathbb{R}\ \forall f, g : A \to \mathbb{R}\ \forall c \in \mathbb{R} :$$
$$\big(\mathrm{UniformlyContinuous}(f, A) \wedge \mathrm{UniformlyContinuous}(g, A)\big) \implies \big(\mathrm{UniformlyContinuous}(f + g, A)$$
$$\exists A \subseteq \mathbb{R}\ \exists f, g : A \to \mathbb{R} :\quad \mathrm{UniformlyContinuous}(f, A) \wedge \mathrm{UniformlyContinuous}(g, A) \wedge \neg\, \mathrm{UniformlyCon}$$

Remark (Interpretation). Uniform continuity is stable under linear combinations on any common domain, so sums, differences, and scalar multiples inherit the same global control on oscillation from the original functions. The proof decomposes the linear expressions into epsilon-delta bounds that add and subtract without upsetting the uniform response. In contrast, products can amplify local variation when one factor lacks a bounded range, so even uniformly continuous multiplicands can produce a product that fails the global tolerance requirement. The theorem explains both the algebraic closure that analysts routinely exploit and the standard obstruction, guiding expectations when building uniformly continuous functions from simpler pieces.

Remark (Dependencies).

- Uniform Continuity
- Linear Combination of Real-Valued Functions
- Pointwise Difference of Functions

Theorem (Algebra of Uniform Continuity on a Bounded Domain)

Theorem 2.4.3 (Algebra of Uniform Continuity on a Bounded Domain)**.** *Let $A \subseteq \mathbb{R}$ be bounded, and let $f, g : A \to \mathbb{R}$ be uniformly continuous on A.*

(a) The product $x \mapsto f(x)g(x)$ is uniformly continuous on A.

(b) If there exists $k > 0$ such that $|g(x)| \geq k$ for all $x \in A$, then the quotient $x \mapsto f(x)/g(x)$ is uniformly continuous on A.

Go to proof.

Remark (Standard quantified statement).

$$\begin{aligned}
&\text{Let } A \subseteq \mathbb{R} \text{ be bounded and } f, g : A \to \mathbb{R}.\\
&\big[\forall \varepsilon > 0\ \exists \delta > 0\ \forall x, y \in A\ (|x - y| < \delta \Rightarrow |f(x) - f(y)| < \varepsilon)\big]\\
&\wedge \big[\forall \varepsilon > 0\ \exists \delta > 0\ \forall x, y \in A\ (|x - y| < \delta \Rightarrow |g(x) - g(y)| < \varepsilon)\big]\\
&\Rightarrow \forall \varepsilon > 0\ \exists \delta > 0\ \forall x, y \in A\ (|x - y| < \delta \Rightarrow |f(x)g(x) - f(y)g(y)| < \varepsilon),\\
&\big[\forall \varepsilon > 0\ \exists \delta > 0\ \forall x, y \in A\ (|x - y| < \delta \Rightarrow |f(x) - f(y)| < \varepsilon)\big]\\
&\wedge \big[\forall \varepsilon > 0\ \exists \delta > 0\ \forall x, y \in A\ (|x - y| < \delta \Rightarrow |g(x) - g(y)| < \varepsilon)\big]\\
&\wedge \big[\exists k > 0\ \forall x \in A\ (|g(x)| \geq k)\big]\\
&\Rightarrow \forall \varepsilon > 0\ \exists \delta > 0\ \forall x, y \in A\ (|x - y| < \delta \Rightarrow |f(x)/g(x) - f(y)/g(y)| < \varepsilon).
\end{aligned}$$

Remark (Predicate reading).

$$\begin{aligned}
&\text{Assume } A \subseteq \mathbb{R} \text{ is bounded and } f, g : A \to \mathbb{R}.\\
&\mathrm{UniformlyContinuous}(f, A) \wedge \mathrm{UniformlyContinuous}(g, A)\\
&\qquad \Longrightarrow \mathrm{UniformlyContinuous}(fg, A),\\
&\mathrm{UniformlyContinuous}(f, A) \wedge \mathrm{UniformlyContinuous}(g, A) \wedge \exists k > 0\ \forall x \in A\ (|g(x)| \geq k)\\
&\qquad \Longrightarrow \mathrm{UniformlyContinuous}(f/g, A).
\end{aligned}$$

Remark (Failure modes). The product conclusion can fail only when one of the hypotheses used to control the product is missing: the domain is not bounded, f is not uniformly continuous on A, or g is not uniformly continuous on A. The quotient conclusion can also fail if the denominator is not bounded away from zero on A.

Remark (Failure mode decomposition).

$$\underbrace{\neg\,\mathrm{BoundedSet}(A)}_{\text{unbounded domain}} \vee \underbrace{\neg\,\mathrm{UniformlyContinuous}(f, A)}_{\text{first factor not uniformly continuous}} \vee \underbrace{\neg\,\mathrm{UniformlyContinuous}(g, A)}_{\text{second factor not uniformly continuous}}$$
$$\vee\ \underbrace{\neg\exists k > 0\ \forall x \in A :\ |g(x)| \geq k}_{\text{denominator not bounded away from zero}}\ .$$

Remark (Interpretation). Bounded subsets of $\mathbb{R}$ prevent the product of two uniformly continuous functions from developing large oscillations, so the shared modulus of continuity for f and g can be combined to control fg. The quotient is uniformly continuous whenever g never approaches zero, because the bounded-away-from-zero hypothesis allows inversion to act as another uniformly continuous operation that preserves the common modulus after scaling. Failure of boundedness or a loss of a positive lower bound for $|g|$ are the only ways these closure properties can break, and in either case the oscillation of the algebraic combination can no longer be controlled uniformly across the entire set.

Remark (Dependencies).

Definition 2.4.1.

Proposition (Sequential Characterization of Uniform Continuity)

Proposition 2.4.4 (Sequential Characterization of Uniform Continuity)**.** *Let $A \subseteq \mathbb{R}$ and let $f : A \to \mathbb{R}$. Then f is uniformly continuous on A if and only if for every pair of sequences (x_n) and (y_n) in A with $\lim_{n\to\infty}(x_n - y_n) = 0$ we have $\lim_{n\to\infty}(f(x_n) - f(y_n)) = 0$. Go to proof.*

Remark (Standard quantified statement).

$$\Big[\forall \varepsilon > 0\ \exists \delta > 0\ \forall x, y \in A\ \big(|x - y| < \delta \Rightarrow |f(x) - f(y)| < \varepsilon\big)\Big] \iff \forall x_n, y_n : \mathbb{N} \to A\ \big(|x_n - y_n| \to 0 \Rightarrow |f$$

Remark (Predicate reading).

$$\text{UniformlyContinuous}(f, A) \iff \forall x_n, y_n : \mathbb{N} \to A\ \big(|x_n - y_n| \to 0 \Rightarrow |f(x_n) - f(y_n)| \to 0\big).$$

Remark (Negated quantified statement).

$$\begin{aligned}
&\exists A \subseteq \mathbb{R}\ \exists f : A \to \mathbb{R} :\\
&\Big(\big[\forall \varepsilon > 0\ \exists \delta > 0\ \forall x, y \in A\ (|x - y| < \delta \Rightarrow |f(x) - f(y)| < \varepsilon)\big]\\
&\qquad \wedge \exists x_n, y_n : \mathbb{N} \to A\ \big(|x_n - y_n| \to 0 \wedge \neg(|f(x_n) - f(y_n)| \to 0)\big)\Big)\\
&\vee \Big(\big[\exists \varepsilon_0 > 0\ \forall \delta > 0\ \exists x, y \in A\ (|x - y| < \delta \wedge |f(x) - f(y)| \geq \varepsilon_0)\big] \wedge \forall x_n, y_n : \mathbb{N} \to A\ \big(|x_n - y_n| \to 0 =
\end{aligned}$$

Remark (Negation predicate reading).

$$\begin{aligned}
&\exists A \subseteq \mathbb{R}\ \exists f : A \to \mathbb{R} :\\
&\Big(\text{UniformlyContinuous}(f, A) \wedge \exists x_n, y_n : \mathbb{N} \to A\ \big(|x_n - y_n| \to 0 \wedge \neg(|f(x_n) - f(y_n)| \to 0)\big)\Big)\\
&\vee \Big(\neg\, \text{UniformlyContinuous}(f, A) \wedge \forall x_n, y_n : \mathbb{N} \to A\ \big(|x_n - y_n| \to 0 \Rightarrow |f(x_n) - f(y_n)| \to 0\big)\Big).
\end{aligned}$$

Remark (Failure modes). The equivalence can fail in exactly two ways: either f is uniformly continuous while some pair of asymptotically coincident sequences in A yields outputs that do not converge together, or every such pair of sequences does converge together while f nonetheless fails to be uniformly continuous on A.

Remark (Failure mode decomposition).

$$\begin{aligned}
&\underbrace{\text{UniformlyContinuous}(f, A)}_{\text{uniform continuity holds}} \wedge \underbrace{\exists x_n, y_n : \mathbb{N} \to A\ \big(|x_n - y_n| \to 0 \wedge \neg(|f(x_n) - f(y_n)| \to 0)\big)}_{\text{sequential test fails}}\\
&\vee\ \underbrace{\neg\, \text{UniformlyContinuous}(f, A)}_{\text{uniform continuity fails}} \wedge \underbrace{\forall x_n, y_n : \mathbb{N} \to A\ \big(|x_n - y_n| \to 0 \Rightarrow |f(x_n) - f(y_n)| \to 0\big)}_{\text{sequential test holds}}.
\end{aligned}$$

Remark (Interpretation). Uniform continuity prevents the function from pulling together inputs that are asymptotically identical while letting their images drift apart. The sequential formulation packages this requirement into a tail test that ignores explicit epsilon-delta bookkeeping: whenever two points of A are eventually arbitrarily close, their function values must also be eventually arbitrarily close. Proving uniform continuity via sequences is often simpler, and conversely, producing two sequences that collide in A but whose images do not collide gives a concrete witness that uniform continuity fails. The two failure modes correspond to breaking one direction or the other of the claimed equivalence.

Remark (Dependencies).

Definition 2.4.1

Theorem (Heine–Cantor Theorem)

Theorem 2.4.5 (Heine–Cantor Theorem). *Let $a, b \in \mathbb{R}$ with $a < b$ and let $f : [a, b] \to \mathbb{R}$ be continuous on $[a, b]$. Then f is uniformly continuous on $[a, b]$.* *Go to proof.*

Remark (Standard quantified statement).

$$\begin{aligned}
&\text{Let } a, b \in \mathbb{R} \text{ with } a < b \text{ and let } f : [a, b] \to \mathbb{R}.\\
&\Big[\forall c \in [a, b]\ \forall \varepsilon > 0\ \exists \delta > 0\ \forall x \in [a, b]\ (|x - c| < \delta \Rightarrow |f(x) - f(c)| < \varepsilon)\Big]\\
&\implies \forall \varepsilon > 0\ \exists \delta > 0\ \forall x, y \in [a, b]\ (|x - y| < \delta \Rightarrow |f(x) - f(y)| < \varepsilon).
\end{aligned}$$

Remark (Predicate reading).

$$\mathrm{ContinuousOn}(f, [a, b]) \Rightarrow \mathrm{UniformlyContinuous}(f, [a, b]).$$

Remark (Negated quantified statement).

$$\begin{aligned}
&\text{Let } a, b \in \mathbb{R} \text{ with } a < b \text{ and let } f : [a, b] \to \mathbb{R}.\\
&\Big[\forall c \in [a, b]\ \forall \varepsilon > 0\ \exists \delta > 0\ \forall x \in [a, b]\ (|x - c| < \delta \Rightarrow |f(x) - f(c)| < \varepsilon)\Big]\\
&\wedge \Big[\exists \varepsilon_0 > 0\ \forall \delta > 0\ \exists x, y \in [a, b]\ (|x - y| < \delta \wedge |f(x) - f(y)| \geq \varepsilon_0)\Big].
\end{aligned}$$

Remark (Negation predicate reading).

$$\mathrm{ContinuousOn}(f, [a, b]) \wedge \neg\, \mathrm{UniformlyContinuous}(f, [a, b]).$$

Remark (Interpretation). Continuity on the compact interval $[a, b]$ forces a single modulus of continuity that works simultaneously at every point, so oscillations of f cannot sharpen when inspecting smaller neighborhoods. Proofs usually argue by contradiction: assuming the negated form produces sequences of points whose distances shrink while their images stay apart, contradicting compactness-driven convergence. The theorem shows that closed bounded domains prevent the pathological behavior that obstructs uniform control on open or unbounded sets.

Remark (Dependencies).

Definition 2.2.2; Definition 2.4.1.

Proposition (Uniform Continuity Preserves Cauchy Sequences)

Proposition 2.4.6 (Uniform Continuity Preserves Cauchy Sequences)**.** *Let $S \subseteq \mathbb{R}$, let $f : S \to \mathbb{R}$, and let (x_n) be a sequence in S. If f is uniformly continuous on S and (x_n) is a Cauchy sequence in S, then $(f(x_n))$ is a Cauchy sequence in $\mathbb{R}$. Go to proof.*

Remark (Standard quantified statement)*.*

$$\begin{aligned}
&\text{Let } S \subseteq \mathbb{R},\ f : S \to \mathbb{R},\ (x_n)_{n\in\mathbb{N}} \subseteq S.\\
&\Big[\forall \varepsilon > 0\ \exists \delta > 0\ \forall x, y \in S\ (|x-y| < \delta \Rightarrow |f(x) - f(y)| < \varepsilon)\Big]\\
&\wedge \Big[\forall \varepsilon > 0\ \exists N \in \mathbb{N}\ \forall m, n \geq N\ (|x_n - x_m| < \varepsilon)\Big]\\
&\Rightarrow \forall \varepsilon > 0\ \exists N \in \mathbb{N}\ \forall m, n \geq N\ (|f(x_n) - f(x_m)| < \varepsilon).
\end{aligned}$$

Remark (Predicate reading)*.*

$$\mathrm{UniformlyContinuous}(f, S) \wedge \mathrm{CauchySequence}(x_n) \ \Rightarrow\ \mathrm{CauchySequence}(f(x_n)).$$

Remark (Interpretation)*.* Uniform continuity controls the change in $f(x)$ solely through the distance between inputs, so any Cauchy sequence in the domain is sent to a sequence whose tails remain uniformly close. Ordinary pointwise continuity cannot guarantee this because the necessary input tolerance could shrink along the sequence, so uniformity is exactly the additional strength needed to transport the Cauchy property. The standard failure mode occurs when f is merely continuous and the sequence approaches a boundary point where continuity loses uniform control, producing large oscillations in the image sequence. Geometrically, uniform continuity supplies a single modulus of continuity that works for the entire path traced by (x_n), ensuring that eventually all image terms lie in an arbitrarily small common interval.

Remark (Dependencies)*.*

Definition 2.4.1; Definition **??**.

Definition (Lipschitz Condition)

Definition 2.4.7 (Lipschitz Condition)**.** Let $A \subseteq \mathbb{R}$, let $f : A \to \mathbb{R}$, and let $K > 0$. The function f satisfies the Lipschitz condition on A with constant K if and only if $|f(x) - f(u)| \leq K|x - u|$ for every $x, u \in A$.

Remark (Standard quantified statement)*.*

$$\begin{aligned}
&(A \subseteq \mathbb{R}) \wedge (f : A \to \mathbb{R}) \wedge (K > 0)\\
&\qquad \Longrightarrow \forall x, u \in A\ \big(|f(x) - f(u)| \leq K|x - u|\big).
\end{aligned}$$

Remark (Definition predicate reading)*.*

$$\mathrm{Lipschitz}(f, A, K) :\!= (K > 0) \wedge \forall x, u \in A\ \big(|f(x) - f(u)| \leq K|x - u|\big).$$

Remark (Negated quantified statement)*.*

$$\begin{aligned}
&\neg(A \subseteq \mathbb{R}) \vee \neg(f : A \to \mathbb{R}) \vee (K \leq 0)\\
&\qquad\qquad\qquad\qquad \vee \exists x, u \in A\ \big(|f(x) - f(u)| > K|x - u|\big).
\end{aligned}$$

Remark (Negation predicate reading).

$$\neg\,\mathrm{Lipschitz}(f,A,K) \iff \neg(A\subseteq\mathbb{R})\vee\neg(f:A\to\mathbb{R})\vee(K\le 0)\vee\exists x,u\in A\ \big(|f(x)-f(u)|>K|x-u|\big).$$

Remark (Failure modes). The Lipschitz condition fails when the ambient data are malformed, when the proposed constant is not positive, or when there exists a pair of points in A whose images separate faster than K times the separation of the points themselves.

Remark (Failure mode decomposition).

$$\underbrace{\neg(A\subseteq\mathbb{R})}_{\text{bad domain}}\vee\underbrace{\neg(f:A\to\mathbb{R})}_{\text{bad function type}}\vee\underbrace{K\le 0}_{\text{bad constant}}\vee\underbrace{\exists x,u\in A:\ |f(x)-f(u)|>K|x-u|}_{\text{excessive separation of outputs}}.$$

Remark (Interpretation). The Lipschitz condition requires a single constant K that simultaneously bounds the rate at which f can stretch distances across the entire set A. It encapsulates a global slope cap: every secant line through the graph has absolute slope at most K. Failure occurs once a single pair of points forces a steeper secant, signaling that no uniform control constant works on the whole domain. Geometrically, the graph of a Lipschitz function lies inside a cone of opening determined by K, so nearby points cannot be blown apart faster than K times their original spacing.

Remark (Dependencies).

- Function

Proposition (Lipschitz Implies Uniform Continuity)

Proposition 2.4.8 (Lipschitz Implies Uniform Continuity)**.** *Let $A\subseteq\mathbb{R}$ and $f:A\to\mathbb{R}$. If there exists $K\ge 0$ such that $|f(x)-f(y)|\le K|x-y|$ for all $x,y\in A$, then f is uniformly continuous on A.* *Go to proof.*

Remark (Standard quantified statement).

$$\begin{aligned}&\text{Let } A\subseteq\mathbb{R},\ f:A\to\mathbb{R},\ \text{and suppose } \exists K\ge 0\ \forall x,y\in A\ (|f(x)-f(y)|\le K|x-y|).\\&\forall\varepsilon>0\ \exists\delta>0\ \forall x,y\in A\ (|x-y|<\delta\Rightarrow|f(x)-f(y)|<\varepsilon).\end{aligned}$$

Remark (Predicate reading).

$$\big[\exists K\ge 0\ \mathrm{Lipschitz}(f,A,K)\big]\Rightarrow\mathrm{UniformlyContinuous}(f,A).$$

Remark (Contrapositive quantified statement). If f fails to be uniformly continuous on A, then f is not Lipschitz on A with any constant.

$$\begin{aligned}&\text{Let } A\subseteq\mathbb{R},\ f:A\to\mathbb{R}.\\&\Big[\exists\varepsilon_0>0\ \forall\delta>0\ \exists x,y\in A\ (|x-y|<\delta\wedge|f(x)-f(y)|\ge\varepsilon_0)\Big]\\&\implies\forall K\ge 0\ \exists x,y\in A\ (|f(x)-f(y)|>K|x-y|).\end{aligned}$$

Remark (Contrapositive predicate reading).

$$\neg\,\mathrm{UniformlyContinuous}(f,A)\implies\neg\exists K\ge 0:\ \mathrm{Lipschitz}(f,A,K).$$

Remark (Interpretation). A Lipschitz bound controls the variation of f by the distance between inputs with a single constant K, so the same K yields a usable $\delta = \varepsilon/K$ for every pair of points at once, proving uniform continuity. When $K = 0$ the function is constant and trivially uniform. Failure of uniform continuity would force two points arbitrarily close together whose images stay a fixed distance apart, contradicting the Lipschitz inequality.

Remark (Strict implication note). The implication Lipschitz $\Rightarrow$ uniform continuity is strict. The canonical counterexample is $f(x) = \sqrt{x}$ on $[0, 1]$, which is uniformly continuous by the Heine–Cantor theorem but satisfies $|f(x) - f(0)|/|x - 0| = 1/\sqrt{x} \to \infty$ as $x \to 0^+$, so no Lipschitz constant exists at the origin. This distinction is relevant to GPU Lipschitz-constant estimation: uniform continuity of a simulation function does not supply a computable K for the Lipschitz condition; only an explicit Lipschitz bound does.

Remark (Dependencies).

Definition 2.4.7, Definition 2.4.1.

Theorem (Algebra of Lipschitz Functions)

Theorem 2.4.9 (Algebra of Lipschitz Functions). *Let $A \subseteq \mathbb{R}$ and let $f, g : A \to \mathbb{R}$ be Lipschitz with constants $K_f, K_g \geq 0$ respectively.*

(a) For every $c \in \mathbb{R}$, the function cf is Lipschitz on A with constant $|c|K_f$.

(b) The functions $f + g$ and $f - g$ are Lipschitz on A with constant $K_f + K_g$.

(c) If $B \subseteq \mathbb{R}$, $h : B \to \mathbb{R}$ is Lipschitz with constant $K_h \geq 0$, and $f(A) \subseteq B$, then $h \circ f$ is Lipschitz on A with constant $K_h K_f$.

(d) If $|f(x)| \leq M_f$ and $|g(x)| \leq M_g$ for all $x \in A$, then fg is Lipschitz on A with constant $M_g K_f + M_f K_g$.

(e) If $|f(x)| \leq M_f$ for all $x \in A$, if $|g(x)| \geq k > 0$ for all $x \in A$, and if g is Lipschitz on A with constant K_g, then f/g is Lipschitz on A with constant $K_f/k + M_f K_g/k^2$.

Go to proof.

Remark (Standard quantified statement).

Let $A \subseteq \mathbb{R},\ f, g : A \to \mathbb{R},\ K_f, K_g \geq 0,$
$\forall x, y \in A :\ |f(x) - f(y)| \leq K_f|x - y|$ and $|g(x) - g(y)| \leq K_g|x - y|$.
(a) $\forall c \in \mathbb{R},\ \forall x, y \in A :\ |cf(x) - cf(y)| \leq |c|K_f|x - y|$.
(b) $\forall x, y \in A :\ |(f \pm g)(x) - (f \pm g)(y)| \leq (K_f + K_g)|x - y|$.
(c) If $B \subseteq \mathbb{R},\ f(A) \subseteq B,\ h : B \to \mathbb{R},\ \forall u, v \in B :\ |h(u) - h(v)| \leq K_h|u - v|$,
then $\forall x, y \in A :\ |h(f(x)) - h(f(y))| \leq K_h K_f|x - y|$.
(d) If $|f(x)| \leq M_f,\ |g(x)| \leq M_g\ \forall x \in A$, then $\forall x, y \in A :$
$|f(x)g(x) - f(y)g(y)| \leq (M_g K_f + M_f K_g)|x - y|$.
(e) If $|f(x)| \leq M_f,\ |g(x)| \geq k > 0\ \forall x \in A$, then $\forall x, y \in A :$

$$\left| \frac{f(x)}{g(x)} - \frac{f(y)}{g(y)} \right| \leq \left(\frac{K_f}{k} + \frac{M_f K_g}{k^2} \right) |x - y|.$$

Remark (Predicate reading).

$$\begin{aligned}
&\mathrm{Lipschitz}(f, A, K_f) \wedge \mathrm{Lipschitz}(g, A, K_g) \\
&\implies \Big[\forall c \in \mathbb{R} : \mathrm{Lipschitz}(cf, A, |c|K_f)\Big] \\
&\qquad \wedge \mathrm{Lipschitz}(f+g, A, K_f + K_g) \wedge \mathrm{Lipschitz}(f-g, A, K_f + K_g) \\
&\qquad \wedge \Big(\mathrm{Lipschitz}(h, B, K_h) \wedge f(A) \subseteq B \Rightarrow \mathrm{Lipschitz}(h \circ f, A, K_h K_f)\Big) \\
&\qquad \wedge \Big[(|f| \le M_f) \wedge (|g| \le M_g) \Rightarrow \mathrm{Lipschitz}(fg, A, M_g K_f + M_f K_g)\Big] \\
&\qquad \wedge \Big[(|f| \le M_f) \wedge (|g| \ge k > 0) \Rightarrow \mathrm{Lipschitz}\Big(\frac{f}{g}, A, \frac{K_f}{k} + \frac{M_f K_g}{k^2}\Big)\Big].
\end{aligned}$$

Remark (Interpretation). Lipschitz functions are stable under the standard algebraic operations: scaling magnifies the constant proportionally, addition and subtraction add the errors, and composition multiplies constants because distortions accumulate multiplicatively. Products and quotients require uniform control of magnitudes; bounds on the factors prevent growth in the coefficients that accompany the derivatives of the product rule heuristic, while a positive lower bound on the denominator guarantees the quotient does not amplify oscillations uncontrollably. Together, these clauses justify building rich classes of Lipschitz maps from simpler ones without losing quantitative control.

Remark (Dependencies).
Definition 2.4.7, Definition 2.3.1

Definition (Bi-Lipschitz Map)

Definition 2.4.10 (Bi-Lipschitz Map)**.** Let $A \subseteq \mathbb{R}$ and let $f : A \to \mathbb{R}$. The map f is bi-Lipschitz on A if and only if there exist constants $0 < \alpha \le K$ such that for every $x, u \in A$,

$$\alpha|x-u| \le |f(x) - f(u)| \le K|x-u|.$$

Any such pair (α, K) is called a bi-Lipschitz constant pair for f on A.

Remark (Standard quantified statement).

$$\begin{aligned}
&(A \subseteq \mathbb{R}) \wedge (f : A \to \mathbb{R}) \\
&\quad \implies \exists \alpha, K \in \mathbb{R} \Big(0 < \alpha \le K \wedge \forall x, u \in A :\ \alpha|x-u| \le |f(x) - f(u)| \le K|x-u|\Big).
\end{aligned}$$

Remark (Definition predicate reading).

$$\mathrm{BiLipschitz}(f, A, \alpha, K) :\equiv (0 < \alpha \le K) \wedge \forall x, u \in A\ \ (\alpha|x-u| \le |f(x) - f(u)| \le K|x-u|)\,.$$ ∎

Remark (Negated quantified statement).

$$\begin{aligned}
&\neg(A \subseteq \mathbb{R}) \vee \neg(f : A \to \mathbb{R}) \\
&\qquad\qquad\qquad \vee \forall \alpha, K \in \mathbb{R} \Big(0 < \alpha \le K \Rightarrow \exists x, u \in A :\ |f(x) - f(u)| < \alpha|x-u| \vee |f(
\end{aligned}$$

Remark (Negation predicate reading).

$$\neg\,\mathrm{BiLipschitz}(f, A, \alpha, K) \iff \neg(0 < \alpha \le K) \vee \exists x, u \in A\ \ (|f(x) - f(u)| < \alpha|x-u| \vee |f(x) - f(u)| > K$$

Remark (Failure modes)*.* The property fails either because f collapses some pair of points faster than the allowed lower constant α, or because it stretches some pair more than the upper constant K permits. Both distortions prevent a uniform two-sided control of distances.

Remark (Failure mode decomposition)*.*

$$\exists x, u \in A : \quad \underbrace{|f(x) - f(u)| < \alpha|x - u|}_{\text{lower-distortion failure}} \quad \text{or} \quad \underbrace{|f(x) - f(u)| > K|x - u|}_{\text{upper-distortion failure}}.$$

Remark (Interpretation)*.* A bi-Lipschitz map distorts every distance in A by factors lying between α and K, so it is simultaneously Lipschitz and has a Lipschitz inverse defined on its image. This two-sided control guarantees injectivity and preserves geometric features such as relative spacing up to uniform scaling, while failures arise precisely from excessive collapsing or stretching of some pair of points.

Remark (Dependencies)*.*

- Function

Theorem (Inverse of a Bi-Lipschitz Map Is Lipschitz)

Theorem 2.4.11 (Inverse of a Bi-Lipschitz Map Is Lipschitz)**.** *Let $A \subseteq \mathbb{R}$, let $f : A \to \mathbb{R}$, and suppose there exist constants $\alpha, K > 0$ such that $\alpha|x - y| \leq |f(x) - f(y)| \leq K|x - y|$ for all $x, y \in A$. Then f is injective, the inverse mapping $f^{-1} : f(A) \to A$ is well defined, and*

$$\frac{1}{K}|u - v| \leq |f^{-1}(u) - f^{-1}(v)| \leq \frac{1}{\alpha}|u - v| \quad \textit{for all } u, v \in f(A).$$

Go to proof.

Remark (Standard quantified statement)*.*

$$\begin{aligned}
&\text{Let } A \subseteq \mathbb{R},\ f : A \to \mathbb{R},\ \alpha, K > 0.\\
&\Big[\forall x, y \in A\ \big(\alpha|x - y| \leq |f(x) - f(y)| \leq K|x - y|\big)\Big]\\
&\implies \Big[\forall x, y \in A\ (f(x) = f(y) \Rightarrow x = y)\\
&\qquad \wedge \exists g : f(A) \to A\ \big(\forall u \in f(A)\ (f(g(u)) = u)\big)\\
&\qquad \wedge \forall u, v \in f(A)\ \Big(\frac{1}{K}|u - v| \leq |g(u) - g(v)| \leq \frac{1}{\alpha}|u - v|\Big)\Big].
\end{aligned}$$

Remark (Predicate reading)*.*

$$\mathrm{BiLipschitz}(f, A, \alpha, K) \Rightarrow \big(\mathrm{Injective}(f) \wedge \mathrm{BiLipschitz}(f^{-1}, f(A), 1/K, 1/\alpha)\big).$$

Remark (Interpretation)*.* A bi-Lipschitz bound simultaneously controls the expansion and contraction of f, so no two points of A can share the same image and the inverse is automatically a well-defined function on $f(A)$. The inequalities for f^{-1} follow by applying the original bi-Lipschitz bounds to pairs of preimages, showing that reversing the mapping preserves the same type of two-sided Lipschitz control with the reciprocals of the original constants. Failure would require collapsing two distinct points or losing the two-sided bounds, either of which contradicts the defining hypothesis.

Remark (Dependencies)*.*

Definition 2.4.7; Definition 2.4.10.

2.5 Monotone Functions

Toolkit — Monotone Functions

- Every monotone function has LeftLimit and RightLimit at every interior point.
- $\text{ContinuousAt}(f, c) \iff f(c^-) = f(c) = f(c^+) \iff j_f(c) = 0$.
- All discontinuities of a monotone function are of first kind — jump only, no oscillation.
- At most countably many discontinuities: disjoint jump intervals inject into $\mathbb{Q}$.
- Strictly monotone and continuous on $I \Rightarrow f^{-1}$ exists, is strictly monotone, and is continuous.
- $\text{ContinuousOn}(f, [a, b]) \wedge \text{Injective}(f) \iff f$ strictly monotone.

Toolkit — Limits Superior and Inferior of Functions

- $\limsup_{x \to x_0} f(x) = \inf_{\delta > 0} \sup\{f(x) : 0 < |x - x_0| < \delta,\ x \in E\}$.
- Both always exist in $[-\infty, +\infty]$.
- $\limsup \geq \liminf$ always.
- $\text{FunctionLimit}(f, x_0, L)$ exists $\iff \limsup = \liminf = L \in \mathbb{R}$.
- Direct analogue of sequence $\limsup/\liminf$: replace tail $\{a_n : n \geq N\}$ with punctured neighbourhood $\{f(x) : \text{InputTolerance}(x, x_0, \delta)\}$.

Monotone Functions and Continuity

Theorem (One-Sided Limits of Monotone Functions)

Theorem 2.5.1 (One-Sided Limits of Monotone Functions)**.** *Let $I \subseteq \mathbb{R}$ be an interval, let c be interior to I, and let $f : I \to \mathbb{R}$ be increasing. Define*

$$L_- := \sup\{f(x) : x \in I,\ x < c\}, \qquad L_+ := \inf\{f(x) : x \in I,\ c < x\}.$$

Then both one-sided limits exist and satisfy

$$\lim_{x \to c^-} f(x) = L_-, \qquad \lim_{x \to c^+} f(x) = L_+.$$

Go to proof.

Remark (Standard quantified statement).

$$
\begin{aligned}
&I \subseteq \mathbb{R} \wedge \big(\forall x, y \in I\ \forall z \in \mathbb{R}\ (x < z < y \Rightarrow z \in I)\big) \wedge f : I \to \mathbb{R} \\
&\wedge \big(\forall x, y \in I\ (x < y \Rightarrow f(x) \le f(y))\big) \wedge c \in I \wedge \exists a, b \in I\ (a < c < b) \\
&\wedge L_- = \sup\{f(x) : x \in I,\ x < c\} \wedge L_+ = \inf\{f(x) : x \in I,\ c < x\} \\
&\Longrightarrow \\
&\forall \varepsilon > 0\ \exists \delta_- > 0\ \forall x \in I\ \big((0 < c - x < \delta_- \wedge x < c) \Rightarrow |f(x) - L_-| < \varepsilon\big) \\
&\wedge \forall \varepsilon > 0\ \exists \delta_+ > 0\ \forall x \in I\ \big((0 < x - c < \delta_+ \wedge c < x) \Rightarrow |f(x) - L_+| < \varepsilon\big).
\end{aligned}
$$

Remark (Predicate reading).

Let $I \subseteq \mathbb{R}$ be an interval, $f : I \to \mathbb{R}$ increasing, and c interior to I.

$$\mathrm{LeftLimit}\Big(f, c, \sup\{f(x) : x \in I,\ x < c\}\Big) \wedge \mathrm{RightLimit}\Big(f, c, \inf\{f(x) : x \in I,\ x > c\}\Big).$$

Remark (Interpretation). For an increasing function, the values just to the left of c form a monotone increasing set bounded above by any right neighborhood, so their supremum matches the left-hand limit. Symmetrically, the values just to the right form a monotone increasing tail bounded below, making the infimum coincide with the right-hand limit. The theorem shows that monotonicity alone guarantees both one-sided limits exist and equals these extremal values. Failure occurs when c is not an interior point or f is not monotone, because the comparison sets may be empty or oscillatory, preventing the suprema or infima from governing the boundary behavior.

Remark (Dependencies).

- Completeness of $\mathbb{R}$
- Monotone Function
- Supremum
- Infimum

Corollary (Continuity Criterion for Monotone Functions)

Corollary 2.5.2 (Continuity Criterion for Monotone Functions)**.** *Let $I \subseteq \mathbb{R}$ be an interval, let $f : I \to \mathbb{R}$ be monotone increasing, and let c be an interior point of I. Then f is continuous at c if and only if its one-sided limits satisfy $f(c^-) = f(c) = f(c^+)$, where $f(c^-) = \lim_{x\to c^-} f(x)$ and $f(c^+) = \lim_{x\to c^+} f(x)$.* *Go to proof.*

Remark (Standard quantified statement).

Let $I \subseteq \mathbb{R}$ be an interval, $f : I \to \mathbb{R}$ monotone increasing, and c an interior point of I.

$$\big(f \text{ is continuous at } c\big) \iff \left(\lim_{x\to c^-} f(x) = f(c) = \lim_{x\to c^+} f(x)\right).$$

Remark (Predicate reading).

$$\mathrm{ContinuousAt}(f, c) \iff \big(\mathrm{LeftLimit}(f, c, f(c)) \wedge \mathrm{RightLimit}(f, c, f(c))\big).$$

Remark (Negated quantified statement).

Let $I \subseteq \mathbb{R}$ be an interval, $f : I \to \mathbb{R}$ monotone increasing, and c an interior point of I.

$$\neg\big(f \text{ is continuous at } c\big) \iff \left(\lim_{x\to c^-} f(x) < f(c)\right) \vee \left(f(c) < \lim_{x\to c^+} f(x)\right).$$

Remark (Negation predicate reading).

$$\neg\,\mathrm{ContinuousAt}(f,c) \iff \big(\exists L^- \left[\mathrm{LeftLimit}(f,c,L^-) \wedge L^- < f(c)\right]\big) \vee \big(\exists L^+ \left[\mathrm{RightLimit}(f,c,L^+) \wedge f(c) <$$

Remark (Failure modes). Continuity can fail only through a jump: either the graph arrives at c from the left strictly below $f(c)$, or it departs to the right strictly above $f(c)$. Both jumps reflect the inability of the one-sided limits to match the function value at the interior point.

Remark (Failure mode decomposition).

$$\neg\,\mathrm{ContinuousAt}(f,c) = \underbrace{\big(\exists L^- \left[\mathrm{LeftLimit}(f,c,L^-) \wedge L^- < f(c)\right]\big)}_{\text{left jump at } c} \vee \underbrace{\big(\exists L^+ \left[\mathrm{RightLimit}(f,c,L^+) \wedge f(c) <\right.}_{\text{right jump at } c}$$

Remark (Interpretation). A monotone graph possesses finite one-sided limits at every interior point. Continuity at c demands that these approach values exactly meet the function value, so the graph neither overshoots nor undershoots when passing through c. Any failure manifests as a jump discontinuity: the graph either leaps upward at c or remains below the right-hand approach. This criterion is indispensable when classifying discontinuities of monotone functions, since it converts continuity into a simple equality check between the function value and the adjacent plateau levels.

Remark (Dependencies).

Definition 2.2.2; Theorem 2.5.1.

Definition (Jump of a Function)

Definition 2.5.3 (Jump of a Function)**.** Let $I \subseteq \mathbb{R}$ be an interval, let $f : I \to \mathbb{R}$ be increasing, and let c be an interior point of I. If the one-sided limits $f(c^-)$ and $f(c^+)$ exist, the jump of f at c is

$$j_f(c) := f(c^+) - f(c^-).$$

Remark (Standard quantified statement).

$$\begin{aligned}&\forall I \subseteq \mathbb{R}\ \forall f : I \to \mathbb{R}\ \forall c \in I\ \forall j \in \mathbb{R}\\ &\Big[\big(\forall x,y \in I\ \forall z \in \mathbb{R}\ (x < z < y \Rightarrow z \in I)\big) \wedge \exists a,b \in I\ (a < c < b)\\ &\qquad \wedge \big(\forall x,y \in I\ (x < y \Rightarrow f(x) \le f(y))\big) \wedge \exists L_-, L_+ \in \mathbb{R}\ \big(\lim_{x\to c^-} f(x) = L_- \wedge \lim_{x\to c^+} f(x) = L_+\big)\Big]\\ &\qquad \Rightarrow \big(j = j_f(c) \iff j = f(c^+) - f(c^-)\big).\end{aligned}$$

Remark (Definition predicate reading).

$$\mathrm{JumpOfFunction}(f,I,c,j) := \mathrm{Interval}(I) \wedge \mathrm{MonotoneIncreasing}(f,I) \wedge \mathrm{InteriorPoint}(c,I) \wedge \exists L_-, L_+ \in \mathbb{R}$$

Remark (Negated quantified statement).

$$\begin{aligned}
&\forall I \subseteq \mathbb{R}\ \forall f : I \to \mathbb{R}\ \forall c \in I\ \forall j \in \mathbb{R} :\\
&\neg\big(j = j_f(c)\big) \iff \neg\big(\forall x, y \in I\ \forall z \in \mathbb{R}\ (x < z < y \Rightarrow z \in I)\big)\\
&\qquad \vee \neg\big(\forall x, y \in I\ (x < y \Rightarrow f(x) \le f(y))\big) \vee \neg\big(\exists a, b \in I\ (a < c < b)\big)\\
&\qquad \vee \forall L_-, L_+ \in \mathbb{R}\ \big(\lim_{x\to c^-} f(x) \ne L_- \vee \lim_{x\to c^+} f(x) \ne L_+ \vee j \ne L_+ - L_-\big).
\end{aligned}$$

Remark (Negation predicate reading).

$\neg\,\text{JumpOfFunction}(f, I, c, j) \iff \neg\,\text{Interval}(I) \vee \neg\,\text{MonotoneIncreasing}(f, I) \vee \neg\,\text{InteriorPoint}(c, I) \vee \forall L_-$

Remark (Failure modes). A proposed jump value fails when a required one-sided limit does not exist, or when both one-sided limits exist but the proposed value is not their difference.

Remark (Failure mode decomposition).

$$\underbrace{\neg\big(f(c^-)\text{ exists}\big)}_{\text{missing left limit}} \vee \underbrace{\neg\big(f(c^+)\text{ exists}\big)}_{\text{missing right limit}} \vee \underbrace{j \ne f(c^+) - f(c^-)}_{\text{incorrect jump magnitude}}.$$

Remark (Interpretation). For a monotone function the one-sided limits exist at every interior point, and their difference measures the vertical gap in the graph at that point. A zero jump means the two limiting heights agree; a positive jump records a discontinuity of first kind.

Remark (Dependencies).

Theorem 2.5.1.

Proposition (Discontinuities of a Monotone Function Are of First Kind)

Proposition 2.5.4 (Discontinuities of a Monotone Function Are of First Kind)**.** *Let $I \subseteq \mathbb{R}$ be an interval, let $f : I \to \mathbb{R}$ be monotone, and let $a \in I$ be such that there exist points of I strictly less and strictly greater than a. If f is discontinuous at a, then both one-sided limits $\lim_{x\uparrow a} f(x)$ and $\lim_{x\downarrow a} f(x)$ exist in $\mathbb{R}$. Go to proof.*

Remark (Standard quantified statement).

$$\text{Let } I \subseteq \mathbb{R} \text{ be an interval, } f : I \to \mathbb{R} \text{ monotone, and } a \in I \text{ interior to } I.$$

$$\big[f \text{ is discontinuous at } a\big] \implies \exists L^-, L^+ \in \mathbb{R} : \begin{cases} \lim\limits_{x\uparrow a} f(x) = L^-,\\ \lim\limits_{x\downarrow a} f(x) = L^+. \end{cases}$$

Remark (Predicate reading).

$\text{MonotoneFunction}(f, I) \wedge \text{DiscontinuousAt}(f, a) \implies \exists L^-, L^+ \in \mathbb{R}\ \big(\text{LeftLimit}(f, a, L^-) \wedge \text{RightLimit}(f, a$

Remark (Interpretation). A monotone graph can jump only by settling on distinct finite one-sided limits; it cannot oscillate wildly or blow up to infinity at a point of discontinuity. Thus every discontinuity of a monotone function is a jump discontinuity of the first kind, which explains why such functions are easy to integrate and why their exceptional sets are well behaved. Geometrically, the graph approaches a finite height from the left and another finite height from the right before making the jump.

Remark (Dependencies).

Definition 2.2.5

Corollary (Jump Intervals Are Disjoint)

Corollary 2.5.5 (Jump Intervals Are Disjoint)**.** *Let $I \subseteq \mathbb{R}$ be an interval and let $f : I \to \mathbb{R}$ be nondecreasing. If $a, b \in I$ are distinct points of discontinuity of f, then the open intervals $(f(a^-), f(a^+))$ and $(f(b^-), f(b^+))$ are disjoint.Go to proof.*

Remark (Standard quantified statement).

$$\forall a, b \in I \; \Big[a \neq b \wedge \big(\exists \varepsilon_a > 0 \; \forall \delta > 0 \; \exists x \in I \; (|x - a| < \delta \wedge |f(x) - f(a)| \geq \varepsilon_a)\big) \wedge \big(\exists \varepsilon_b > 0 \; \forall \delta > 0 \; \exists y \in I \; (|y-$$

Remark (Predicate reading).

$$\text{MonotoneFunction}(f, I) \wedge \begin{cases} a, b \in I, \\ a \neq b, \\ \text{DiscontinuousAt}(f, a), \\ \text{DiscontinuousAt}(f, b) \end{cases} \implies (f(a^-), f(a^+)) \cap (f(b^-), f(b^+)) = \varnothing.$$

Remark (Interpretation). The corollary states that a nondecreasing function assigns disjoint open value intervals to distinct jump discontinuities. Because the function never reverses direction, the lower one-sided limit at the later point cannot drop below the upper one-sided limit at the earlier point, so the corresponding ranges cannot overlap. This distinction of jump intervals ensures that the cumulative vertical jumps of a monotone function stay separated, preventing the graph from revisiting the same band of values via different discontinuities.

Remark (Dependencies).

Definition 2.5.3; Proposition 2.5.4

Theorem (Discontinuities of a Monotone Function Are Countable)

Theorem 2.5.6 (Discontinuities of a Monotone Function Are Countable)**.** *Let $I \subseteq \mathbb{R}$ be an interval and let $f : I \to \mathbb{R}$ be monotone. Then the set $D_f := \{c \in I : f \text{ is discontinuous at } c\}$ is at most countable. Go to proof.*

Remark (Standard quantified statement).

$$\text{Let } I \subseteq \mathbb{R} \text{ be an interval and } f : I \to \mathbb{R} \text{ be monotone.}$$
$$\#\{c \in I : f \text{ fails to be continuous at } c\} \leq \aleph_0.$$

Remark (Predicate reading).

$$\text{MonotoneFunction}(f, I) \implies \#\{c \in I : \text{DiscontinuousAt}(f, c)\} \leq \aleph_0.$$

Remark (Interpretation). Every discontinuity of a monotone function is a jump discontinuity with a nondegenerate vertical gap $(f(c^-), f(c^+))$. Distinct discontinuities yield pairwise disjoint gaps, and each such interval contains a rational number. Associating to every discontinuity the rational number inside its gap produces an injection of the discontinuity set into $\mathbb{Q}$, forcing the set to be countable. The only way the conclusion could fail would be the existence of uncountably many disjoint jump intervals, which is impossible because the rationals are countable.

Remark (Dependencies).

Theorem 2.5.1 and Corollary 2.5.5.

Corollary (Monotone Function Is Continuous on a Dense Set)

Corollary 2.5.7 (Monotone Function Is Continuous on a Dense Set)**.** *Let $f : [a,b] \to \mathbb{R}$ be monotone. Then the set $\{x \in [a,b] : f$ is continuous at $x\}$ is dense in $[a,b]$. Go to proof.*

Remark (Standard quantified statement).

$$\text{Let } f : [a,b] \to \mathbb{R} \text{ be monotone.}$$
$$\forall u,v \in [a,b]\ (u < v \Rightarrow \exists c \in (u,v) \text{ such that } f \text{ is continuous at } c).$$

Remark (Predicate reading).

$$\text{MonotoneFunction}(f,[a,b]) \implies \text{DenseSubset}\big(\{x \in [a,b] : \text{ContinuousAt}(f,x)\},[a,b]\big).$$

Remark (Negated quantified statement).

$$\exists u,v \in [a,b]\ \Big(u < v \wedge \forall c \in (u,v)\ \exists \varepsilon_c > 0\ \forall \delta > 0\ \exists x \in [a,b]\ \big(|x-c| < \delta \wedge |f(x)-f(c)| \geq \varepsilon_c\big)\Big).$$

Remark (Negation predicate reading).

$$\exists u,v \in [a,b]\ \Big(u < v \wedge \forall c \in (u,v)\ \neg\,\text{ContinuousAt}(f,c)\Big).$$

Remark (Interpretation). A monotone function on a closed interval can only jump at most countably many times, so the points of discontinuity leave room inside every subinterval for a continuity point. Thus, although the function may fail to be continuous at isolated or countably many locations, continuity still occurs throughout any interval one inspects. This property ensures that monotone functions retain many features of continuous functions when arguments depend only on dense sets of continuity points.

Remark (Dependencies).

Discontinuities of a Monotone Function Are Countable

Proposition (Continuous Injective Is Strictly Monotone)

Proposition 2.5.8 (Continuous Injective Is Strictly Monotone)**.** *Let $f : [a,b] \to \mathbb{R}$ be continuous on $[a,b]$. Then f is injective if and only if f is strictly monotone on $[a,b]$. Go to proof.*

Remark (Standard quantified statement).

$$\text{Let } f : [a,b] \to \mathbb{R} \text{ be continuous on } [a,b].$$
$$\big[\forall x,y \in [a,b]\ (f(x) = f(y) \Rightarrow x = y)\big] \iff \big[\forall x,y \in [a,b]\ (x < y \Rightarrow f(x) < f(y))\big]$$
$$\vee \big[\forall x,y \in [a,b]\,(x < y \Rightarrow f(y) < f(x))\big].$$

Remark (Predicate reading).

$$\text{ContinuousOn}(f,[a,b]) \implies \big(\text{Injective}(f) \iff \text{StrictlyMonotone}(f,[a,b])\big).$$

Remark (Interpretation). A continuous function on a closed interval cannot reverse direction without repeating a value, so injectivity enforces a single increasing or decreasing trend across the entire domain. Conversely any strictly monotone map is automatically injective. The result emphasizes that topological constraints supplied by continuity interact with order-theoretic injectivity to force global monotonicity; failure occurs precisely when the graph turns back on itself, producing repeated values. Geometrically the continuous injective graph must cross each horizontal line at most once, which is only possible when the graph rises everywhere or falls everywhere.

Remark (Dependencies).

Bolzano's Intermediate Value Theorem.

Theorem (Continuous Inverse Theorem)

Theorem 2.5.9 (Continuous Inverse Theorem)**.** *Let $I \subseteq \mathbb{R}$ be an interval and let $f : I \to \mathbb{R}$ be strictly monotone and continuous on I. Then the inverse function $f^{-1} : f(I) \to I$ exists, is strictly monotone, and is continuous on $f(I)$. Go to proof.*

Remark (Standard quantified statement).

Let $I \subseteq \mathbb{R}$ be an interval and $f : I \to \mathbb{R}$.

$$\Big[(\forall x, y \in I)(x < y \Rightarrow f(x) < f(y)) \wedge \forall c \in I\ \forall \varepsilon > 0\ \exists \delta > 0\ \forall x \in I\big(|x - c| < \delta \Rightarrow |f(x) - f(c)| < \varepsilon\big)\Big]$$
$$\Longrightarrow \Big[\exists g : f(I) \to I \text{ such that } (\forall x \in I)\, g(f(x)) = x \wedge (\forall y \in f(I))\, f(g(y)) = y$$
$$\wedge\, (\forall u, v \in f(I))(u < v \Rightarrow g(u) < g(v)) \wedge \forall y \in f(I)\ \forall \varepsilon > 0\ \exists \eta > 0\ \forall u \in f(I)\big(|u - y| < \eta \Rightarrow |g(u) -$$

Remark (Interpretation). A continuous strictly monotone function on an interval never reverses the order of points, so it is automatically bijective onto its image. This theorem records that the inverse map retains the same order orientation and inherits continuity, reflecting the fact that no sudden jumps can occur when the original graph crosses each horizontal line exactly once. Failure would require either loss of monotonicity or a discontinuity, either of which would create a flat spot or a jump preventing the existence of a continuous inverse. Geometrically, the graph of f^{-1} is the reflection of the graph of f across the diagonal line $y = x$, so the smoothness and strict increase or decrease are visibly preserved by symmetry.

Remark (Dependencies).

- Bolzano's Intermediate Value Theorem
- Strictly Increasing Function
- Strictly Decreasing Function
- Continuity at a Point

Limits Superior and Inferior of Functions

Definition (Limits Superior and Inferior of a Function)

Definition 2.5.10 (Limits Superior and Inferior of a Function)**.** Let $f : E \to \mathbb{R}$ and let x_0 be an accumulation point of E. For each $\delta > 0$ set

$$U(\delta) = \{\, f(x) : x \in E,\ 0 < |x - x_0| < \delta \,\}.$$

The extended real numbers $\limsup_{x\to x_0} f(x)$ and $\liminf_{x\to x_0} f(x)$ are defined by

$$\limsup_{x\to x_0} f(x) = \inf_{\delta>0} \sup U(\delta), \qquad \liminf_{x\to x_0} f(x) = \sup_{\delta>0} \inf U(\delta),$$

where each inner supremum or infimum is taken in $\overline{\mathbb{R}}$ and the outer infimum or supremum ranges over all $\delta > 0$.

Remark (Standard quantified statement)*.*

Let $f : E \to \mathbb{R},\ x_0$ an accumulation point of $E,$ and $U(\delta) = \{f(x) : x \in E,\ 0 < |x - x_0| < \delta\}.$

$$\limsup_{x\to x_0} f(x) = \inf_{\delta>0} \sup U(\delta),$$

$$\liminf_{x\to x_0} f(x) = \sup_{\delta>0} \inf U(\delta).$$

Remark (Definition predicate reading)*.*

$$\mathrm{FunctionLimsup}(L, f, E, x_0) :\equiv L = \inf_{\delta>0} \sup U(\delta),$$
$$\mathrm{FunctionLiminf}(l, f, E, x_0) :\equiv l = \sup_{\delta>0} \inf U(\delta),$$

where $U(\delta) = \{f(x) : x \in E,\ 0 < |x - x_0| < \delta\}$.

Remark (Negated quantified statement)*.*

Let $f : E \to \mathbb{R},\ x_0$ an accumulation point of $E,$ and $U(\delta) = \{f(x) : x \in E,\ 0 < |x - x_0| < \delta\}.$

$$\neg\Big[\limsup_{x\to x_0} f(x) = \inf_{\delta>0} \sup U(\delta) \ \wedge\ \liminf_{x\to x_0} f(x) = \sup_{\delta>0} \inf U(\delta)\Big]$$
$$\iff \limsup_{x\to x_0} f(x) \neq \inf_{\delta>0} \sup U(\delta) \vee \liminf_{x\to x_0} f(x) \neq \sup_{\delta>0} \inf U(\delta).$$

Remark (Negation predicate reading)*.*

$$\neg\big(\mathrm{FunctionLimsup}(L, f, E, x_0) \wedge \mathrm{FunctionLiminf}(l, f, E, x_0)\big)$$
$$\iff L \neq \inf_{\delta>0} \sup U(\delta) \vee l \neq \sup_{\delta>0} \inf U(\delta),$$

where $U(\delta) = \{f(x) : x \in E,\ 0 < |x - x_0| < \delta\}$.

Remark (Failure modes). The definition can fail only in structurally distinct ways: punctured neighborhoods might be empty despite the accumulation-point hypothesis, or the upper envelopes may blow up while a finite limsup is asserted, or the lower envelopes may dive to negative infinity while a finite liminf is asserted.

Remark (Failure mode decomposition).

$$\underbrace{\exists\delta>0:\ U(\delta)=\varnothing}_{\text{missing punctured samples}} \vee \underbrace{\Big(\inf_{\delta>0}\sup U(\delta)=+\infty \wedge L<+\infty\Big)}_{\text{upper envelope blow-up}} \vee \underbrace{\Big(\sup_{\delta>0}\inf U(\delta)=-\infty \wedge l>-\infty\Big)}_{\text{lower envelope collapse}}.$$

Remark (Interpretation). The limit superior records the smallest height that eventually dominates all nearby function values, while the limit inferior records the largest depth that eventually underestimates all nearby values. Taking suprema within punctured neighborhoods captures every oscillation that can be witnessed arbitrarily close to x_0, and the outer infimum or supremum squeezes these oscillations to the sharpest possible bounds. Failure occurs only if the neighborhood sampling breaks down or if the oscillations race off to infinite extremal heights, signaling that no finite envelope can control the function near the accumulation point.

Remark (Dependencies).

- Function
- Supremum
- Infimum
- Epsilon Neighbourhood

Proposition

Proposition 2.5.11 ($\limsup \geq \liminf$). *Let $A \subseteq \mathbb{R}$, let x_0 be a limit point of A, and let $f : A \to \mathbb{R}$. Then*

$$\limsup_{x\to x_0} f(x) \geq \liminf_{x\to x_0} f(x).$$

Go to proof.

Remark (Standard quantified statement).

$$\forall A \subseteq \mathbb{R}\ \forall f : A \to \mathbb{R}\ \forall x_0 \in \mathbb{R} :\\ \Big(\forall \eta > 0\ \exists x \in A \setminus \{x_0\}\ (|x - x_0| < \eta)\Big) \implies \limsup_{x\to x_0} f(x) \geq \liminf_{x\to x_0} f(x).$$

Remark (Interpretation). The limsup records the largest cluster value that f attains near x_0, while the liminf records the smallest cluster value. Because every subsequential limit contributing to the liminf also contributes to the limsup, the supremum of all cluster values cannot lie below their infimum. The only way equality holds is when all cluster values coincide, i.e. when the ordinary limit exists.

Remark (Dependencies).
Definition (Limits Superior and Inferior of a Function).

Proposition

Proposition 2.5.12 (Limit Exists iff $\limsup = \liminf$). *Let $A \subseteq \mathbb{R}$, let $f : A \to \mathbb{R}$, let x_0 be a limit point of A with $x_0 \in \mathbb{R}$, and let $L \in \mathbb{R}$. Then $\lim_{x\to x_0} f(x)$ exists and equals L if and only if $\limsup_{x\to x_0} f(x) = \liminf_{x\to x_0} f(x) = L$.* *Go to proof.*

Remark (Standard quantified statement).

$$\begin{aligned}&\text{Let } A \subseteq \mathbb{R},\ f\colon A \to \mathbb{R},\ x_0 \text{ a limit point of } A,\ L \in \mathbb{R}.\\&(\forall \varepsilon > 0\ \exists \delta > 0\ \forall x \in A \setminus \{x_0\}\ (0 < |x - x_0| < \delta \Rightarrow |f(x) - L| < \varepsilon))\\&\iff \qquad \left(\limsup_{x \to x_0} f(x) = L = \liminf_{x \to x_0} f(x)\right).\end{aligned}$$

Remark (Predicate reading).

$$\text{FunctionLimit}(f, A, x_0, L) \iff \big(\limsup_{x \to x_0} f(x) = L = \liminf_{x \to x_0} f(x)\big).$$

Remark (Negated quantified statement).

$$\begin{aligned}&\text{Let } A \subseteq \mathbb{R},\ f\colon A \to \mathbb{R},\ x_0 \text{ a limit point of } A,\ L \in \mathbb{R}.\\&\Big[\exists \varepsilon_0 > 0\ \forall \delta > 0\ \exists x \in A \setminus \{x_0\}\ \big(0 < |x - x_0| < \delta \land |f(x) - L| \geq \varepsilon_0\big)\Big]\\&\iff \qquad \big(\limsup_{x \to x_0} f(x) \neq L \lor \liminf_{x \to x_0} f(x) \neq L\big).\end{aligned}$$

Remark (Negation predicate reading).

$$\neg\, \text{FunctionLimit}(f, A, x_0, L) \iff \big(\limsup_{x \to x_0} f(x) \neq L \lor \liminf_{x \to x_0} f(x) \neq L\big).$$

Remark (Failure modes). The criterion fails only when at least one side of the equivalence breaks: either the function fails to approach L in the epsilon-delta sense, or one of the two extremal envelopes near x_0 does not equal L.

Remark (Failure mode decomposition).

$$\underbrace{\exists \varepsilon_0 > 0\ \forall \delta > 0\ \exists x \in A \setminus \{x_0\}\ \big(0 < |x - x_0| < \delta \land |f(x) - L| \geq \varepsilon_0\big)}_{\text{limit condition fails}} \lor \underbrace{\limsup_{x \to x_0} f(x) \neq L}_{\text{upper envelope mismatch}} \lor \underbrace{\liminf_{x \to x_0} f(x}_{\text{lower envelope}}$$

Remark (Interpretation). The two-sided limit of a real-valued function at a finite limit point exists precisely when the largest subsequential accumulation value and the smallest subsequential accumulation value coincide, in which case they both equal the limit. If the limsup exceeds the liminf, as happens for $f(x) = \sin(1/x)$ near $x_0 = 0$, the function oscillates between distinct cluster heights and therefore has no finite limit. The theorem isolates the limit as the unique height compatible with every oscillatory envelope and shows that agreeing envelopes are the exact signature of convergence in this local setting.

Remark (Dependencies).

Definition 2.5.10

2.6 Limits at Infinity

Toolkit — Limits at Infinity

- $+\infty$ is adherent to X iff $\neg$ BoundedSet(X) above ($\sup X = +\infty$).
- LimitAtInfinity$(f, +\infty, L)$: replace InputTolerance(x, c, δ) with $x > M$. Same OutputTolerance$(f(x), L, \varepsilon)$.
- All standard limit laws carry over unchanged.
- Sequential characterisation: LimitAtInfinity$(f, +\infty, L) \iff f(x_n) \to L$ for every $x_n \to +\infty$ in X.

Limits at Infinity

Definition (Infinite Adherent Points)

Definition 2.6.1 (Infinite Adherent Points)**.** Let $X \subseteq \mathbb{R}$. The point $+\infty$ is adherent to X if and only if for every $M \in \mathbb{R}$ there exists $x \in X$ such that $x > M$. The point $-\infty$ is adherent to X if and only if for every $M \in \mathbb{R}$ there exists $x \in X$ such that $x < M$.

Remark (Standard quantified statement).

$$\begin{aligned} +\infty \text{ is adherent to } X &\iff \forall M \in \mathbb{R}\ \exists x \in X\ (x > M), \\ -\infty \text{ is adherent to } X &\iff \forall M \in \mathbb{R}\ \exists x \in X\ (x < M). \end{aligned}$$

Remark (Definition predicate reading).

$$\begin{aligned} \mathrm{PlusInfAdherent}(X) &:= \forall M \in \mathbb{R}\ \exists x \in X\ (x > M), \\ \mathrm{MinusInfAdherent}(X) &:= \forall M \in \mathbb{R}\ \exists x \in X\ (x < M). \end{aligned}$$

Remark (Negated quantified statement).

$$\begin{aligned} \neg(+\infty \text{ is adherent to } X) &\iff \exists M \in \mathbb{R}\ \forall x \in X\ (x \le M), \\ \neg(-\infty \text{ is adherent to } X) &\iff \exists M \in \mathbb{R}\ \forall x \in X\ (x \ge M). \end{aligned}$$

Remark (Negation predicate reading).

$$\begin{aligned} \neg\,\mathrm{PlusInfAdherent}(X) &\iff \exists M \in \mathbb{R}\ \forall x \in X\ (x \le M), \\ \neg\,\mathrm{MinusInfAdherent}(X) &\iff \exists M \in \mathbb{R}\ \forall x \in X\ (x \ge M). \end{aligned}$$

Remark (Failure modes)*.* Infinite adherence fails at $+\infty$ exactly when X is bounded above, and fails at $-\infty$ exactly when X is bounded below.

Remark (Failure mode decomposition).

$$\underbrace{\exists M \in \mathbb{R}\ \forall x \in X\ (x \le M)}_{\text{finite upper bound}} \qquad \underbrace{\exists M \in \mathbb{R}\ \forall x \in X\ (x \ge M)}_{\text{finite lower bound}}.$$

Remark (Interpretation)*.* Both finite and infinite adherence are instances of adherence in $\overline{\mathbb{R}}$: finite adherence uses balls; infinite adherence uses rays.

Definition (Limit at Infinity)

Definition 2.6.2 (Limit at Infinity)**.** Let $X \subseteq \mathbb{R}$ be such that for every $M \in \mathbb{R}$ there exists $x \in X$ with $x > M$, let $f : X \to \mathbb{R}$, and let $L \in \mathbb{R}$. The limit of f at $+\infty$ equals L if and only if for every $\varepsilon > 0$ there exists $M \in \mathbb{R}$ such that for every $x \in X$, $x > M$ implies $|f(x) - L| < \varepsilon$.

Remark (Standard quantified statement)*.*

$$\begin{aligned}&\text{Let } X \subseteq \mathbb{R},\ \forall M \in \mathbb{R}\ \exists x \in X\ (x > M),\ \ f : X \to \mathbb{R},\ \ L \in \mathbb{R}.\\ &\lim_{x\to+\infty} f(x) = L \iff \forall \varepsilon > 0\ \exists M \in \mathbb{R}\ \forall x \in X\\ &\qquad (x > M \Rightarrow |f(x) - L| < \varepsilon).\end{aligned}$$

Remark (Definition predicate reading)*.*

$$\mathrm{LimitAtInfinity}(f, +\infty, L) :\equiv \forall \varepsilon > 0\ \exists M \in \mathbb{R}\ \forall x \in X\ (x > M \Rightarrow |f(x) - L| < \varepsilon).$$

Remark (Negated quantified statement)*.*

$$\begin{aligned}&\text{Let } X \subseteq \mathbb{R},\ \forall M \in \mathbb{R}\ \exists x \in X\ (x > M),\ \ f : X \to \mathbb{R},\ \ L \in \mathbb{R}.\\ &\neg\big(\lim_{x\to+\infty} f(x) = L\big) \iff \exists \varepsilon_0 > 0\ \forall M \in \mathbb{R}\ \exists x \in X\\ &\qquad (x > M \land |f(x) - L| \geq \varepsilon_0).\end{aligned}$$

Remark (Negation predicate reading)*.*

$$\neg\,\mathrm{LimitAtInfinity}(f, +\infty, L) \iff \exists \varepsilon_0 > 0\ \forall M \in \mathbb{R}\ \exists x \in X\ (x > M \land |f(x) - L| \geq \varepsilon_0).$$

Remark (Failure modes)*.* The definition fails only when there is a fixed tolerance ε_0 that the function cannot satisfy on any tail of X, meaning every attempt to choose M leaves some point $x > M$ with $|f(x) - L| \geq \varepsilon_0$.

Remark (Failure mode decomposition)*.*

$$\underbrace{\exists \varepsilon_0 > 0}_{\text{fixed tolerance}}\ \underbrace{\forall M \in \mathbb{R}}_{\text{no tail succeeds}}\ \underbrace{\exists x \in X\ (x > M \land |f(x) - L| \geq \varepsilon_0)}_{\text{far-point violation}}.$$

Remark (Interpretation)*.* The limit at $+\infty$ demands that eventually the function values stay within any prescribed accuracy of L. Intuitively, once x is large enough, the graph of f must lie in the horizontal band of height ε around L. Failure occurs when some positive tolerance is violated infinitely often by points escaping to $+\infty$. This definition allows us to treat unbounded domains using the same epsilon control that governs finite limits, capturing the long-range behavior of f with respect to the ideal point at infinity.

Remark (Dependencies)*.*

Definition 2.6.1.

2.7 Approximation

Toolkit — Approximation

- Hierarchy: step functions $\subset$ piecewise linear $\subset$ polynomials. All three classes are dense in $(C([a,b]), \|\cdot\|_\infty)$.
- All three results use Heine–Cantor: UniformlyContinuous$(f,[a,b])$ provides the controlling δ.
- Bernstein polynomials: $B_n(x) = \sum_{k=0}^{n} f(k/n)\binom{n}{k}x^k(1-x)^{n-k}$.
- Step function approximation feeds directly into Riemann integrability of continuous functions.

Approximation of Continuous Functions

Theorem (Step Function Approximation)

Theorem 2.7.1 (Step Function Approximation)**.** *Let $f:[a,b]\to\mathbb{R}$ be continuous and let $\varepsilon>0$. Then there exists a step function $s_\varepsilon:[a,b]\to\mathbb{R}$ such that*

$$\sup_{x\in[a,b]} |f(x)-s_\varepsilon(x)| < \varepsilon.$$

Go to proof.

Remark (Standard quantified statement)*.*

$$\begin{gathered}\text{Let } f:[a,b]\to\mathbb{R} \text{ be continuous and let } \varepsilon>0.\\ \exists s_\varepsilon:[a,b]\to\mathbb{R} \text{ step function } \forall x\in[a,b]\ (|f(x)-s_\varepsilon(x)|<\varepsilon).\end{gathered}$$

Remark (Predicate reading)*.*

$$\mathrm{StepApproximation}(f,s_\varepsilon,[a,b],\varepsilon) :\equiv \mathrm{StepFunction}(s_\varepsilon,[a,b]) \wedge \forall x\in[a,b]\ (|f(x)-s_\varepsilon(x)|<\varepsilon).$$ ∎

Remark (Negated quantified statement)*.*

$$\begin{gathered}\text{Let } f:[a,b]\to\mathbb{R} \text{ be continuous and let } \varepsilon>0.\\ \forall s:[a,b]\to\mathbb{R} \text{ step function } \exists x\in[a,b]\ (|f(x)-s(x)|\ge\varepsilon).\end{gathered}$$

Remark (Negation predicate reading)*.*

$$\neg\exists s_\varepsilon\ \mathrm{StepApproximation}(f,s_\varepsilon,[a,b],\varepsilon) \iff \forall s:[a,b]\to\mathbb{R}\ \Big(\mathrm{StepFunction}(s,[a,b]) \Rightarrow \exists x\in[a,b]\ (|f(x)$$

Remark (Failure modes)*.* The statement fails precisely when a continuous f and tolerance ε are fixed but every step function leaves an error of at least ε at some point in $[a,b]$.

Remark (Failure mode decomposition)*.*

$$\underbrace{\forall s:[a,b]\to\mathbb{R} \text{ step function}}_{\text{all approximants}}\ \underbrace{\exists x\in[a,b]\ (|f(x)-s(x)|\ge\varepsilon)}_{\text{witnessed large deviation}}.$$

Remark (Interpretation). Uniform continuity on $[a, b]$ guarantees that f does not oscillate rapidly at small scales, so subdividing the interval finely enough allows a step function to track f within any prescribed uniform error. The typical construction samples f at one point of each subinterval and keeps that value constant across the piece. Failure would mean that no matter how the interval is partitioned, some subinterval forces an error at least ε, contradicting uniform continuity. The result is a foundational approximation fact used to show that continuous functions on compact intervals are Riemann integrable and to connect continuous functions with simple combinatorial models.

Remark (Dependencies).

Heine–Cantor Theorem.

Definition (Piecewise Linear Function)

Definition 2.7.2 (Piecewise Linear Function)**.** Let $a, b \in \mathbb{R}$ with $a < b$ and let $g : [a, b] \to \mathbb{R}$. The function g is piecewise linear if and only if there exist $m \in \mathbb{N}$ with $m \geq 1$ and points $a = x_0 < x_1 < \cdots < x_m = b$ such that $g|_{[x_{k-1}, x_k]}$ is affine for every $k \in \{1, \ldots, m\}$.

Remark (Standard quantified statement).

$$\begin{aligned}
&\text{Let } a, b \in \mathbb{R} \text{ with } a < b \text{ and } g : [a, b] \to \mathbb{R}.\\
&g \text{ is piecewise linear}\\
&\iff \exists m \in \mathbb{N}\ \Big(m \geq 1 \land \exists x_0, \ldots, x_m \in [a, b]\ \big[a = x_0 < x_1 < \cdots < x_m = b\\
&\qquad\qquad \land \forall k \in \{1, \ldots, m\}\ g|_{[x_{k-1}, x_k]} \text{ is affine}\big]\Big).
\end{aligned}$$

Remark (Definition predicate reading).

$$\text{PiecewiseLinear}(g, [a, b]) :\equiv \exists m \in \mathbb{N}\ \Big(m \geq 1 \land \exists x_0, \ldots, x_m \in [a, b]\ \big[a = x_0 < \cdots < x_m = b \land \forall k \in \{1, \ldots,$$

Remark (Negated quantified statement).

$$\begin{aligned}
&\text{Let } a, b \in \mathbb{R} \text{ with } a < b \text{ and } g : [a, b] \to \mathbb{R}.\\
&g \text{ is not piecewise linear}\\
&\iff \forall m \in \mathbb{N}\ \Big(m \geq 1 \Rightarrow \forall x_0, \ldots, x_m \in [a, b]\ \big[a = x_0 < \cdots < x_m = b\\
&\qquad\qquad \Rightarrow \exists k \in \{1, \ldots, m\}\ g|_{[x_{k-1}, x_k]} \text{ is not affine}\big]\Big).
\end{aligned}$$

Remark (Negation predicate reading).

$$\neg\,\text{PiecewiseLinear}(g, [a, b]) \iff \forall m \in \mathbb{N}\ \Big(m \geq 1 \Rightarrow \forall x_0, \ldots, x_m \in [a, b]\ \big[a = x_0 < \cdots < x_m = b \Rightarrow \exists k \in$$

Remark (Failure modes). A proposed piecewise-linear structure can fail because no finite ordered partition of $[a, b]$ is available, or because for every such partition at least one subinterval has a restriction of g that is not affine.

Remark (Failure mode decomposition).

$$\underbrace{\forall m \in \mathbb{N}\ (m \geq 1 \Rightarrow \forall x_0, \ldots, x_m \in [a, b]\ (a = x_0 < \cdots < x_m = b \Rightarrow \cdots))}_{\text{no valid finite affine subdivision}} \lor \underbrace{\exists k \in \{1, \ldots, m\}\ g|_{[x_{k-1}, x_k]} \text{ is}}_{\text{non-affine segment}}$$

Remark (Interpretation). A piecewise linear function on $[a, b]$ has a graph made from finitely many straight line segments. The definition permits different affine rules on different subintervals, but requires a finite ordered list of nodes that covers the whole interval.

Remark (Dependencies).

- Function
- Closed Interval
- Natural Numbers
- Order on $\mathbb{R}$

Theorem

Theorem 2.7.3 (Piecewise Linear Approximation). *Let $f : [a, b] \to \mathbb{R}$ be continuous and let $\varepsilon > 0$. Then there exists a continuous piecewise linear function $\ell_\varepsilon : [a, b] \to \mathbb{R}$ such that $\sup_{x\in[a,b]} |f(x) - \ell_\varepsilon(x)| < \varepsilon$.* *Go to proof.*

Remark (Standard quantified statement).

$$\begin{aligned}&\text{Let } f : [a, b] \to \mathbb{R},\ f \text{ continuous},\ \varepsilon > 0.\\ &\exists \text{ continuous piecewise linear } \ell_\varepsilon : [a, b] \to \mathbb{R} : \sup_{x\in[a,b]} |f(x) - \ell_\varepsilon(x)| < \varepsilon.\end{aligned}$$

Remark (Interpretation). Every continuous function on a compact interval can be uniformly approximated by a polygonal curve matching finitely many sampled values of the original graph. The construction refines the interval into a sufficiently fine partition guaranteed by compactness, then connects the sampled values with straight segments, producing a uniformly small error over the entire domain. Failure would require a uniform oscillation exceeding ε on some subinterval, which cannot occur once the partition is fine enough. Geometrically, the theorem states that continuous graphs admit arbitrarily close polyline shadows, providing a bridge from rough step approximations to smooth polynomial approximations.

Remark (Dependencies).

- Heine–Cantor Theorem
- Piecewise Linear Function
- Continuity at a Point

Theorem (Weierstrass Approximation Theorem)

Theorem 2.7.4 (Weierstrass Approximation Theorem). *Let $f : [a, b] \to \mathbb{R}$ be continuous. For every $\varepsilon > 0$ there exists a polynomial $p_\varepsilon \in \mathbb{R}[x]$ such that $\|f - p_\varepsilon\|_{\infty,[a,b]} < \varepsilon$.* *Go to proof.*

Remark (Standard quantified statement).

$$\begin{aligned}&\text{Let } a < b \text{ in } \mathbb{R}, \text{ and let } f : [a, b] \to \mathbb{R} \text{ be continuous.}\\ &\forall \varepsilon > 0\ \exists p \in \mathbb{R}[x]\ (\|f - p\|_{\infty,[a,b]} < \varepsilon).\end{aligned}$$

Remark (Predicate reading).

$$\text{ContinuousOn}(f, [a, b]) \Longrightarrow \forall\varepsilon > 0\ \exists p \in \mathbb{R}[x]\ \big(\|f - p\|_{\infty,[a,b]} < \varepsilon\big).$$

Remark (Interpretation). Every continuous function on a closed interval can be uniformly approximated by polynomials, so polynomials are dense in the space of continuous functions under the supremum norm. The result holds because convolutions with Bernstein polynomials preserve continuity and converge uniformly to the original function. Failure would mean there exists a uniform tolerance that no polynomial can match, contradicting the approximation scheme built from Bernstein polynomials. Geometrically, the theorem states that the graph of a continuous function can be matched arbitrarily well by the graph of a polynomial, so polynomial graphs form a flexible mesh that can shadow any continuous curve on a compact interval.

Remark (Dependencies).

Depends on thm:bernstein-approximation.

Definition (Bernstein Polynomial)

Definition 2.7.5 (Bernstein Polynomial)**.** Let $n \in \mathbb{N}$ and let $f : [0, 1] \to \mathbb{R}$. A function $B_n : [0, 1] \to \mathbb{R}$ is called the nth Bernstein polynomial of f if and only if, for every $x \in [0, 1]$,

$$B_n(x) = \sum_{k=0}^{n} f\left(\frac{k}{n}\right)\binom{n}{k} x^k (1 - x)^{n-k}.$$

Remark (Standard quantified statement).

$$\begin{aligned}&\text{Let } n \in \mathbb{N},\ f : [0, 1] \to \mathbb{R},\ B_n : [0, 1] \to \mathbb{R}.\\ &B_n \text{ is the } n\text{th Bernstein polynomial of } f\\ &\Longleftrightarrow \forall x \in [0, 1]\ \Big(B_n(x) = \sum_{k=0}^{n} f(k/n)\binom{n}{k} x^k (1 - x)^{n-k}\Big).\end{aligned}$$

Remark (Definition predicate reading).

$$\text{BernsteinPolynomial}(B_n, f, n) :\equiv \Big(B_n : [0, 1] \to \mathbb{R}\Big) \wedge \forall x \in [0, 1]\ \Big(B_n(x) = \sum_{k=0}^{n} f(k/n)\binom{n}{k} x^k (1-x)^{n-k}\Big)$$

Remark (Interpretation). The nth Bernstein polynomial samples f at the uniformly spaced nodes k/n, weights those samples by the binomial probabilities of order n, and produces for each x an average that depends polynomially on x. This construction smooths f into a polynomial that agrees with f at the endpoints and approximates f uniformly as n grows, so describing it explicitly is essential for subsequent approximation theorems. Failure arises only when the candidate polynomial disagrees with the binomial averaging recipe at some point of the unit interval, reflecting an incorrect coefficient or evaluation in the polynomial expression.

Remark (Dependencies).

- Function
- Natural Numbers

- Multiplication on $\mathbb{R}$
- Addition on $\mathbb{R}$

Theorem (Bernstein Approximation Theorem)

Theorem 2.7.6 (Bernstein Approximation Theorem)**.** *Let $f : [0,1] \to \mathbb{R}$ be continuous, and for each $n \in \mathbb{N}$ let B_n denote the nth Bernstein polynomial of f. Then for every $\varepsilon > 0$ there exists $n_\varepsilon \in \mathbb{N}$ such that $\|f - B_n\|_\infty < \varepsilon$ for all $n \geq n_\varepsilon$.* *Go to proof.*

Remark (Standard quantified statement).

Let $f : [0,1] \to \mathbb{R}$ be continuous, and for each $n \in \mathbb{N}$ let B_n be the nth Bernstein polynomial of f. $\forall \varepsilon > 0\ \exists n_\varepsilon \in \mathbb{N}\ \forall n \geq n_\varepsilon : \ \|f - B_n\|_\infty < \varepsilon$.

Remark (Interpretation). The theorem asserts that the Bernstein polynomials of a continuous function on $[0,1]$ converge uniformly to the original function, so the entire graph of f can be approximated to any prescribed sup-norm accuracy by explicit degree-n polynomials once n is large enough. The proof works by showing that the Bernstein operators preserve positivity and linear growth while averaging values of f, which forces the uniform error to vanish as $n \to \infty$. This matters because it gives an elementary constructive route from continuous functions to polynomials, strengthening the Weierstrass approximation theorem with a concrete approximating sequence. The approximation can fail only when f is not continuous on $[0,1]$, in which case the Bernstein polynomials may smooth out jumps and no longer capture the original function in the sup norm. Geometrically, each B_n is a convex combination of the samples $f(k/n)$, so the polygons formed by these samples become finer and eventually hug the graph of f everywhere on the interval.

Remark (Dependencies).

Definition (Continuity at a Point), Definition (Bernstein Polynomial).

2.8 Gauge

Toolkit — Gauges and Tagged Partitions

- A **gauge** $\delta : I \to (0,\infty)$ is a variable-width tolerance function allowing the mesh to depend on position.
- A tagged partition $\dot{\mathcal{P}}$ is δ-fine if each subinterval I_i lies inside the $\delta(t_i)$-neighbourhood of its tag t_i: Within$(x, t_i, \delta(t_i))$ for all $x \in [x_{i-1}, x_i]$.
- **Cousin's theorem**: every gauge on $[a,b]$ admits a δ-fine tagged partition.
- Gauge proofs of Boundedness, EVT, Location of Roots, and Heine–Cantor directly construct the required constants from a δ-fine partition.

Gauges and Tagged Partitions

Definition (Partition)

Definition 2.8.1 (Partition)**.** Let $a < b$ and set $I = [a, b]$. A finite ordered set $P = \{x_0, \dots, x_n\}$ with $n \in \mathbb{N}$ and $n \geq 1$ is a partition of I if and only if $x_0 = a$, $x_n = b$, and $x_0 < x_1 < \cdots < x_n$. In this case the closed subintervals $[x_{i-1}, x_i]$ for $i = 1, \dots, n$ subdivide I.

Remark (Standard quantified statement).

$$\begin{aligned}&\text{Let } a < b,\ I = [a, b],\ n \in \mathbb{N},\ n \geq 1, \text{ and } P = \{x_0, \dots, x_n\} \subseteq I.\\ &P \text{ is a partition of } I\\ &\iff \big(x_0 = a \wedge x_n = b \wedge \forall i \in \{1, \dots, n\}\ (x_{i-1} < x_i)\big).\end{aligned}$$

Remark (Definition predicate reading).

$$\text{PartitionOfInterval}(P, I) :\equiv \exists n \in \mathbb{N}\ \Big(n \geq 1 \wedge P = \{x_0, \dots, x_n\} \subseteq I \wedge x_0 = \min I \wedge x_n = \max I \wedge \forall i \in \{1, .$$

Remark (Negated quantified statement).

$$\begin{aligned}&\text{Let } a < b,\ I = [a, b],\ n \in \mathbb{N},\ n \geq 1, \text{ and } P = \{x_0, \dots, x_n\} \subseteq I.\\ &P \text{ is not a partition of } I\\ &\iff \big((x_0 \neq a) \vee (x_n \neq b) \vee \exists i \in \{1, \dots, n\}\ (x_{i-1} \geq x_i)\big).\end{aligned}$$

Remark (Negation predicate reading).

$$\neg\,\text{PartitionOfInterval}(P, I) \iff \forall n \in \mathbb{N}\ \Big(n \geq 1 \Rightarrow \big(P \neq \{x_0, \dots, x_n\} \vee P \nsubseteq I \vee x_0 \neq \min I \vee x_n \neq \max I$$

Remark (Failure modes). The definition fails precisely in three ways: the left endpoint is omitted, the right endpoint is omitted, or the sequence of nodes ceases to be strictly increasing, leading to repeated or out-of-order points.

Remark (Failure mode decomposition).

$$\underbrace{x_0 \neq a}_{\text{left endpoint missing}} \quad \vee \quad \underbrace{x_n \neq b}_{\text{right endpoint missing}} \quad \vee \ \underbrace{\exists i \in \{1, \dots, n\}\ (x_{i-1} \geq x_i)}_{\text{nodes not strictly increasing}}.$$

Remark (Interpretation). A partition records the finite list of nodes that slice a closed interval into adjacent subintervals. The strict ordering ensures that every interior point appears exactly once in this linear traversal, so the induced subintervals $[x_{i-1}, x_i]$ tile $[a, b]$ without overlap or gaps. Computations of Riemann sums, Darboux sums, and gauge integrals use partitions as the combinatorial scaffolding that supports sampling and measurement. Failure arises when the endpoints are not captured or when the nodes regress or stagnate, which would leave gaps or redundancies in the subdivision.

Remark (Dependencies).

- Closed Interval
- Natural Numbers

- Order on $\mathbb{R}$

Definition (Tagged Partition)

Definition 2.8.2 (Tagged Partition)**.** Let $a, b \in \mathbb{R}$ with $a < b$. A list $\dot{\mathcal{P}} = \{([x_{i-1}, x_i], t_i)\}_{i=1}^n$ is a tagged partition of $[a, b]$ if and only if the subintervals determine a partition $P = \{[x_{i-1}, x_i]\}_{i=1}^n$ of $[a, b]$ with $a = x_0 < x_1 < \cdots < x_n = b$, and each tag satisfies $t_i \in [x_{i-1}, x_i]$ for $i = 1, \ldots, n$. Equivalently, a tagged partition is exactly a partition P of $[a, b]$ together with one tag chosen inside every subinterval of P.

Remark (Standard quantified statement)*.*

$$\begin{aligned}
&\text{Let } a, b \in \mathbb{R} \text{ with } a < b, \text{ and } \dot{\mathcal{P}} = \{([x_{i-1}, x_i], t_i)\}_{i=1}^n. \\
&\dot{\mathcal{P}} \text{ is a tagged partition of } [a, b] \\
&\iff \big(a = x_0 < \cdots < x_n = b \land \forall i \in \{1, \ldots, n\}\ (t_i \in [x_{i-1}, x_i])\big).
\end{aligned}$$

Remark (Definition predicate reading)*.*

$$\mathrm{TaggedPartition}(\dot{\mathcal{P}}, [a, b]) :\!= \Big(\mathrm{Partition}(P, [a, b]) \land \dot{\mathcal{P}} = \{([x_{i-1}, x_i], t_i)\}_{i=1}^n \land \forall i\ (t_i \in [x_{i-1}, x_i])\Big),$$

where $\mathrm{Partition}(P, [a, b])$ abbreviates $a = x_0 < \cdots < x_n = b$ for $P = \{[x_{i-1}, x_i]\}_{i=1}^n$.

Remark (Negated quantified statement)*.*

$$\begin{aligned}
&\text{Let } a, b \in \mathbb{R} \text{ with } a < b, \text{ and } \dot{\mathcal{P}} = \{([x_{i-1}, x_i], t_i)\}_{i=1}^n. \\
&\dot{\mathcal{P}} \text{ is not a tagged partition of } [a, b] \\
&\iff \big(x_0 \neq a \lor x_n \neq b \lor \exists j \in \{1, \ldots, n\}\ (x_{j-1} \geq x_j) \lor \exists i \in \{1, \ldots, n\}\ (t_i \notin [x_{i-1}, x_i])\big).
\end{aligned}$$

Remark (Negation predicate reading)*.*

$$\neg\,\mathrm{TaggedPartition}(\dot{\mathcal{P}}, [a, b]) \iff (x_0 \neq a) \lor (x_n \neq b) \lor \Big(\exists j\ (x_{j-1} \geq x_j)\Big) \lor \Big(\exists i\ (t_i \notin [x_{i-1}, x_i])\Big).$$

Remark (Failure modes)*.* A list fails to be a tagged partition either because the subintervals do not form a valid partition (the mesh is disordered or has gaps) or because at least one chosen tag lies outside its designated subinterval.

Remark (Failure mode decomposition)*.*

$$\underbrace{x_0 \neq a}_{\text{left endpoint defect}} \quad \lor \quad \underbrace{x_n \neq b}_{\text{right endpoint defect}} \quad \lor\ \underbrace{\exists j\ (x_{j-1} \geq x_j)}_{\text{ordering defect}} \lor \underbrace{\exists i\ (t_i \notin [x_{i-1}, x_i])}_{\text{tag misplaced}}.$$

Remark (Interpretation)*.* A tagged partition augments the usual subdivision of a closed interval by selecting a representative sampling point inside each component subinterval. These tags serve as evaluation points for Riemann sums and gauge integrals. The construction encapsulates both the geometric slicing of the interval and the analytic data needed for sample values. Failure occurs when the slices no longer tile the interval in order or when a sample point is chosen outside its slice, breaking the correspondence between tags and subintervals.

Remark (Dependencies).

Definition 2.8.1.

Definition (Gauge)

Definition 2.8.3 (Gauge)**.** Let $a, b \in \mathbb{R}$ with $a < b$ and set $I = [a, b]$. A gauge on I is a function $\delta : I \to (0, \infty)$.

Remark (Standard quantified statement).

$$\begin{aligned}&\text{Let } a, b \in \mathbb{R} \text{ with } a < b, \text{ put } I = [a, b], \text{ and let } \delta : I \to \mathbb{R}.\\ &\delta \text{ is a gauge on } I \iff \forall x \in I\ (\delta(x) > 0).\end{aligned}$$

Remark (Definition predicate reading).

$$\mathrm{Gauge}(\delta, I) :\equiv \big(\delta : I \to \mathbb{R}\big) \land \forall x \in I\ (\delta(x) > 0).$$

Remark (Negated quantified statement).

$$\begin{aligned}&\text{Let } a, b \in \mathbb{R} \text{ with } a < b, \text{ put } I = [a, b], \text{ and let } \delta : I \to \mathbb{R}.\\ &\delta \text{ fails to be a gauge on } I \iff \exists x \in I\ (\delta(x) \le 0).\end{aligned}$$

Remark (Negation predicate reading).

$$\neg\,\mathrm{Gauge}(\delta, I) \iff \neg\big(\delta : I \to \mathbb{R}\big) \lor \exists x \in I\ (\delta(x) \le 0).$$

Remark (Failure modes). A purported gauge fails either because it is not a function from I to $\mathbb{R}$, or because it takes on a nonpositive value somewhere in I.

Remark (Failure mode decomposition).

$$\underbrace{\neg\big(\delta : I \to \mathbb{R}\big)}_{\text{wrong function type}} \lor \underbrace{\exists x \in I\ (\delta(x) \le 0)}_{\text{nonpositive gauge value}}$$

Remark (Interpretation). A gauge assigns to each point of the closed interval a positive radius, encoding how tightly a tagged subinterval must cluster around that point in fine partition arguments. The sole obstruction is positivity: if any value is zero or negative, the construction cannot deliver legitimate local scales, so the gauge structure breaks down.

Remark (Dependencies).

- Function
- Closed Interval
- Order on $\mathbb{R}$

Definition (δ-Fine Partition)

Definition 2.8.4 (δ-Fine Partition)**.** Let $[a, b]$ be a closed interval, let $\delta : [a, b] \to (0, \infty)$ be a gauge, and let $\dot{\mathcal{P}} = \{([x_{i-1}, x_i], t_i)\}_{i=1}^{n}$ be a tagged partition of $[a, b]$. The tagged partition $\dot{\mathcal{P}}$ is δ-fine if and only if for every $i \in \{1, \ldots, n\}$ and every $x \in [x_{i-1}, x_i]$ the inequality $|x - t_i| < \delta(t_i)$ holds.

Remark (Standard quantified statement).

Let $[a,b]$ be a closed interval, $\delta : [a,b] \to (0,\infty)$, and $\dot{\mathcal{P}} = \{([x_{i-1},x_i],t_i)\}_{i=1}^n$ a tagged partition of $[a,b$
$\dot{\mathcal{P}}$ is δ-fine $\iff \forall i \in \{1,\dots,n\}\ \forall x \in [x_{i-1},x_i]\ (|x-t_i| < \delta(t_i))$.

Remark (Definition predicate reading).

$$\mathrm{DeltaFine}(\dot{\mathcal{P}},\delta) := \forall i \in \{1,\dots,n\}\ \forall x \in [x_{i-1},x_i]\ (|x-t_i| < \delta(t_i)).$$

Remark (Negated quantified statement).

$$\exists i \in \{1,\dots,n\}\ \exists x \in [x_{i-1},x_i]\ \big(|x-t_i| \ge \delta(t_i)\big).$$

Remark (Negation predicate reading).

$$\neg\,\mathrm{DeltaFine}(\dot{\mathcal{P}},\delta) \iff \exists i \in \{1,\dots,n\}\ \exists x \in [x_{i-1},x_i]\ (|x-t_i| \ge \delta(t_i)).$$

Remark (Failure modes). A tagged partition fails to be δ-fine precisely when some subinterval contains a point whose distance from its tag reaches or exceeds the available radius $\delta(t_i)$, so the corresponding subinterval is not contained in the prescribed open neighborhood around its tag.

Remark (Failure mode decomposition).

$$\underbrace{\exists i \in \{1,\dots,n\}\ \exists x \in [x_{i-1},x_i]}_{\text{choose offending subinterval and point}}\ \underbrace{(|x-t_i| \ge \delta(t_i))}_{\text{distance not controlled by } \delta}.$$

Remark (Interpretation). A δ-fine tagged partition refines the interval so that each subinterval sits entirely inside the open ball centered at its tag with radius dictated by the gauge. This ensures that when summing local contributions, every evaluation point remains within a tolerance chosen for that location, a key requirement for gauge-based integral constructions. Failure occurs when a subinterval is too wide for the tolerance prescribed at its tag, so some point of the subinterval lies on or outside the allowed neighborhood.

Remark (Dependencies).

Definition 2.8.3; Definition 2.8.2

Lemma

Lemma 2.8.5 (Every Point Is Covered by a Tag). *Let $\dot{\mathcal{P}} = \{([x_{i-1},x_i],t_i)\}_{i=1}^n$ be a tagged partition of $[a,b]$ that is δ-fine on $[a,b]$, and let $x \in [a,b]$. Then there exists $i \in \{1,\dots,n\}$ such that $|x-t_i| < \delta(t_i)$. Go to proof.*

Remark (Standard quantified statement).

Let $a<b$, $\delta : [a,b] \to (0,\infty)$, and $\dot{\mathcal{P}} = \{([x_{i-1},x_i],t_i)\}_{i=1}^n$ be δ-fine on $[a,b]$.
$\forall x \in [a,b]\ \exists i \in \{1,\dots,n\}\ (|x-t_i| < \delta(t_i))$.

Remark (Interpretation). A δ-fine tagged partition matches each subinterval to a tag that lies inside the interval and records the gauge radius allowed at that tag. The lemma asserts that every point of the underlying interval falls within the gauge ball centered at one of the tags, so the gauge control at that tag can be applied to the point. This coverage property is the mechanism that transfers local control encoded by the gauge to arbitrary points of the interval.

Remark (Dependencies).
Definition 2.8.2, Definition 2.8.4.

Theorem (Cousin's Theorem)

Theorem 2.8.6 (Cousin's Theorem)**.** *Every gauge* $\delta : [a,b] \to (0,\infty)$ *on a closed interval* $[a,b]$ *admits a* δ*-fine tagged partition of* $[a,b]$*. Go to proof.*

Remark (Standard quantified statement).

$$\begin{aligned}&\text{Let } a,b \in \mathbb{R} \text{ with } a<b \text{ and let } \delta : [a,b] \to (0,\infty).\\ &\exists m \in \mathbb{N}\ \exists x_0,\dots,x_m\ \exists t_1,\dots,t_m\\ &\qquad \big(a = x_0 < x_1 < \cdots < x_m = b \wedge \forall j \in \{1,\dots,m\}\ (t_j \in [x_{j-1},x_j])\\ &\qquad \wedge \forall j \in \{1,\dots,m\}\ [x_{j-1},x_j] \subseteq (t_j - \delta(t_j), t_j + \delta(t_j))\big).\end{aligned}$$

Remark (Predicate reading).

$$\begin{aligned}\forall \delta : [a,b] \to (0,\infty) \quad \exists \dot{\mathcal{P}} \Big(&\mathrm{TaggedPartition}(\dot{\mathcal{P}}, [a,b])\\ &\wedge \mathrm{DeltaFine}(\dot{\mathcal{P}}, \delta)\Big).\end{aligned}$$

Remark (Negated quantified statement).

$$\begin{aligned}&\text{Let } a,b \in \mathbb{R} \text{ with } a<b \text{ and let } \delta : [a,b] \to (0,\infty).\\ &\neg\Big(\text{there exists a } \delta\text{-fine tagged partition of } [a,b]\Big)\\ &\iff \forall m \in \mathbb{N}\ \forall x_0,\dots,x_m\ \forall t_1,\dots,t_m\\ &\qquad \Big[\big(a = x_0 < x_1 < \cdots < x_m = b \wedge \forall j\ (t_j \in [x_{j-1},x_j])\big)\\ &\qquad \Rightarrow \exists j \in \{1,\dots,m\}\ [x_{j-1},x_j] \not\subseteq (t_j - \delta(t_j), t_j + \delta(t_j))\Big].\end{aligned}$$

Remark (Negation predicate reading).

$$\exists \delta : [a,b] \to (0,\infty)\ \forall \dot{\mathcal{P}}\ \Big(\mathrm{TaggedPartition}(\dot{\mathcal{P}}, [a,b]) \Rightarrow \neg\,\mathrm{DeltaFine}(\dot{\mathcal{P}}, \delta)\Big).$$

Remark (Failure modes). Failure would mean that a positive gauge δ defeats every tagged partition: no matter how the interval is subdivided and tagged, at least one subinterval is not contained in the gauge neighborhood around its tag.

Remark (Failure mode decomposition).

$$\underbrace{\forall \dot{\mathcal{P}}\ \ \mathrm{TaggedPartition}(\dot{\mathcal{P}}, [a,b])}_{\text{all tagged partitions}} \quad \Rightarrow \quad \underbrace{\neg\,\mathrm{DeltaFine}(\dot{\mathcal{P}}, \delta)}_{\text{some interval escapes its gauge ball}}\ .$$

Remark (Interpretation). The theorem is the gauge analogue of compactness for a closed interval. Although the allowed radius may vary from point to point, finitely many tagged subintervals can still be chosen so that each lies inside the neighborhood dictated by its own tag. The standard bisection proof uses the Nested Interval Property to rule out a gauge that defeats all finite tagged partitions.

Remark (Dependencies).

Definition 2.8.3, Definition 2.8.2, Definition 2.8.4.

Remark (Gauge proofs of the four classical theorems)*.* Each proof follows the same pattern: define a gauge δ from the continuity hypothesis, invoke Cousin's theorem for a δ-fine partition, apply the Covering Lemma to reduce to finitely many tags, derive the global conclusion.

Boundedness. At each t, continuity gives $\delta(t)$ with $\mathrm{Within}(x, t, \delta(t)) \Rightarrow |f(x) - f(t)| \leq 1$. A δ-fine partition with tags $t_1, \ldots, t_n$ gives $K = \max_i |f(t_i)|$; the Covering Lemma yields $|f(x)| \leq 1 + K$ for all x.

EVT. Let $M = \sup f(I)$; suppose $f(x) < M$ everywhere. At each t, take $\delta(t)$ with $\mathrm{Within}(x, t, \delta(t)) \Rightarrow f(x) < \frac{1}{2}(M + f(t))$. A δ-fine partition produces $\widetilde{M} = \frac{1}{2}(M + \max_i f(t_i)) < M$ bounding f everywhere — contradiction.

Location of Roots. Assume $f(t) \neq 0$ for all t. At each t, take $\delta(t)$ preserving $\mathrm{sgn}(f(t))$. A δ-fine partition forces consistent sign from $f(a) < 0$ to $f(b) > 0$ — contradiction.

Heine–Cantor. Given $\varepsilon > 0$, at each t let $\delta(t)$ satisfy $\mathrm{Within}(x, t, 2\delta(t)) \Rightarrow |f(x) - f(t)| \leq \varepsilon/2$. A δ-fine partition yields $\delta_\varepsilon = \min_i \delta(t_i) > 0$. For $|x - u| < \delta_\varepsilon$: find t_i covering x; then $|u - t_i| \leq |u - x| + |x - t_i| < 2\delta(t_i)$; so $|f(x) - f(u)| \leq \varepsilon$.

Tagged Partitions and Mesh

Definition (Mesh of a Partition)

Definition 2.8.7 (Mesh of a Partition)**.** Let $a, b \in \mathbb{R}$ with $a < b$, and let $P = \{x_0, \ldots, x_n\}$ be a partition of $[a, b]$. A real number m is the *mesh* (or *norm*) of P if and only if

$$m = \max_{1 \leq i \leq n} (x_i - x_{i-1}).$$

The mesh of P is denoted by $\|P\|$ and is the unique m satisfying this equivalence.

Remark (Standard quantified statement).

$$\text{Let } a, b \in \mathbb{R} \text{ with } a < b, \text{ and let } P = \{x_0, \ldots, x_n\} \text{ be a partition of } [a, b].$$
$$m \text{ is the mesh of } P \iff m = \max_{1 \leq i \leq n} (x_i - x_{i-1}).$$

Remark (Definition predicate reading).

$$\mathrm{NormOfPartition}(P, m) :\equiv m = \max_{1 \leq i \leq n} (x_i - x_{i-1}).$$

Remark (Negated quantified statement).

$$\text{Let } a, b \in \mathbb{R} \text{ with } a < b, \text{ and let } P = \{x_0, \ldots, x_n\} \text{ be a partition of } [a, b].$$
$$\neg\big(m \text{ is the mesh of } P\big) \iff \big(m < \max_{1 \leq i \leq n} (x_i - x_{i-1})\big) \lor \big(m > \max_{1 \leq i \leq n} (x_i - x_{i-1})\big).$$

Remark (Negation predicate reading).

$$\neg\,\mathrm{NormOfPartition}(P, m) \iff \big(m < \max_{1 \leq i \leq n} (x_i - x_{i-1})\big) \lor \big(m > \max_{1 \leq i \leq n} (x_i - x_{i-1})\big).$$

Remark (Failure modes). Failure occurs either when m underestimates the largest subinterval of P or when m overshoots that largest subinterval.

Remark (Failure mode decomposition).

$$\neg\,\mathrm{NormOfPartition}(P, m) = \underbrace{\Big(m < \max_{1\le i\le n}(x_i - x_{i-1})\Big)}_{\text{Underestimate}} \lor \underbrace{\Big(m > \max_{1\le i\le n}(x_i - x_{i-1})\Big)}_{\text{Overestimate}}.$$

Remark (Interpretation). The mesh of a finite partition records the length of the widest subinterval and thus measures the resolution of the partition. Refining a partition strictly decreases or preserves this value, so a small mesh signals uniformly short subintervals, a critical input for Riemann and gauge integral estimates. Failure arises when a proposed number does not coincide with the actual maximum subinterval length, either because some subinterval is longer than the proposal or because all subintervals are shorter.

Remark (Dependencies).

Definition 2.8.1.

Definition (Refinement)

Definition 2.8.8 (Refinement)**.** Let $[a, b] \subseteq \mathbb{R}$ and let P and Q be partitions of $[a, b]$. Then Q is a refinement of P if and only if every point of P is also a point of Q; equivalently, $P \subseteq Q$.

Remark (Standard quantified statement).

$$\forall x\ (x \in P \Rightarrow x \in Q).$$

Remark (Definition predicate reading).

$$\mathrm{Refinement}(Q, P) := P \subseteq Q.$$

Remark (Negated quantified statement).

$$\exists x\ (x \in P \land x \notin Q).$$

Remark (Negation predicate reading).

$$\neg\,\mathrm{Refinement}(Q, P) \iff \exists x \in P\ (x \notin Q).$$

Remark (Failure modes). Failure occurs precisely when some point listed in P fails to appear in Q, meaning Q omits at least one of the partition points it was supposed to inherit from P.

Remark (Failure mode decomposition).

$$\underbrace{\exists x \in P\ (x \notin Q)}_{\text{point of } P \text{ omitted by } Q}.$$

Remark (Interpretation). Refinement formalizes the idea of subdividing an interval more finely without losing any previously chosen partition points. The condition $P \subseteq Q$ guarantees that every subinterval defined by P is still represented when one passes to Q, allowing comparisons of sums or integrals computed on different meshes. Failure means that Q discards at least one point of P, so information tied to that breakpoint would be lost when transitioning between partitions.

Remark (Dependencies).

Definition 2.8.1.

Definition (Common Refinement)

Definition 2.8.9 (Common Refinement)**.** Let $a < b$ and let P and Q be partitions of $[a, b]$. Their common refinement is the partition R obtained by taking the union of their node sets, so $R = P \cup Q$ as sets of points in $[a, b]$.

Remark (Standard quantified statement).

$$\begin{aligned}&\text{Let } a < b \text{ in } \mathbb{R}, \text{ and let } P, Q \text{ be partitions of } [a, b].\\ &\forall R \text{ partition of } [a, b] : \; R \text{ is the common refinement of } P \text{ and } Q \iff R = P \cup Q.\end{aligned}$$

Remark (Definition predicate reading).

$$\mathrm{CommonRefinement}(R, P, Q) \coloneqq (R = P \cup Q).$$

Remark (Negated quantified statement).

$$\begin{aligned}&a < b \wedge P, Q \subseteq [a, b] \wedge R \subseteq [a, b]\\ &\implies \neg\big(\forall x \in \mathbb{R}\ (x \in R \iff x \in P \cup Q)\big)\\ &\iff \big(\exists x \in R\ (x \notin P \cup Q)\big) \vee \big(\exists x \in P \cup Q\ (x \notin R)\big).\end{aligned}$$

Remark (Negation predicate reading).

$$\neg\,\mathrm{CommonRefinement}(R, P, Q) \iff R \neq P \cup Q.$$

Remark (Failure modes). The claim that a partition R is the common refinement of P and Q fails exactly when R either contains a node not present in $P \cup Q$ or omits a node that $P \cup Q$ contributes. These are the only two structural ways in which R can differ from $P \cup Q$.

Remark (Failure mode decomposition).

$$R \neq P \cup Q \iff \underbrace{\exists x \in R \setminus (P \cup Q)}_{\text{extra node}} \vee \underbrace{\exists x \in (P \cup Q) \setminus R}_{\text{missing node}}.$$

Remark (Interpretation). Forming the common refinement of two partitions simply collects every partition point that either partition already uses, producing the unique partition whose nodes are exactly those existing ones. No new nodes are created beyond those already in P or Q, and none are discarded; the construction merely merges the lists so that later comparisons, such as evaluating tagged sums on both partitions simultaneously, can be performed without further adjustment.

Remark (Dependencies).

Definition 2.8.1

Bibliography